

Functions

Unit 1-3 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 1-3 slides and follow them section by section. They go further than the slides: every idea comes with numbered worked examples, most of them from the clinic or the data table, and the traps students actually fall into are flagged in boxes titled **Watch out**. Every exercise is followed by a worked solution; try each one before reading on.

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1 Why Functions?

1.1 A Problem to Motivate Us

Suppose the pharmacy asks you to mix an IV bag:

“We need a 500 mL bag of saline at a concentration of 4 mg/mL. How many mg of drug go in?”

The rule is one you already know: the mass of drug dissolved in a liquid is its concentration times the volume,

$$m = c \cdot V.$$

Here m is the mass in mg, c is the concentration in mg/mL, and V is the volume in mL. The units agree: mg/mL times mL leaves mg.

Now fix the bag at $V = 500$ mL. The rule stops being a relationship among three quantities and becomes a **machine**: feed it a target concentration, and it returns the mass to weigh out,

$$m(c) = 500 c.$$

For the order above, $m(4) = 500 \times 4 = 2000$ mg.

That machine is a **function**, and functions are the subject of these notes. Running the machine forward, from $c = 4$ to $m = 2000$, is called **evaluation**. Running it backward is just as natural a question: “the vial holds 1500 mg; what concentration will that give?” Now the output is known and the input is wanted ($1500 = 500 c$, so $c = 3$ mg/mL). Running a function backward is called taking its **inverse**. Both ideas are in this lecture, and inverses get a section of their own (Section 7).

Before any definitions, it is worth running the machine a few times, because every idea in these notes is already visible in this one small example.

Example 1.1: Running the IV machine forward and backward. Keep the 500 mL bag, so $m(c) = 500 c$, and fill in an order sheet. Forward, each concentration is multiplied by 500:

concentration c (mg/mL)	0.5	1	2.5	4
mass $m(c)$ (mg)	250	500	1250	2000

Backward, a known mass is divided by 500: a 750 mg vial gives $c = 750/500 = 1.5$ mg/mL, and a 3000 mg vial gives $c = 6$ mg/mL. So the forward machine **multiplies** by 500 and the backward machine **divides** by 500. A rule paired with the rule that undoes it is exactly what Section 7 is about.

Example 1.2: Which quantity is the input is a choice. The relationship $m = c \cdot V$ involves three quantities, and fixing a *different* one produces a different function. Suppose the pharmacy stocks a single premixed concentration, $c = 4$ mg/mL, and the question is how much drug is in a given volume. Now the volume is the input:

$$m(V) = 4 V, \quad m(250) = 1000 \text{ mg}, \quad m(1000) = 4000 \text{ mg}.$$

Same law of nature, same units, but a different machine: this one takes a volume in mL, not a concentration. Deciding which quantity goes in and which comes out is the first modeling decision in any problem, and it comes back in Section 9 as “which column plays which role.”

Example 1.3: Weight-based dosing. Many drugs are ordered per kilogram of body weight. A drug ordered at 15 mg/kg gives the function

$$D(w) = 15 w,$$

where w is the patient’s weight in kg and $D(w)$ is the dose in mg. A 70 kg patient gets $D(70) = 15 \times 70 = 1050$ mg. Run it backward and a 900 mg dose turns out to be the right dose for a patient of $900/15 = 60$ kg. Each weight produces one dose, never two, and that single fact is what will make D a **function** in the sense of Section 2.

1.2 Where This Shows Up in Data Science

Almost everything you will build in data science is a function:

- **A model is a function:** it takes a patient’s measurements and returns a prediction.
- **Every data transformation is a function:** rescaling a column, taking a log, converting units.
- **Error is a function:** “how wrong is this model?” is a number that depends on the model’s settings.
- **Training is a search over functions:** pick a family of candidate functions, then pick the best member of it.
- **Reading a result backward,** asking “what input would have given me this output?”, is an inverse.

Example 1.4: Three functions hiding in one cleaning step. An electronic health record (EHR) stores a patient’s weight as 176 lb and height as 1.75 m, and an analyst wants a column flagging obesity. Each step of the work is a machine with one input and one output:

1. **Convert units:** pounds in, kilograms out, by dividing by 2.20462. Here $176/2.20462 \approx 79.83$ kg.
2. **Compute BMI:** weight in kg in (height held at 1.75 m), BMI out, by dividing by $1.75^2 = 3.0625$. Here $79.83/3.0625 \approx 26.07$.
3. **Flag:** BMI in, a yes or no out: “yes” when BMI is at least 30. Here the answer is “no.”

Each step’s output is the next step’s input. That chaining is called **composition**, and it gets its own section (Section 5). The last step is also a function whose outputs are not numbers at all, which is perfectly allowed.

Unit 1-2 built **sets**. A function is what connects two sets: it sends each element of one set to an element of the other. So the language of Unit 1-2 (elements, $\in$, intervals, intersections, ordered pairs) is exactly the language we need to describe functions precisely.

1.3 What We Will Build

The plan for these notes, in order:

1. **Relations**, and what makes one a **function**.
2. **Domain** and **range**: which inputs are valid, and which outputs occur.
3. The **families**: transformations, polynomials, roots, reciprocals, exponentials, logarithms, trigonometric and piecewise rules.
4. **Combining** functions, especially **composition**.
5. **Quantifiers**: the “for all” and “there exists” that our definitions are written with.
6. **Inverses**: undoing a function, and when that is possible.

7. Functions of **several variables**: the **Cartesian product** their inputs live in, and the **surfaces** they graph as.
8. Functions in **machine learning**: from a data table to a prediction.

Five notebooks go with this lecture:

- `U1-3_Functions-1_DomainRangeAndFamilies`: evaluating functions, domain and range, and the four main families (Sections 2 to 4).
- `U1-3_Functions-2_Composition`: composing functions, and why the order matters (Section 5).
- `U1-3_Functions-3_Inverse`: one-to-one functions, the horizontal line test, and finding inverses (Section 7).
- `U1-3_Functions-4_FormulaExplorer`: type in any formula and get a slider for each of its constants (Section 4).
- `U1-3_Functions-5_SeveralVariables`: functions of several variables and machine learning (Sections 8 and 9). Its section numbers match the notebook pointers printed in those two sections.

2 From Relations to Functions

2.1 Pairing Things Up

Unit 1-2 ended with **ordered pairs**: a pair (a, b) in which the order matters, so $(1, 2) \neq (2, 1)$. An ordered pair is the simplest way to record that two things go together, such as a patient and a date of birth, or an x and a y . A whole collection of such pairings is called a relation.

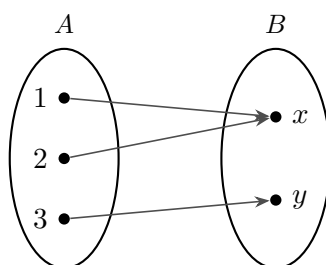
Definition: Relation

A **relation** R from a set A to a set B is any set of ordered pairs (a, b) with $a \in A$ and $b \in B$. If $(a, b) \in R$ we say “ a is related to b .”

A relation is just a **set of links**. Drawing one arrow per pair turns the list into a picture, called a **mapping diagram**. Here is the relation

$$R = \{(1, x), (2, x), (3, y)\}$$

from $A = \{1, 2, 3\}$ to $B = \{x, y\}$:

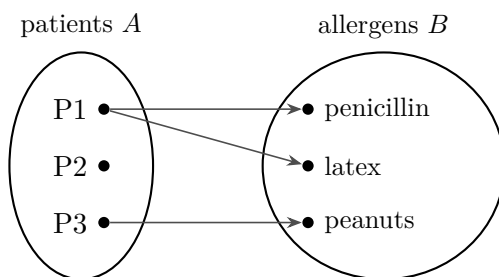


Nothing in a relation is required to be tidy. An element of A may link to many elements of B , to exactly one, or to none at all, and an element of B may receive any number of arrows. A function, as we will see shortly, is a relation that obeys a particular tidiness rule.

Example 2.1: A relation from the clinic: patients and allergies. Let $A = \{P1, P2, P3\}$ be three patients and $B = \{\text{penicillin}, \text{latex}, \text{peanuts}\}$ three allergens. The clinic's allergy list is

$$R = \{(P1, \text{penicillin}), (P1, \text{latex}), (P3, \text{peanuts})\}.$$

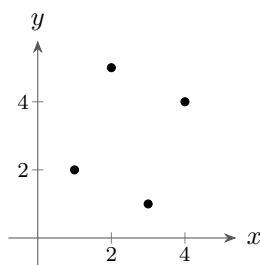
This is a relation from A to B : a set of ordered pairs whose first entry is a patient and whose second entry is an allergen. As a mapping diagram:



The relation is untidy in both of the ways a relation is allowed to be. P1 is linked to **two** allergens, and P2 is linked to **none** (no known allergies). Both situations are completely normal for an allergy list, and, as the next subsections show, each of them on its own is enough to stop the relation from being a function.

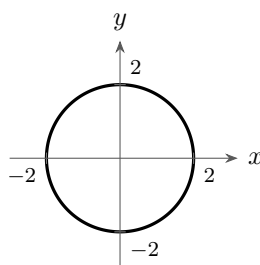
2.2 Relations Live in the Plane

When A and B are both the real numbers $\mathbb{R}$, a pair (x, y) is a **point** in the plane, with the x -coordinate first and the y -coordinate second. So a relation between real numbers is a **picture**: every set of points you can draw is a relation. Here are three.



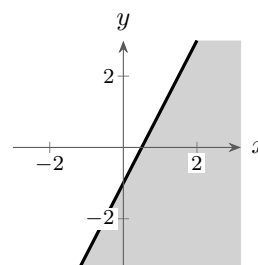
a handful of points

$$R = \{(1, 2), (2, 5), (3, 1), (4, 4)\}$$



a curve

$$R = \{(x, y) \mid x^2 + y^2 = 4\}$$



a whole region

$$R = \{(x, y) \mid y \leq 2x - 1\}$$

The last two are written in the set-builder notation of Unit 1-2: $\{(x, y) \mid x^2 + y^2 = 4\}$ is read “the set of all pairs (x, y) such that $x^2 + y^2 = 4$,” which is the circle of radius 2 centered at the origin. The third is every point on or below the line $y = 2x - 1$.

Example 2.2: Is this point in the relation? A relation written in set-builder notation comes with a membership test: substitute the point's coordinates into the condition and see whether it holds.

- For the circle $x^2 + y^2 = 4$: the point $(0, 2)$ gives $0 + 4 = 4$, so it is **in**. The point $(1, 1)$ gives $1 + 1 = 2 \neq 4$, so it is **not** (it lies inside the circle, not on it). The point $(\sqrt{2}, \sqrt{2})$ gives $2 + 2 = 4$, so it is in.
- For the half-plane $y \leq 2x - 1$: the point $(3, 4)$ gives $4 \leq 5$, **in**. The point $(0, 0)$ gives $0 \leq -1$, which is false, so it is **not** in. The point $(1, 1)$ gives $1 \leq 1$, which is true, so it is in: it sits exactly on the boundary line, and “ $\leq$ ” includes the boundary.

Reading a relation this way, as a yes-or-no test on pairs, is exactly how a database filter works: each row is a tuple, and the condition decides whether it is kept.

All three are perfectly good relations: a finite list, a curve, and a shaded half-plane. Only one of them will turn out to be a **function**; once we have the vertical line test (Section 2.9) you will be able to say which.

2.3 When Is a Relation a Function?

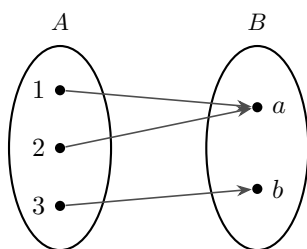
Definition: Function

A **function** f from A to B is a relation in which **every** element of A is related to **exactly one** element of B . In symbols,

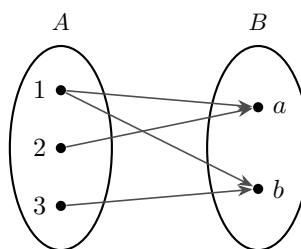
$$\forall a \in A, \exists! b \in B \text{ with } (a, b) \in f.$$

The symbols in that last line are shorthand that Section 6 studies properly: $\forall$ is read “for every” and $\exists!$ is read “there exists exactly one.” So the line says, word for word, “for every a in A there is exactly one b in B with (a, b) in f .”

Compare these two mapping diagrams.



Is a function:
one arrow out of each input



Not a function:
1 has two arrows out

The pictures are identical except for **one extra arrow**, $1 \rightarrow b$, and that alone breaks the second one: the input 1 now has two outputs. This is the distinction to keep:

- Two inputs sharing an output is fine. On the left, 1 and 2 both go to a ; this is called **many-to-one**.
- One input with two outputs is not. On the right, 1 goes to both a and b ; this is called **one-to-many**, and it is exactly what the definition forbids.

In plain words: a function is a relation you can **look things up in**. Hand it any input and it hands back one definite answer, never “it depends” and never “nothing.”

Example 2.3: Functions and non-functions in a patient record. Test each pairing against the definition: is every input used, and does each input get exactly one output?

- **Patient** $\mapsto$ **blood type** is a function. Every patient has a blood type, and only one. Many patients share type O+, but sharing an output is many-to-one, which is allowed.
- **Patient** $\mapsto$ **medical record number (MRN)** is a function, and an especially tidy one: every patient has one MRN, and no two patients share one.
- **Patient** $\mapsto$ **allergen**, the relation of Example 2.1, is **not** a function, and it fails twice over: P1 has two outputs, and P2 has none.
- **Patient** $\mapsto$ **current medication** is not a function either: one patient may be on five drugs, another on none.

A failing pairing can often be repaired by changing what counts as the output. “Patient $\mapsto$ *the set of that patient’s allergens*” is a function: P1 goes to {penicillin, latex}, P2 goes to the empty set $\emptyset$ of Unit 1-2, and P3 goes to {peanuts}. Every patient now has exactly one output, which happens to be a set. This is the same trick a database uses when it stores a list in a single cell.

Example 2.4: Function or not, from a list of pairs. Decide whether each set of pairs is a function from $A = \{-1, 0, 1, 2\}$ to $\mathbb{R}$.

1. $\{(-1, 1), (0, 0), (1, 1), (2, 4)\}$. The first entries are $-1, 0, 1, 2$, each exactly once, so every input has exactly one output: **yes**. The output 1 appears twice, from -1 and from 1 , but that is many-to-one. (These are the pairs of $y = x^2$ on A .)
2. $\{(-1, 3), (0, 3), (1, 3), (2, 3)\}$. Every input once, all sharing the output 3: **yes**. This is a **constant function**, as many-to-one as a function can be.
3. $\{(-1, 0), (0, 1), (1, 2), (1, 5), (2, 3)\}$. The input 1 appears twice, with outputs 2 and 5: **no**, “at most one” fails.
4. $\{(-1, 0), (0, 1), (2, 3)\}$. The input 1 never appears: **no**, “at least one” fails.

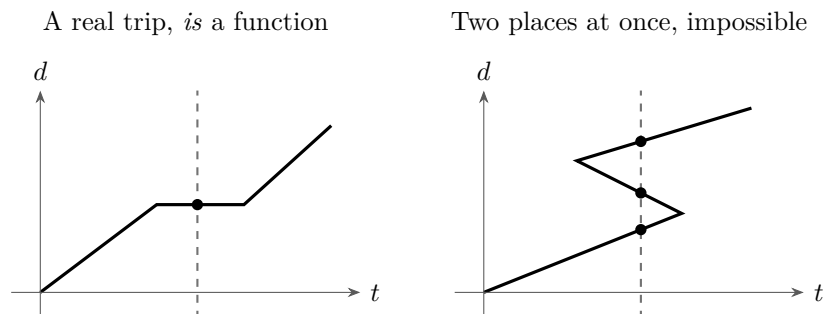
For a finite list, the whole check is a scan of the **first** entries: every element of A must appear there exactly once. The second entries can repeat freely.

Watch out: repeated outputs are not the problem

The most common wrong answer is to reject a relation because two inputs share an output, as in #1 and #2 above. That is many-to-one, and it is legal. Only a repeated **input** with different outputs, or a missing input, disqualifies a relation. Look down the *input* column, not the output column.

2.4 Why “Exactly One”? A Drive Across Town

Why should a function be allowed to repeat outputs but not inputs? A physical example makes the rule feel natural. You get in the car and drive. Let t be the **time** since you pulled out, and $d(t)$ the **distance** you have covered by then. At any instant you are somewhere on the road, and you are somewhere **once**.

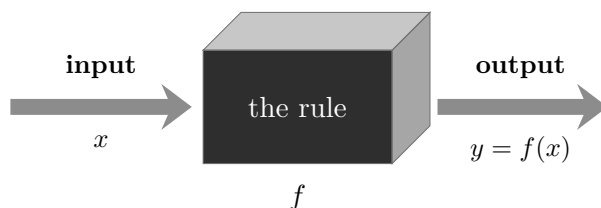


The dashed vertical line in each graph marks a single instant t . Read the two graphs along it.

- On the left, the **flat stretch** is the red light you sat at: time ran on while d stayed put, so many different times share one distance. That is **many-to-one**, and it is perfectly legal. The vertical line meets the graph once.
- The right-hand graph doubles back on itself. The vertical line meets it at **three** points, so that one instant would have three different distances: the car would be in three places at once. That is **one-to-many**, and it is not a function (nor a possible trip).

The rule to remember: repeats read **horizontally** (several times giving the same distance) are fine; repeats read **vertically** (one time giving several distances) are not.

2.5 A Function Is a Machine



The most useful mental picture of a function is a machine. Feed in an x , the rule runs, and one y comes out, which we write $y = f(x)$. The phrase “exactly one” in the definition hides two separate demands, and it is worth naming them, because a relation can fail either one:

- **At least one:** no input is left without an output. Feed the machine a valid input and something must come out.
- **At most one:** no input produces two different outputs. Feed it the same input twice and you get the same answer twice.

Example 2.5: Is “patient $\mapsto$ systolic blood pressure” a function? Not as stated, and it is the “at most one” demand that fails. Measure the same patient at 8 am and again at 2 pm and you will get two different readings, say 128 and 141 mmHg; the input “this patient” has produced two outputs. The fix is to make the input more specific. The rule

(patient, time of measurement) $\mapsto$ the reading taken then

is a function: a given patient at a given moment has one reading. Its input is now a *pair*, which makes it a function of two variables, the subject of Section 8. The lesson carries over to every data set: if the “same” input seems to give different outputs, something that matters has been left out of the input.

The IV bag rule $m(c) = 500c$ from Section 1 is exactly this: a box you drop a concentration into and read a mass out of. The rest of these notes ask three questions about the box: what may go **in** (the domain, Section 3), what comes **out** (the range, in the same section), and can we run it **backward** (the inverse, Section 7)?

2.6 Notation

Reading $f : A \rightarrow B$

“ f is a function from A to B .” Here A is the **domain**, the set of valid inputs, and the outputs land in B .

If $f(x) = y$, then x is the **input** (or **argument**) and y is the **output**, also called the **value of f at x** . For the IV bag, $m(4) = 2000$: the argument is 4, and the value of m at 4 is 2000. Notice that B is only a set the outputs are guaranteed to land in. Nothing says that every element of B is actually produced; which outputs really occur is the question of the *range*, in Section 3.

A distinction worth keeping straight: f is the **rule**, and $f(x)$ is the **number** that rule produces at x . So writing “ $f = x^2$ ” is sloppy, while “ $f(x) = x^2$ ” is right: the first sets a rule equal to a number, and the second says which number the rule produces for each x . Functions are usually named f , g , h , but any name works, and in code they get descriptive names like **normalize** or **predict**.

2.7 Evaluating a Function

Evaluating f at an input means running the machine once: replace *every* x in the formula by the input, wrapped in parentheses, and do the arithmetic. The parentheses are not decoration. They keep the whole input together, which matters most when the input is negative or is itself an expression.

Example 2.6: Reading $f(x) = y$ aloud. A common rule of thumb estimates a person’s maximum heart rate, in beats per minute, from age a in years:

$$H(a) = 220 - a.$$

The statement $H(40) = 180$ reads “the value of H at 40 is 180,” and in context, “a 40-year-old’s estimated maximum heart rate is 180 bpm.” Likewise $H(65) = 155$. Asked backward, “whose estimated maximum is 170?”, you solve $220 - a = 170$ and get $a = 50$. Being able to translate between the symbols and the sentence is most of what reading a function means.

Example 2.7: Evaluating at numbers. Let $f(x) = x^2 - 3x + 2$. Replace each x by the input in parentheses:

$$\begin{aligned} f(0) &= (0)^2 - 3(0) + 2 = 2, \\ f(-2) &= (-2)^2 - 3(-2) + 2 = 4 + 6 + 2 = 12, \\ f\left(\frac{1}{2}\right) &= \left(\frac{1}{2}\right)^2 - 3\left(\frac{1}{2}\right) + 2 = \frac{1}{4} - \frac{3}{2} + 2 = \frac{3}{4}. \end{aligned}$$

The parentheses earn their keep at $x = -2$: $(-2)^2 = 4$, but -2^2 written without them means $-(2^2) = -4$; and $-3(-2) = +6$, where dropping the parentheses invites the slip $-3 - 2 = -5$. Both are sign errors, and both are prevented by writing the parentheses every time.

Example 2.8: Evaluating at an expression. The input can be a whole expression rather than a number. For $f(a + 1)$, every x becomes $(a + 1)$:

$$\begin{aligned} f(a + 1) &= (a + 1)^2 - 3(a + 1) + 2 \\ &= a^2 + 2a + 1 - 3a - 3 + 2 \\ &= a^2 - a. \end{aligned}$$

Spot check at $a = 2$: the simplified formula gives $4 - 2 = 2$, and directly, $f(3) = 9 - 9 + 2 = 2$. ✓ A spot check at one convenient number is the cheapest insurance there is against algebra slips. Substituting an expression into a function is also exactly the move that **composition** is built from (Section 5), so it is worth becoming fluent now.

Watch out: $f(\dots)$ is not multiplication

$f(a + 1)$ is **not** $f(a) + 1$, and $f(2x)$ is **not** $2f(x)$. With $f(x) = x^2 - 3x + 2$ at $x = 3$:

$$f(2 \cdot 3) = f(6) = 36 - 18 + 2 = 20, \quad \text{but} \quad 2f(3) = 2 \cdot 2 = 4.$$

The parentheses in $f(\dots)$ mean “apply the rule to this.” They do not mean “multiply f by this,” so you cannot distribute f over a sum or pull a number out of it.

2.8 The Same Function, Four Ways

Take $f(x) = 2x + 1$ on all of $\mathbb{R}$. The same function can be written down in four ways:

As a formula	$f(x) = 2x + 1$			compact and exact	
As a set of pairs	$\{ (x, 2x + 1) \mid x \in \mathbb{R} \}$			<i>every</i> pair, not a list	
As a table	x	0	1	2	a sample of pairs
	$f(x)$	1	3	5	
As a graph	the line through all the pairs				the whole picture

The four are not equally complete. The domain here is all of $\mathbb{R}$, so the function really pairs *every* real number with its output, including in-between values such as $f(0.5) = 2$ and $f(-\pi) = 1 - 2\pi$. The formula, the set of pairs, and the graph all capture every one of those pairs. Only the **table** is incomplete: it can only ever show finitely many rows, which is why we still write the formula.

That gap is the whole job of modeling. Real data arrives as a **table**, a finite sample of input and output pairs, and modeling is guessing the formula behind it. Section 9 comes back to this.

Example 2.9: From a table to a formula. A nurse logs the volume left in a 1000 mL IV bag every hour:

hours since start, t	0	1	2	3
volume left, V (mL)	1000	875	750	625

Each hour the volume drops by the same 125 mL, so a formula that reproduces every row is

$$V(t) = 1000 - 125t.$$

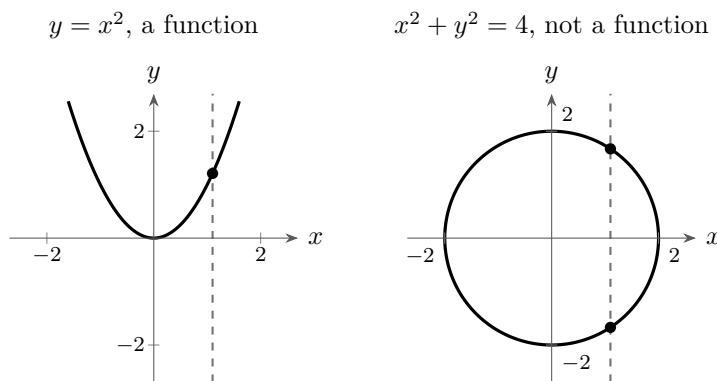
Check a row: $V(2) = 1000 - 250 = 750$. ✓ Now the formula does what the table cannot. It answers for times between the logged ones, such as $V(1.5) = 1000 - 187.5 = 812.5$ mL, and it answers questions in reverse: the bag reaches 250 mL when $1000 - 125t = 250$, that is at $t = 6$ hours. The same function now exists in all four forms: this formula; the set of pairs $\{(t, 1000 - 125t) \mid 0 \leq t \leq 8\}$, stopping at $t = 8$ because that is when the bag runs empty; the logged table, as a sample of those pairs; and a falling straight line as its graph.

2.9 The Vertical Line Test

The graph of f is a picture of all the pairs $(x, f(x))$. Every point on a vertical line has the same x -coordinate, so a vertical line crossing the graph twice would mean one input with two outputs. “One output per input” therefore has a visual meaning.

The vertical line test

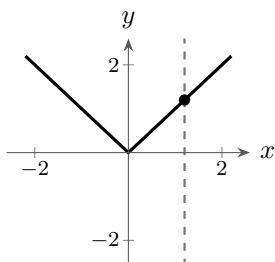
No vertical line may cross the graph of a function twice. A curve that some vertical line crosses twice is not the graph of y as a function of x .



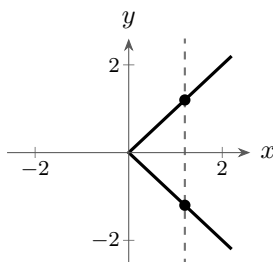
The parabola $y = x^2$ passes: every vertical line meets it exactly once. The circle fails: the dashed line at $x = 1.1$ meets it at two heights, one above the axis and one below. That one input meets **two** outputs, so the circle does not define y as a function of x . The circle is a perfectly good *relation*, just not a function. It is the same reason a position-versus-time graph may never run vertically: one input, one output.

Now go back to the three relations of Section 2.2. The circle fails the test, and so does the shaded half-plane, since a vertical line through it meets infinitely many shaded points. The finite list $\{(1, 2), (2, 5), (3, 1), (4, 4)\}$ passes, because no two of its points share an x -coordinate. It is the one function among the three.

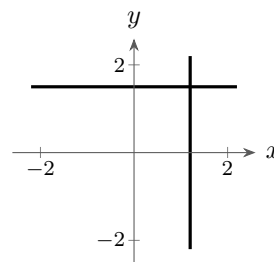
Example 2.10: Three more graphs through the test. In each panel the dashed line is one vertical test line; the dots mark where it meets the graph.



$y = |x|$: passes



$x = |y|$: fails



$x = 1.2$ fails; $y = 1.5$ passes

- $y = |x|$, the V-shape, meets every vertical line exactly once: each x has one absolute value. **A function.**
- $x = |y|$ is the same V turned on its side. The line at $x = 1.2$ meets it at $y = 1.2$ and $y = -1.2$, since both have absolute value 1.2. Not a function of x .
- The vertical line $x = 1.2$ is the extreme failure: a vertical test line placed on top of it meets it at *every* height, so one input has infinitely many outputs. The horizontal line $y = 1.5$, by contrast, passes: every x gets the single output 1.5. It is a constant function.

Watch out: vertical, not horizontal

The test uses **vertical** lines because a vertical line holds the *input* fixed. A **horizontal** line crossing a graph several times is fine: that is several inputs sharing one output, many-to-one, and $y = x^2$ does it at every positive height. (Horizontal lines do have a job of their own. They test whether a function can be run backward, in Section 7.)

Exercise 2.1

Which of these are functions from A to B ? (The arrow $\mapsto$ is read “is sent to.”)

1. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (2, 4), (3, 5)\}$
2. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (1, 5), (2, 4), (3, 5)\}$
3. $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (3, 5)\}$
4. Each patient $\mapsto$ that patient’s date of birth.
5. Each date of birth $\mapsto$ the patient born on it.
6. $y^2 = x$, taking x as the input and y as the output.

For each one that fails, say *which* of the two demands (“at least one” or “at most one”) it breaks.

Solution to Exercise 2.1

For each input, ask two questions: does it have an arrow out (at least one), and does it have only one (at most one)?

1. **Yes.** All three inputs are used, with one arrow each. The output 4 is hit twice, by 1 and by 2, but that is many-to-one, which is allowed.
2. **No.** The input 1 is paired with both 4 and 5, so “at most one” fails.

3. **No.** The input 2 has no output at all, so “at least one” fails.
4. **Yes.** A person has exactly one date of birth. Many patients may share a birthday, but that is many-to-one again.
5. **No.** Many patients share a birthday, so one date leads to many patients: one-to-many, and “at most one” fails. (A date on which none of the patients was born would have no output either, so “at least one” can fail too.)
6. **No.** The input $x = 4$ gives $y = 2$ **and** $y = -2$, since both square to 4, so “at most one” fails. The graph of $y^2 = x$ is a parabola lying on its side, and it fails the vertical line test. (If every real x counts as an input, a negative one such as $x = -1$ has no y at all, since no real number squares to a negative, so “at least one” fails as well.)

Compare #4 and #5: they are the same pairing with the arrows reversed, yet one is a function and the other is not. Reversing a function’s arrows does **not** always give a function back. That single observation is the whole story of **inverses**, in Section 7.

2.10 More Practice: Relations and Functions

Exercise 2.2

Let $g(x) = 2x^2 - x$. Compute $g(-1)$, $g(3)$, and $g(a - 1)$, simplifying the last one. Then spot-check your formula for $g(a - 1)$ at one value of a .

Solution to Exercise 2.2

Replace every x by the input, in parentheses:

$$g(-1) = 2(-1)^2 - (-1) = 2 + 1 = 3,$$

$$g(3) = 2(3)^2 - 3 = 18 - 3 = 15,$$

$$g(a - 1) = 2(a - 1)^2 - (a - 1) = 2(a^2 - 2a + 1) - a + 1 = 2a^2 - 5a + 3.$$

Spot check at $a = 4$: the formula gives $2(16) - 20 + 3 = 15$, and $g(4 - 1) = g(3) = 15$ from the line above. ✓ Notice the minus sign in front of $(a - 1)$: it flips both terms, giving $-a + 1$. Forgetting to flip the second term is the usual slip.

Exercise 2.3

Which of these equations define y as a function of x (for real x and y)? Justify each answer with a specific input where possible.

1. $y = 3 - x^2$

2. $x = y^2 + 1$

3. $y^3 = x$

4. $x = 5$

Solution to Exercise 2.3

1. **Yes.** The formula hands back one y for every real x : square, then subtract from 3. (It is an upside-down parabola, and every vertical line meets it once.)
2. **No.** Take $x = 2$: then $y^2 = 1$, so $y = 1$ **and** $y = -1$ both work. One input, two outputs. (Also, $x = 0$ gives $y^2 = -1$, which has no real solution, so “at least one” fails too.)
3. **Yes.** Every real number has exactly one real cube root, so $y = \sqrt[3]{x}$ is determined by x . For example $x = -8$ gives only $y = -2$. Cubing, unlike squaring, does not lose the sign, so there is no second candidate.
4. **No.** This is a vertical line. The single input $x = 5$ is paired with every y at once, and every other input is paired with nothing.

The pattern: an equation that must be solved for y fails as soon as solving it produces a $\pm$ (#2) or produces nothing.

Exercise 2.4

A hospital lab system maps each specimen ID to its glucose result. Some specimens were rerun after an instrument error, and the system stored both results for them. Is “specimen ID $\mapsto$ glucose result” a function? If not, which demand fails, and suggest two ways a data analyst might repair it.

Solution to Exercise 2.4

No. A rerun specimen has two results, so one input has two outputs and “at most one” fails. Two standard repairs:

- **Keep one output per input** by a stated rule, for example “keep the most recent result” or “keep the result flagged as valid.” The relation is trimmed back to a function.
- **Enlarge the input** to the pair (specimen ID, run number). Each run has one result, so this is a function of two inputs, just like the blood pressure example earlier in this section.

Which repair is right is a scientific decision, not a mathematical one. The mathematics only says that until you make one, “the glucose of specimen 1047” has no single answer.

3 Domain and Range

3.1 Definitions, and the Letters We Will Use

Every function comes with two sets attached: the inputs it accepts, and the outputs it actually produces.

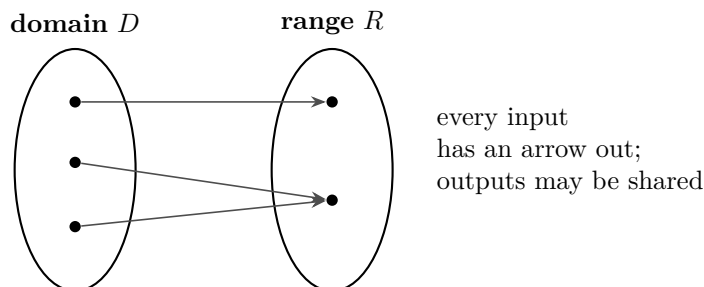
Definition: Domain and range

For a function f ,

- the **domain** D is the set of every input f accepts;
- the **range** $R = \{f(x) \mid x \in D\}$ is the set of outputs that actually **occur**.

When more than one function is in play, tag the letter with the function's name: D_f , R_f , D_g , R_g .

Read the set-builder notation for R aloud: “the set of all values $f(x)$, as x runs through the domain.” So every element of the range was produced by at least one input. (Section 2 used the letter R for a relation; from here on it means the range.)



D asks *what may go in*, R asks *what comes out*, and the second is usually the harder question.

Example 3.1: Domain and range of a small function. Take the function from $A = \{1, 2, 3\}$ to $B = \{4, 5, 6\}$ given by the pairs

$$f = \{(1, 4), (2, 4), (3, 5)\}.$$

Its domain is the set of first entries, $D = \{1, 2, 3\}$. Its range is the set of second entries, each listed once: $R = \{4, 5\}$. The value 6 is in B , but no input produces it, so 6 is **not** in the range.

Example 3.2: The range of a lookup on a cohort. Five patients have blood types O+, A+, O+, B−, O+. For the function “patient $\mapsto$ blood type” on this cohort, D is the five patients and

$$R = \{O+, A+, B-\}.$$

The range has three elements, not five: a set lists each element once, however many patients share it. Types such as AB+ are perfectly possible outputs in general, but they do not occur in this cohort, so they are not in this range. In a data table, the range of a column is what **unique()** returns.

Watch out: the range is not the target set

In $f : A \rightarrow B$, the set B only says where outputs are *allowed* to land. The range is where they *actually* land, and it can be much smaller. Declaring $g(x) = x^2$ as a function $\mathbb{R} \rightarrow \mathbb{R}$ is correct, but its range is $[0, \infty)$, not $\mathbb{R}$.

3.2 D Is Chosen, R Is Discovered

Take $g(x) = x^2$. The function g accepts every real number, so $D = \mathbb{R}$. But no square is negative, so $R = [0, \infty)$: the entire negative half of the number line is never produced. (Every non-negative number does occur, as the square of its own square root.) Recall the interval notation of Unit 1-2: $[0, \infty)$ is every real number from 0 upward, where the square bracket says 0 itself is included and the round bracket at ∞ says there is no largest element.

The two sets play different roles.

- D is **chosen**. It is part of saying what the function *is*: declaring g on all of $\mathbb{R}$, or only on $x \geq 0$, gives two different functions from the same formula.
- R is **discovered**. Once the rule and the domain are fixed, the range is whatever comes out; you do not get to pick it.

That difference is why the two are found in different ways. A domain is usually read off the **formula**: what would break? A range is usually read off the **graph**: how high does it go, how low, and does it skip anything in between?

Example 3.3: Choosing the domain changes the range. Return to the IV bag, $m(c) = 500c$. As a bare formula it accepts any real c , and its range is all of $\mathbb{R}$. But the pharmacy only ever mixes concentrations from 0 up to 10 mg/mL, so the sensible choice is $D = [0, 10]$. On that domain the smallest output is $m(0) = 0$ mg and the largest is $m(10) = 5000$ mg, and since m climbs steadily in between, every mass in between occurs:

$$D = [0, 10] \text{ mg/mL}, \quad R = [0, 5000] \text{ mg}.$$

Same formula, two different functions. We *chose* the domain; the range was then *discovered* by running the rule over it. This is typical: when a formula comes from a real situation, its domain is decided by the situation and its range follows.

3.3 What Restricts a Domain: the Mathematics

Often D is all of $\mathbb{R}$, because most formulas can be evaluated at any real number. Three operations spoil that.

The three usual suspects

1. **Division:** $\frac{1}{x}$ requires $x \neq 0$, because no number times 0 gives 1. Exclude *every* value that makes the denominator zero: $\frac{1}{x^2 - 4}$ loses two points, $x = \pm 2$.
2. **Even roots:** $\sqrt{x}$, $\sqrt[4]{x}$, $\sqrt[6]{x}$, $\dots$ require $x \geq 0$. An **even** power of a real number is never

negative, so nothing squares (or raises to the fourth power, and so on) to a negative. Note $\geq$, not $>$: $\sqrt{0} = 0$ is legal.

3. **Logarithms:** $\log(x)$ requires $x > 0$, **strictly**. The log asks “what exponent gives x ?”, and b^y is always positive, so no exponent yields 0 or a negative.

A word on the first rule. Dividing 1 by x means finding the number that, multiplied by x , gives 1. When $x = 0$ no such number exists, since anything times 0 is 0. Logarithms are covered properly in Section 4.9; for now the rule is enough.

Odd roots are not a problem. $\sqrt[3]{x}$, $\sqrt[5]{x}$, ... are defined for **every** real number: for example $\sqrt[3]{-8} = -2$, since $(-2)^3 = -8$. An odd power keeps the sign of its base, so an odd root can undo a negative. **Only even indices restrict D** (the index is the small number on the root sign): $\sqrt{x}$ has domain $\mathbb{R}$, with no conditions at all.

Example 3.4: One suspect at a time. Each of these formulas contains exactly one of the three operations, so each has exactly one condition.

- $f(x) = \frac{1}{x+5}$. The denominator vanishes when $x+5 = 0$, that is at $x = -5$. So $D = \mathbb{R} \setminus \{-5\}$: every real number except -5 .
- $g(x) = \sqrt{2x-6}$. The radicand must be non-negative: $2x-6 \geq 0$, so $2x \geq 6$ and $x \geq 3$. So $D = [3, \infty)$, with the square bracket because $x = 3$ gives $\sqrt{0} = 0$, which is fine.
- $h(x) = \log(x-1)$. The argument must be strictly positive: $x-1 > 0$, so $x > 1$ and $D = (1, \infty)$, with a round bracket because $x = 1$ would ask for $\log(0)$.
- $k(x) = \frac{1}{x^2+1}$. There is a division, but $x^2 \geq 0$ means $x^2+1 \geq 1$: the denominator is never zero. So the division excludes nothing and $D = \mathbb{R}$. Spotting a suspect is only step one; you still have to check whether it actually bites.

Watch out: two traps with fractions

- **Only the denominator matters.** In $\frac{x-2}{x+1}$, the input $x = 2$ makes the *numerator* zero, giving $\frac{0}{3} = 0$, a perfectly good output. Only $x = -1$ is excluded.
- **Find the domain before simplifying.** The formula $\frac{x^2-1}{x-1}$ simplifies to $x+1$ whenever $x \neq 1$, since $x^2-1 = (x-1)(x+1)$. But at $x = 1$ the original formula asks for $\frac{0}{0}$, which is undefined. So its domain is $\mathbb{R} \setminus \{1\}$, even though the simplified $x+1$ looks as if it accepts everything. Simplifying can hide an excluded point; it cannot remove it.

3.4 What Restricts a Domain: the Context

Those three rules give the **mathematical** domain: the inputs for which the algebra is legal. The **practical** domain is usually smaller, because reality has rules the algebra never heard of.

- **Physical impossibility.** A dose or a concentration cannot be negative, and if t is hours

since admission then $t \geq 0$. The formula will still evaluate at $t = -3$, and report on a time that never happened.

- **Discreteness.** A count of patients is a whole number: $f(2.7)$ computes fine and means nothing.
- **Plausible range.** An age of 150 or a systolic blood pressure of 400 is a data-entry error, not an input. So a domain doubles as a **validation rule**: most pipeline “bugs” are values that should never have been let in.
- **Units.** $\text{BMI}(w, h) = w/h^2$ expects kilograms and meters. Feed it pounds and inches and every step is valid arithmetic on the way to a meaningless answer.

A practical domain is part of what the function *is*, and it marks a model’s honest limit: outside it, the model’s answers mean nothing, however confidently they are printed.

Example 3.5: A practical domain for a dosing rule. Take the weight-based dose $D(w) = 15 w$ mg from Section 1. Mathematically, D accepts every real number. Practically:

- a weight must be positive, which rules out $w \leq 0$;
- suppose the dosing table was validated only for patients from 3 kg (a small infant) up to 120 kg.

Then the practical domain is $[3, 120]$ kg. On it, D runs from $D(3) = 45$ mg up to $D(120) = 1800$ mg, so the practical range is $[45, 1800]$ mg. A request to dose a 150 kg patient is not a question this function answers. The formula will happily return 2250 mg, but that number was never checked by anyone, which is why real protocols add a **maximum dose**, a piecewise rule we meet in Section 4.

Example 3.6: Validation rules are domains. A data pipeline receives these systolic blood pressures, in mmHg: 128, 141, 0, 119, 1400, -5 . Suppose the analysis declares the plausible domain for systolic pressure to be $[50, 300]$ mmHg. Then 0 (probably “not measured” stored as a zero), 1400 (probably 140 with an extra keystroke), and -5 (impossible) all fall outside it and are flagged before any model sees them. Checking whether a value lies in the domain is one line of code, and it catches a large share of real-world data errors.

3.5 Finding a Domain: The Recipe

Procedure: finding a domain

1. Find every division, even root, and logarithm in the formula.
2. Write the condition each one forces: denominator $\neq 0$, radicand ≥ 0 , log argument > 0 .
3. Solve each condition for x .
4. **Intersect** the results: all the conditions must hold at once.
5. Write D in interval notation.

Two words in step 2 may be new: the **radicand** is the expression under a root sign, and the **argument** of a log is the expression it is applied to.

Step 4 is where Units 1-1 and 1-2 pay off. “All conditions at once” is the **AND** of Unit 1-1, and the inputs that satisfy several conditions at once form the **intersection** $\cap$ (Unit 1-2) of the sets of inputs each condition allows.

Every family in Section 4 closes with worked examples of the whole recipe, domain *and* range, once the family’s graph is available to read the range from.

3.6 The Recipe in Action

Here is the recipe run in full on four formulas, from easier to harder. In each one, notice that the work is almost entirely steps 2 and 3, writing and solving the conditions; step 4 is where people slip.

Example 3.7: A denominator with two zeros. Find the domain of $f(x) = \frac{3}{x(x-5)}$.

1. *Suspects*: one division.
2. *Condition*: $x(x-5) \neq 0$.
3. *Solve*: a product is zero exactly when one of its factors is zero, so $x(x-5) = 0$ at $x = 0$ and at $x = 5$. Both are excluded.
4. *Intersect*: only one condition, nothing to combine.
5. *Interval notation*: $D = (-\infty, 0) \cup (0, 5) \cup (5, \infty)$.

The number line is cut at two points, leaving three pieces, and f is defined on all three.

Example 3.8: A root on top, a division below. Find the domain of $g(x) = \frac{\sqrt{x+3}}{x-2}$.

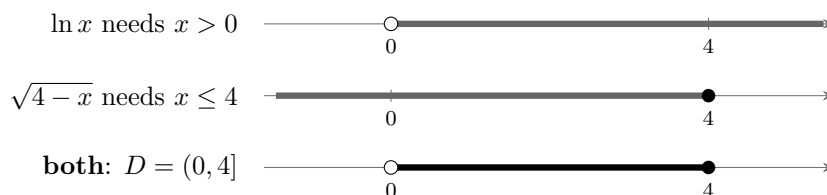
1. *Suspects*: an even root and a division.
2. *Conditions*: radicand $x+3 \geq 0$; denominator $x-2 \neq 0$.
3. *Solve*: $x \geq -3$, and $x \neq 2$.
4. *Intersect*: start from $[-3, \infty)$ and remove the single point 2, which does lie inside it.
5. $D = [-3, 2) \cup (2, \infty)$.

Check a few inputs: $g(-3) = \sqrt{0}/(-5) = 0$ is fine, $g(2)$ would divide by zero, and $g(-4)$ would need $\sqrt{-1}$. All three agree with D .

Example 3.9: A logarithm and a root. Find the domain of $h(x) = \ln(x) + \sqrt{4-x}$.

1. *Suspects*: a logarithm and an even root.
2. *Conditions*: log argument $x > 0$; radicand $4-x \geq 0$.
3. *Solve*: $x > 0$, and $x \leq 4$.
4. *Intersect*: both must hold, so x is above 0 and at most 4.
5. $D = (0, 4]$.

The intersection is easiest to see with one number line per condition, stacked, and the answer is the stretch that is shaded on every line:



A hollow dot marks an endpoint that is **excluded** (round bracket), and a filled dot one that is **included** (square bracket), the same convention the piecewise graphs use in Section 4.

Example 3.10: A root in the denominator. Find the domain of $k(x) = \frac{1}{\sqrt{x-1}}$.

1. *Suspects:* an even root, sitting inside a division.
2. *Conditions:* radicand $x - 1 \geq 0$; denominator $\sqrt{x-1} \neq 0$.
3. *Solve:* $x \geq 1$; and $\sqrt{x-1} = 0$ exactly when $x = 1$, so $x \neq 1$.
4. *Intersect:* $x \geq 1$ with $x = 1$ removed leaves $x > 1$.
5. $D = (1, \infty)$.

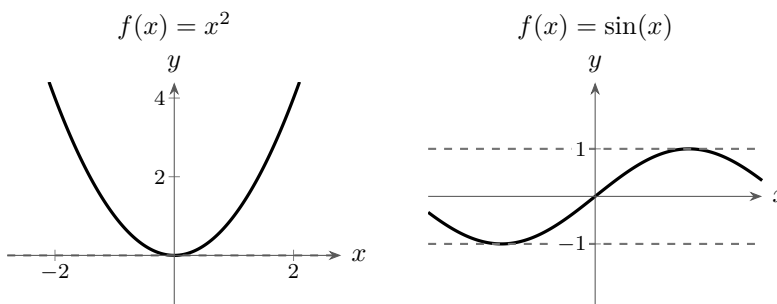
The two conditions merged into one strict inequality. That is the general pattern: an even root **in a denominator** needs its radicand strictly positive, because the root may not be zero either.

Watch out: conditions are intersected, answers are often written with $\cup$

The conditions are combined with AND, which is $\cap$. But the *answer* is frequently written as a union of intervals, as in $[-3, 2) \cup (2, \infty)$. There is no contradiction. The $\cup$ glues together the pieces left over after a point was cut out; it does not mean “either condition will do.” Always ask of the final answer: does every number in it satisfy **every** condition?

3.7 Range Is the Harder Half

D comes from the formula. R usually comes from **looking at the graph** and asking three questions: how high does it go, how low does it go, and does it skip anything in between?



On the left, the parabola bottoms out at 0 (the dashed line) and rises forever without skipping any height, so $R = [0, \infty)$. On the right, the sine wave (Section 4.10) is trapped between the two dashed lines at heights -1 and 1 , and it reaches both, so $R = [-1, 1]$. A function whose outputs are trapped between two fixed numbers like this is called **bounded**.

The definition of range also gives a way to **check** what the graph suggests. A number y is in the range exactly when some input produces it, that is, when the equation $f(x) = y$ has a solution x in the domain. So: try to solve $f(x) = y$, and see which y let you.

Example 3.11: A parabola lifted up. Find the range of $f(x) = x^2 + 1$ on $D = \mathbb{R}$.

From the shape: this is x^2 shifted up 1, so the bottom of the bowl moves from 0 to 1 and the bowl still rises without limit: $R = [1, \infty)$.

Check by solving: $x^2 + 1 = y$ means $x^2 = y - 1$. A square can equal $y - 1$ only if $y - 1 \geq 0$, and then $x = \sqrt{y - 1}$ works. So exactly the $y \geq 1$ are produced, confirming $R = [1, \infty)$. For instance $y = 10$ comes from $x = 3$ (and from $x = -3$), while $y = 0.5$ would need $x^2 = -0.5$, which is impossible.

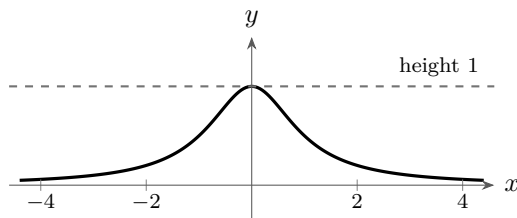
Example 3.12: A parabola that opens downward. Find the range of $f(x) = 5 - (x - 2)^2$.

The square $(x - 2)^2$ is never negative and is zero only at $x = 2$. It is *subtracted* from 5, so $f(x) \leq 5$ always, with $f(2) = 5$ exactly. Moving away from $x = 2$ in either direction makes the square as large as you like, so f falls without limit. The graph is an upside-down bowl with its peak at $(2, 5)$, and

$$R = (-\infty, 5].$$

This “a square is never negative” argument is the same one Section 8 uses to find the best dose pair for two drugs.

Example 3.13: A function trapped between two heights. Find the range of $k(x) = \frac{1}{x^2 + 1}$, whose domain is all of $\mathbb{R}$ (Example 3.4).



The denominator $x^2 + 1$ is smallest, equal to 1, at $x = 0$, so the fraction is largest there: $k(0) = 1$. Everywhere else the denominator is bigger than 1 and the fraction is smaller than 1. As x runs far out in either direction the denominator grows without limit and the fraction sinks toward 0, but a fraction with numerator 1 is never zero, and here it is never negative either. So the curve stays strictly above the x -axis and reaches the dashed line at height 1 exactly once:

$$R = (0, 1].$$

Round bracket at 0 (approached, never reached), square bracket at 1 (reached, at $x = 0$). Like the sine wave, k is **bounded**.

Example 3.14: The same formula on a restricted domain. Find the range of $g(x) = x^2$ on the domain $D = [-1, 2]$.

The tempting shortcut is to evaluate the endpoints, $g(-1) = 1$ and $g(2) = 4$, and answer $[1, 4]$. That is **wrong**. The domain contains $x = 0$, where $g(0) = 0$, which is below both endpoint values. Following the graph from left to right: it starts at height 1, falls to 0 at $x = 0$, then rises to 4 at $x = 2$. Every height from 0 to 4 is passed through, so

$$R = [0, 4].$$

Watch out: the endpoints do not tell the whole story

The lowest and highest outputs of a function on an interval can occur **inside** the interval, at a turning point, rather than at its ends. Before reading a range off the endpoints, ask whether the graph turns around somewhere in between. For a parabola, the place to look is its vertex.

3.8 More Practice: Domain and Range

Exercise 3.1

Give D and R for each function.

1. $f(x) = \sqrt{x} - 2$
2. $g(x) = \frac{1}{x+4}$
3. $h(x) = x^2$ on the domain $[-3, 1]$

Solution to Exercise 3.1

1. The root needs $x \geq 0$, so $D = [0, \infty)$. The graph is $\sqrt{x}$ shifted down 2: $\sqrt{x}$ produces every value in $[0, \infty)$, and subtracting 2 slides that set down, so $R = [-2, \infty)$. The lowest output is $f(0) = 0 - 2 = -2$.
2. The denominator vanishes at $x = -4$, so $D = \mathbb{R} \setminus \{-4\}$. A fraction with numerator 1 is never 0, and every other value is reached: solving $\frac{1}{x+4} = y$ gives $x = \frac{1}{y} - 4$, which works for every $y \neq 0$. So $R = \mathbb{R} \setminus \{0\}$.
3. $D = [-3, 1]$ is given. The endpoints give $h(-3) = 9$ and $h(1) = 1$, but the vertex $x = 0$ lies inside the domain and gives $h(0) = 0$. The graph falls from 9 to 0 and climbs back to 1, so $R = [0, 9]$. (Answering $[1, 9]$ is the endpoint trap from the Watch out box.)

Exercise 3.2

Run the recipe to find D , in interval notation.

1. $f(x) = \frac{\sqrt{x-3}}{x-7}$
2. $g(x) = \ln(10-x) + \frac{1}{x}$

$$3. h(x) = \frac{1}{\sqrt{6-2x}}$$

$$4. k(x) = \frac{\sqrt[3]{x+1}}{x^2+4}$$

Solution to Exercise 3.2

1. Root: $x - 3 \geq 0$, so $x \geq 3$. Denominator: $x \neq 7$. The point 7 lies inside $[3, \infty)$, so remove it: $D = [3, 7) \cup (7, \infty)$.
2. Log: $10 - x > 0$, so $x < 10$. Division: $x \neq 0$. The point 0 lies inside $(-\infty, 10)$, so remove it: $D = (-\infty, 0) \cup (0, 10)$.
3. An even root in a denominator, so the radicand must be strictly positive: $6 - 2x > 0$, so $2x < 6$ and $x < 3$. $D = (-\infty, 3)$.
4. Trick question. The root is a **cube** root, which accepts every real number, and the denominator $x^2 + 4$ is at least 4, never zero. No condition bites, so $D = \mathbb{R}$. Spotting the suspects is only step one; each must still be checked.

Exercise 3.3

A pediatric dosing chart gives $D(w) = 10w$ mg for children weighing 5 to 40 kg.

1. Give the practical domain and the practical range of D .
2. What does the formula return for a 60 kg adolescent, and why should the chart not be used there?

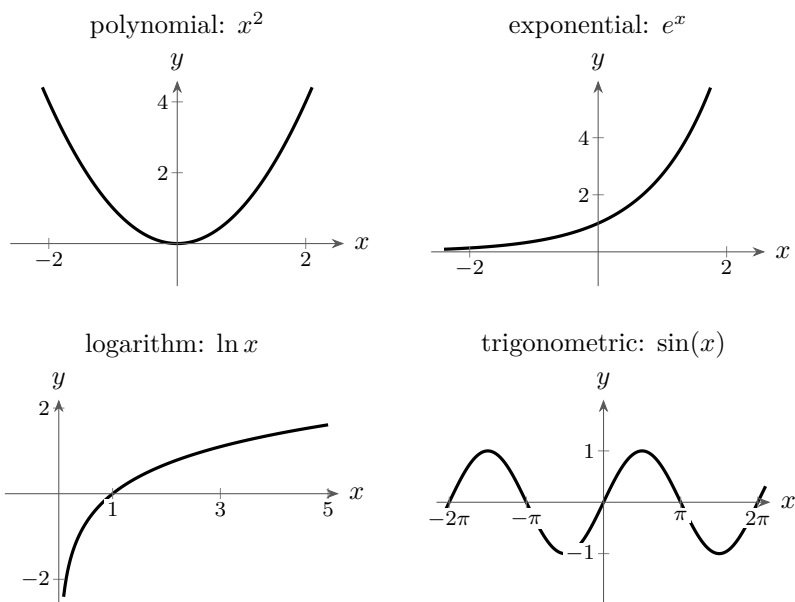
Solution to Exercise 3.3

1. The chart's stated weights give $D = [5, 40]$ kg. The rule multiplies by 10, so its outputs run from $D(5) = 50$ mg to $D(40) = 400$ mg, with every value in between: the practical range is $[50, 400]$ mg.
2. The formula returns $D(60) = 600$ mg. But 60 kg is outside the practical domain, so the chart makes no claim about it: nobody checked that the per-kilogram rule still holds at that size, and adult dosing is usually capped. Using a function outside the domain it was built for is called **extrapolation**, and it is one of the most common ways a correct formula produces a wrong answer.

4 Families of Functions

4.1 Four Shapes Worth Memorizing

Most functions you meet in data belong to a handful of **families**: groups of functions that share a pattern of formula and a shape of graph. Four shapes come up so often that it is worth recognizing them at a glance, because real data keeps producing them.



- The **polynomial** x^2 curves up in a bowl.
- The **exponential** e^x shoots up.
- The **logarithm** $\ln x$ rises slowly, and only exists for $x > 0$.
- The **trigonometric** function $\sin(x)$ repeats forever.

The rest of this section takes the families one at a time. Two notebooks go with it:

- `U1-3_Functions-1_DomainRangeAndFamilies` plots the four families side by side;
- `U1-3_Functions-4_FormulaExplorer` gives you a slider for every constant in a formula, so you can watch the edits of the next two subsections happen.

Example 4.1: Which shape would you expect? Before fitting anything to data, it pays to ask which family the situation suggests. Each of these is described in words; the family follows from the words.

Situation	Family, and the phrase that gives it away
A lab panel costs a fixed \$25 plus \$10 per test ordered.	linear : “the same amount per unit”
The area of a circular wound as its radius grows.	polynomial (quadratic): $\text{area} = \pi r^2$
The amount of drug left in the blood, when the body clears the same <i>fraction</i> of it each hour.	exponential decay: “the same fraction per step”
The pH of a solution as its hydrogen-ion concentration rises tenfold, then tenfold again.	logarithmic : “each tenfold change adds a step”
Flu admissions over several years, peaking every winter.	trigonometric : “repeats on a fixed cycle”

The phrases in the right column are worth memorizing, because problem statements and data dictionaries use them constantly: *adds the same* is linear, *multiplies by the same* is exponential, *each tenfold change adds a step* is logarithmic, and *repeats* is trigonometric.

4.2 Transformations: One Family, Many Members

Starting from any function $f(x)$, five small edits move and reshape its graph. Each edit produces a new function, so one basic shape generates a whole family of relatives. Take $f(x) = x^2$ throughout:

Edit	Effect	Example	Watch out
$f(x) + c$	shift up by c	$x^2 + 2$	vertical, as expected
$f(x + c)$	shift left by c	$(x + 2)^2$	opposite to the sign
$a f(x)$	stretch vertically by a	$3x^2$	taller if $a > 1$
$f(bx)$	squeeze horizontally by b	$(2x)^2$	narrower if $b > 1$
$f(-x)$	reflect across the y -axis	$(-x)^2 = x^2$	unchanged here!

Rows two and four catch people out, for the same reason: an edit *inside* the function acts on the input, so it runs backwards.

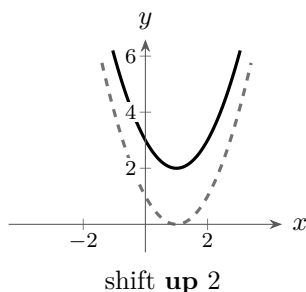
- $f(x + 2)$ moves the graph **left**, not right. The new function reaches the value that f had at 0 when $x + 2 = 0$, that is at $x = -2$: every feature of the graph arrives 2 units early.
- $f(2x)$ makes the graph **narrower**, not wider, since the input reaches any given value twice as fast.

The short version: vertical edits do what they say; horizontal edits do the opposite. The last row is a third surprise: a parabola is **symmetric** about the y -axis, so flipping it left to right changes nothing, and $(-x)^2 = x^2$.

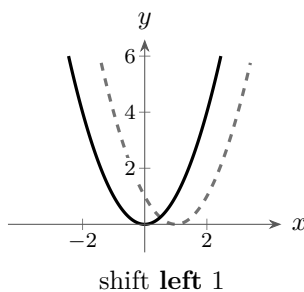
4.3 The Five Edits, Applied to $f(x) = (x - 1)^2$

Because plain x^2 is symmetric, it hides what the horizontal edits do. So take a *shifted* parabola as the base, $f(x) = (x - 1)^2$, whose lowest point is at $x = 1$. In every panel below, the dashed curve is f itself and the solid curve is the edited version.

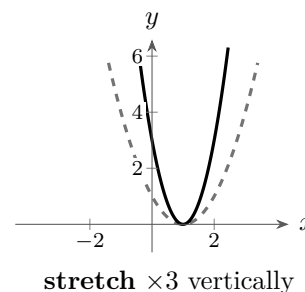
$$f(x) + 2 = (x - 1)^2 + 2$$

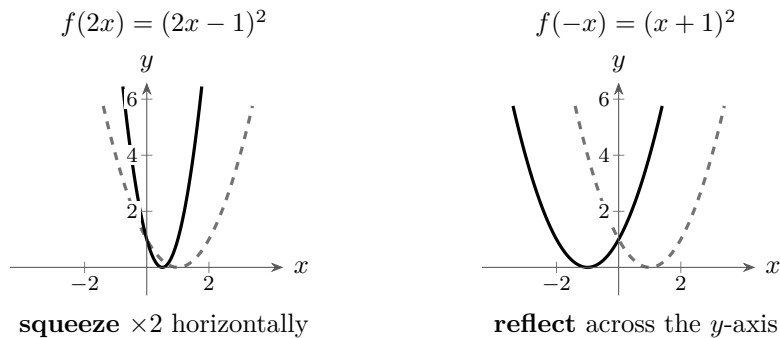


$$f(x + 1) = x^2$$



$$3f(x) = 3(x - 1)^2$$





Each panel follows from substituting into $f(x) = (x - 1)^2$:

- $f(x) + 2 = (x - 1)^2 + 2$: the whole curve moves **up** 2.
- $f(x + 1) = ((x + 1) - 1)^2 = x^2$: the curve moves **left** 1, so the lowest point slides from $x = 1$ to $x = 0$.
- $3f(x) = 3(x - 1)^2$: every height is tripled, a vertical **stretch**; the curve looks narrower because it climbs three times as fast.
- $f(2x) = (2x - 1)^2$: a horizontal **squeeze** by 2, so the lowest point moves from $x = 1$ to $x = \frac{1}{2}$, halfway to the y -axis.
- $f(-x) = (-x - 1)^2 = (x + 1)^2$: a **reflection** across the y -axis, so the lowest point moves from $x = 1$ to $x = -1$.

4.4 Transformations, Worked

The quickest way to make the five edits concrete is to follow a single point of the graph through each of them.

Example 4.2: Following one point through each edit. Suppose all we know about f is that $f(2) = 5$: the point $(2, 5)$ is on its graph. Where does that point go under each edit?

New function	New point	Why	Move
$f(x) + 3$	$(2, 8)$	at $x = 2$: $f(2) + 3 = 8$	up 3
$f(x - 3)$	$(5, 5)$	at $x = 5$: $f(5 - 3) = f(2) = 5$	right 3
$f(x + 3)$	$(-1, 5)$	at $x = -1$: $f(-1 + 3) = f(2) = 5$	left 3
$2f(x)$	$(2, 10)$	at $x = 2$: $2f(2) = 10$	stretch $\times 2$
$f(2x)$	$(1, 5)$	at $x = 1$: $f(2 \cdot 1) = f(2) = 5$	squeeze $\times 2$
$f(-x)$	$(-2, 5)$	at $x = -2$: $f(-(-2)) = f(2) = 5$	reflect

Read the “Why” column for the horizontal edits. To find where the old output 5 appears in $f(x - 3)$, ask which x feeds f the old input 2: that is $x - 3 = 2$, so $x = 5$. The new function has to be given a *larger* x to reach the same place, which is why subtracting inside moves the graph **right**. Every horizontal edit is worked out the same way: set the inside equal to the old input and solve.

Example 4.3: Reading a formula as a list of edits. Describe $g(x) = 2(x + 1)^2 - 3$ as edits of x^2 , and give its lowest point, D , and R .

Work from the inside outward, the order in which the formula acts on x :

1. $(x + 1)^2$: the $+1$ is inside, so shift **left** 1. The lowest point moves from $(0, 0)$ to $(-1, 0)$.
2. $2(x + 1)^2$: multiply the output by 2, a vertical **stretch**. Heights double; the lowest point, at height 0, stays put.
3. $2(x + 1)^2 - 3$: shift **down** 3. The lowest point moves to $(-1, -3)$.

Check: $g(-1) = 2(0)^2 - 3 = -3$, and nearby inputs give more, for example $g(0) = g(-2) = -1$. The domain is untouched by every one of these edits, $D = \mathbb{R}$, and the range starts at the lowest point: $R = [-3, \infty)$.

Example 4.4: Building a formula from a description. Start from $f(x) = \sqrt{x}$. Shift its graph **right** 4 and **up** 1. Give the formula, D , and R .

Right 4 is a horizontal edit, so it goes inside with the *opposite* sign: $\sqrt{x - 4}$. Up 1 is vertical and goes outside with its own sign:

$$g(x) = \sqrt{x - 4} + 1.$$

The root needs $x - 4 \geq 0$, so $D = [4, \infty)$: the domain moved right with the graph. The graph starts at $g(4) = 0 + 1 = 1$ and rises from there (for example $g(8) = 2 + 1 = 3$), so $R = [1, \infty)$: the range moved up with the graph. Each shift dragged exactly one of the two sets along.

Example 4.5: A delayed dose and a change of units. Let $C(t)$ be the concentration of a drug in the blood, in mg/L, t hours after a dose given at time 0. Three variations, each one edit:

- **Same dose, given 2 hours later.** Every feature of the curve happens 2 hours later, so the graph moves **right** 2. The new curve is $C(t - 2)$: at $t = 2$ it gives $C(0)$, the moment of dosing.
- **Double the dose.** For many drugs over their usual dosing range, concentration is proportional to dose, so every height doubles: $2C(t)$, a vertical stretch.
- **Report in a different unit.** A curve for blood glucose in mg/dL is converted to mmol/L by dividing by 18 (a rounded factor). If $G(t)$ is the curve in mg/dL, the converted curve is $\frac{1}{18}G(t)$: a vertical stretch by a factor smaller than 1, which squashes the heights and leaves the times alone.

Watch out: “later” is a minus sign

A curve that happens 2 hours **later** is $C(t - 2)$, not $C(t + 2)$. The formula $C(t + 2)$ reaches every value 2 hours *earlier*. When in doubt, plug in the moment of dosing: the delayed curve must give $C(0)$ at $t = 2$, and only $C(t - 2)$ does.

Watch out: a negative stretch also flips

In $a f(x)$ the table assumes $a > 0$. If a is negative, every height changes sign as well as size, so the graph is also flipped upside down, across the x -axis. For example $-x^2$ is the bowl x^2 turned into a dome, with range $(-\infty, 0]$. The dose-response dome of Section 8 is built exactly this way.

Exercise 4.1

1. Describe how the graph of $(x - 3)^2 + 1$ differs from that of x^2 .
2. Starting from $f(x) = \sqrt{x}$, write the formula that shifts the graph **left** 2 and then stretches it vertically by 3. What happens to D ?
3. Which of the five edits leave the range unchanged, no matter what the function is?

Solution to Exercise 4.1

1. With $f(x) = x^2$, the new function is $f(x - 3) + 1$. The $+1$ outside is a vertical edit: shift **up** 1. The -3 inside is the edit $f(x + c)$ with $c = -3$, which shifts the graph “left by -3 ,” that is **right** 3: horizontal shifts run opposite to their sign. So the parabola moves right 3 and up 1, and its lowest point sits at $(3, 1)$ instead of the origin. As a check, $(x - 3)^2 + 1$ is smallest when $x - 3 = 0$, and its value there is 1.
2. Shifting left 2 replaces x by $x + 2$, giving $\sqrt{x + 2}$. The stretch then multiplies the whole output by 3:

$$3\sqrt{x + 2}.$$

The root still needs a non-negative argument, so $x + 2 \geq 0$, that is $x \geq -2$, and $D = [-2, \infty)$. The original $\sqrt{x}$ had $D = [0, \infty)$, so the horizontal shift dragged the domain 2 units left along with the graph, while the vertical stretch left the domain alone.

3. The three **horizontal** edits: a shift $f(x + c)$, a squeeze $f(bx)$ (any $b \neq 0$), and a reflection $f(-x)$. Each one feeds f exactly the same set of input values, just reached at different x 's, so the set of outputs is untouched.

The two vertical edits act on the outputs themselves, so they **can** change R : a shift slides it and a stretch scales it. For example x^2 has $R = [0, \infty)$, but $x^2 + 2$ has $R = [2, \infty)$. They do not *always* change it, though: the line $f(x) = x$ has range $\mathbb{R}$, and so does $x + 2$. That is why only the horizontal edits make the list of edits that are safe “no matter what the function is.”

Exercise 4.2

1. Starting from $f(x) = |x|$, write a formula for the V-shape whose corner is at $(-2, 3)$ and which is twice as steep. Give its D and R .
2. The point $(4, -1)$ is on the graph of some function f . Name one point on the graph of $f(x + 1) - 2$, and one on the graph of $3f(x/2)$.

Solution to Exercise 4.2

1. The corner of $|x|$ is at the origin. Moving it **left** 2 means $|x + 2|$ (opposite sign, inside); “twice as steep” is a vertical stretch by 2, giving $2|x + 2|$; moving it **up** 3 adds 3 outside:

$$g(x) = 2|x + 2| + 3.$$

Check: $g(-2) = 0 + 3 = 3$, the corner; and one unit to either side, $g(-1) = g(-3) = 5$, a climb of 2 per unit, twice the slope of $|x|$. No edit restricts inputs, so $D = \mathbb{R}$; the lowest output is the corner's height, so $R = [3, \infty)$.

2. For $f(x+1) - 2$: set the inside equal to the old input, $x+1 = 4$, so $x = 3$; the output there is $f(4) - 2 = -1 - 2 = -3$. The point is $(3, -3)$: left 1, down 2. For $3f(x/2)$: set $x/2 = 4$, so $x = 8$; the output is $3f(4) = 3(-1) = -3$. The point is $(8, -3)$. Dividing x by 2 inside *stretches* the graph horizontally, doubling every distance from the y -axis, the opposite of $f(2x)$.

4.5 Linear Functions

Definition: Linear function

$$f(x) = mx + b$$

m is the **slope** (rise over run: how much f changes per unit of x) and b is the **intercept** (the value at $x = 0$).

- The domain is $\mathbb{R}$. The range is $\mathbb{R}$ when $m \neq 0$, since any target value y is reached at $x = (y - b)/m$; when $m = 0$ the range is just the single value $\{b\}$, a flat line.
- **Constant rate of change:** every extra unit of x adds the same amount, m , to the output. That is what makes the graph a straight line.
- This is the entire content of **linear regression**: choose m and b so that the line sits as close as possible to the data. Section 9 sets this up.

Slope carries units, and therefore meaning. If x is body weight in kg and $f(x)$ is blood pressure in mmHg, then “blood pressure rises 0.4 mmHg per additional kg” is a slope: $m = 0.4$ mmHg per kg.

A line is pinned down by any two of its points. The slope is the change in output divided by the change in input between them, and the intercept then follows from either point.

Example 4.6: A line through two measurements. In a small study, patients weighing 60 kg average a systolic pressure of 120 mmHg, and patients weighing 80 kg average 128 mmHg. Find the line through these two points, $(60, 120)$ and $(80, 128)$.

$$m = \frac{\text{change in output}}{\text{change in input}} = \frac{128 - 120}{80 - 60} = \frac{8}{20} = 0.4 \text{ mmHg per kg,}$$

$$b = 120 - 0.4 \times 60 = 120 - 24 = 96 \text{ mmHg.}$$

The intercept came from rearranging $120 = 0.4 \times 60 + b$. So $f(x) = 0.4x + 96$. Check the other point: $f(80) = 32 + 96 = 128$. ✓ The slope is the 0.4 mmHg per kg from the text above, and it predicts, for example, $f(70) = 124$ mmHg for a 70 kg patient.

Watch out: the intercept may be outside the practical domain

The intercept $b = 96$ is “the blood pressure of a patient weighing 0 kg,” which is meaningless. The intercept is always the value at $x = 0$, and when $x = 0$ is outside the practical domain, b is just a number that positions the line, not a fact about patients. Interpret a slope freely; interpret an intercept only if $x = 0$ makes sense.

Example 4.7: Temperature conversion is linear. Water freezes at $0^\circ\text{C} = 32^\circ\text{F}$ and boils at $100^\circ\text{C} = 212^\circ\text{F}$. Using the points $(0, 32)$ and $(100, 212)$:

$$m = \frac{212 - 32}{100 - 0} = \frac{180}{100} = \frac{9}{5}, \quad b = 32 \text{ (the value at } c = 0\text{)}.$$

So $F(c) = \frac{9}{5}c + 32$, the conversion Section 7 starts from. Normal body temperature, 37°C , becomes $F(37) = 66.6 + 32 = 98.6^\circ\text{F}$. Here the intercept *does* mean something, because 0°C is a real temperature.

Example 4.8: An infusion bag, and a negative slope. The IV bag of Section 2 held $V(t) = 1000 - 125t$ mL after t hours. This is linear with slope $m = -125$ mL per hour and intercept $b = 1000$ mL. A negative slope means the output **falls** as the input grows: the bag loses 125 mL every hour. The bag is empty when $1000 - 125t = 0$, at $t = 8$ hours, so the practical domain is $[0, 8]$ and the practical range is $[0, 1000]$ mL. The mathematical range of a non-constant line is all of $\mathbb{R}$, but a bag cannot hold -500 mL: the practical domain has cut the range down to what can happen.

4.6 Polynomials

Definition: Polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i$$

The largest exponent n is the **degree**; the numbers a_i are the **coefficients**.

A polynomial is a sum of whole-number powers of x , each multiplied by a constant. The summation sign $\sum$ is shorthand for “add up”: $\sum_{i=0}^n a_i x^i$ means “let i run through $0, 1, \dots, n$, compute $a_i x^i$ for each, and add the results.” The first few degrees have names:

Degree	Name	Shape	Example
0	constant	flat line	$f(x) = 4$
1	linear	straight line	$f(x) = 2x - 1$
2	quadratic	one bowl (parabola)	$f(x) = x^2 - 3x + 2$
3	cubic	one S-bend	$f(x) = x^3 - x$

Two facts hold for the whole family.

- The domain is always $\mathbb{R}$: a polynomial only adds and multiplies, and those are defined for every number.
- A degree- n polynomial bends (switches between rising and falling) at most $n - 1$ times. So a higher degree can chase the data more closely, and its *noise* with it: a curve with many bends can pass through random wiggles that mean nothing.

Example 4.9: Polynomials and non-polynomials. Which of these are polynomials, and of what degree?

- $q(x) = 4 - x^2 + 3x^5$. **Yes**, degree 5. The terms are out of order, but the degree is the *largest* exponent present, wherever it is written. In standard order, $q(x) = 3x^5 - x^2 + 4$, with $a_5 = 3$, $a_2 = -1$, $a_0 = 4$ and every other coefficient 0.
- $r(x) = (x + 1)(x - 2)$. **Yes**, degree 2: multiplying out gives $x^2 - 2x + x - 2 = x^2 - x - 2$.
- $\frac{1}{x} = x^{-1}$ and $\sqrt{x} = x^{1/2}$. **No**: the exponents -1 and $\frac{1}{2}$ are not whole numbers $0, 1, 2, \dots$. These are the roots and reciprocals of the next subsection.
- 2^x . **No**: here x is the exponent, not the base. That is an exponential.
- The constant function $c(x) = 7$. **Yes**, degree 0: a constant is $7x^0$.

The test is mechanical: after multiplying everything out, every term must be a number times x raised to a whole-number power.

Example 4.10: Evaluating a cubic. Evaluate $p(x) = 2x^3 - 5x + 1$ at $x = -1, 0$, and 2 :

$$\begin{aligned} p(-1) &= 2(-1)^3 - 5(-1) + 1 = -2 + 5 + 1 = 4, \\ p(0) &= 0 - 0 + 1 = 1, \\ p(2) &= 2(8) - 10 + 1 = 7. \end{aligned}$$

Notice $p(0) = a_0 = 1$: a polynomial's value at 0 is always its constant term, because every other term contains a factor of x . That is the polynomial version of “the intercept is the value at $x = 0$.”

Example 4.11: The range of a quadratic, read from its edits. Find the range of $f(x) = 2(x - 3)^2 - 8$.

This is x^2 shifted right 3, stretched by 2, and shifted down 8. The square $(x - 3)^2$ is never negative and equals 0 only at $x = 3$, so the lowest output is $f(3) = -8$, and the bowl opens upward from there:

$$D = \mathbb{R}, \quad R = [-8, \infty).$$

Multiplied out, the same function is $2x^2 - 12x + 10$ (check at $x = 0$: both forms give 10). The multiplied-out form hides the lowest point; the edited form, called **vertex form**, shows it directly. Checking two more inputs, $f(1) = 2(4) - 8 = 0$ and $f(5) = 0$ as well: the bowl is symmetric about $x = 3$.

Exercise 4.3

1. Give D and R for $g(x) = 3x^2 - 1$.
2. State the degree of $p(x) = 2x^3 - 5x + 1$, and say at most how many bends its graph can have.
3. Write p in summation notation, naming each coefficient a_i .

For #1, remember that a domain comes from the formula but a range comes from the shape.

Solution to Exercise 4.3

1. A polynomial only adds and multiplies, so nothing is ever undefined: $D = \mathbb{R}$. For the range, look at the shape. $3x^2 - 1$ is the parabola x^2 stretched vertically by 3 and shifted

down 1, two of the five edits. It opens upward, and since $3x^2 \geq 0$ it bottoms out where $x = 0$, at the value $3(0)^2 - 1 = -1$. From there it rises without limit, so $R = [-1, \infty)$.

2. The largest exponent is 3, so p is a **cubic**, of degree 3. A degree- n polynomial bends at most $n - 1$ times, so p has at most $3 - 1 = 2$ bends.

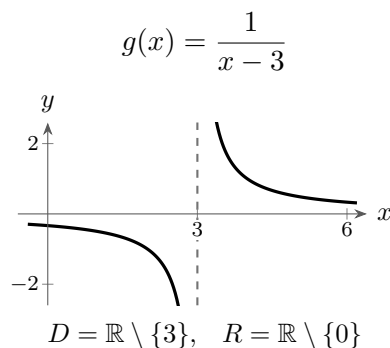
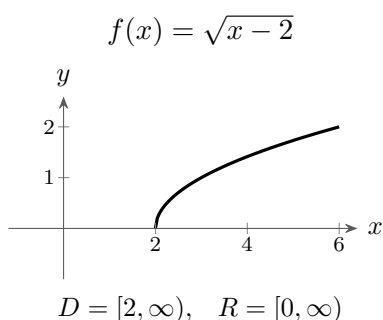
3. Line the terms up by power of x , starting from $x^0 = 1$: $p(x) = 1 + (-5)x + 0 \cdot x^2 + 2x^3$. So

$$p(x) = \sum_{i=0}^3 a_i x^i, \quad a_0 = 1, \quad a_1 = -5, \quad a_2 = 0, \quad a_3 = 2.$$

The missing x^2 term is not absent: it is present with coefficient 0.

4.7 Roots and Reciprocals

Roots and reciprocals are not polynomials, but they are built from them, and they are the two shapes that **restrict** D .



The root. For $f(x) = \sqrt{x-2}$ the recipe forces $x-2 \geq 0$, so $D = [2, \infty)$. For the range: a square root is never negative, and as x runs over $[2, \infty)$ the radicand $x-2$ sweeps all of $[0, \infty)$, so every non-negative output occurs, and $R = [0, \infty)$. The graph starts at the point $(2, 0)$ and rises ever more slowly.

The reciprocal. For $g(x) = \frac{1}{x-3}$ the denominator vanishes at $x = 3$, so that one point is excluded: $D = \mathbb{R} \setminus \{3\}$, the set difference of Unit 1-2, read “all real numbers except 3.” The graph tears apart there. Near $x = 3$ the denominator is tiny, so the output is huge, positive on one side and negative on the other; the dashed vertical line that the graph approaches but never touches is called an **asymptote**. Far from 3 the output gets as close to 0 as you like, but it never equals 0, since 1 divided by anything is never zero. So $R = \mathbb{R} \setminus \{0\}$.

Example 4.12: A root with a stretched inside. Find D and R for $f(x) = \sqrt{2x+6}$.

The radicand must be non-negative: $2x+6 \geq 0$, so $2x \geq -6$ and $x \geq -3$. So $D = [-3, \infty)$. At $x = -3$ the output is $\sqrt{0} = 0$, and as x increases the radicand sweeps through every non-negative number, so every non-negative output occurs: $R = [0, \infty)$. For instance, $f(5) = \sqrt{16} = 4$.

Example 4.13: A root shifted up. Find D and R for $g(x) = \sqrt{x} + 3$.

The only condition is $x \geq 0$, so $D = [0, \infty)$. The $+3$ is outside the root, a vertical shift: the outputs of $\sqrt{x}$ fill $[0, \infty)$, and adding 3 slides that set up to $R = [3, \infty)$. The smallest output is $g(0) = 3$. Compare with the previous example: there a change *inside* the root moved the domain and left the range alone; here a change *outside* the root moved the range and left the domain alone. That is the horizontal-versus-vertical pattern of the five edits.

Example 4.14: A reciprocal, moved. Find D and R for $h(x) = \frac{1}{x+2} + 1$.

Read it as edits of $\frac{1}{x}$: the $+2$ inside shifts the graph **left** 2, and the $+1$ outside shifts it **up** 1. So the tear in the graph moves from $x = 0$ to $x = -2$, and the height the curve approaches but never reaches moves from 0 to 1:

$$D = \mathbb{R} \setminus \{-2\}, \quad R = \mathbb{R} \setminus \{1\}.$$

The algebra agrees. The denominator vanishes at $x = -2$. And $h(x) = 1$ would need $\frac{1}{x+2} = 0$, which never happens, so 1 is the one output that never occurs.

Example 4.15: An odd root needs nothing. Find D for $k(x) = \sqrt[3]{x-8}$.

A cube root is defined for every real number, so there is no condition at all: $D = \mathbb{R}$. For example $k(0) = \sqrt[3]{-8} = -2$ and $k(16) = \sqrt[3]{8} = 2$. It is the **index** of the root, not its contents, that decides whether a condition is needed.

Watch out: roots do not split over sums, and $\sqrt{x^2}$ is not x

- $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$. For example, $\sqrt{9+16} = \sqrt{25} = 5$, but $\sqrt{9} + \sqrt{16} = 7$.
- $\sqrt{x^2} = |x|$, not x . At $x = -3$: $\sqrt{(-3)^2} = \sqrt{9} = 3 = |-3|$. The square root always returns the **non-negative** root, so it cannot give back a negative x . This fact reappears in the composition exercise of Section 5.

Exercise 4.4

Give D in interval notation.

1. $f(x) = \sqrt{5-x}$ (then give R as well)
2. $g(x) = \frac{1}{x^2-4}$
3. $k(x) = \frac{\sqrt{x}}{x-1}$ (careful: this one has **two** conditions)

Run the recipe: spot the roots and divisions, write each condition, solve, then **intersect**.

Solution to Exercise 4.4

1. One even root, so one condition: $5-x \geq 0$. Adding x to both sides gives $x \leq 5$, so $D = (-\infty, 5]$. For the range: a square root is never negative, and as x runs down from 5 the radicand $5-x$ covers all of $[0, \infty)$, so every non-negative value occurs, and $R = [0, \infty)$.
2. One division, so one condition: $x^2-4 \neq 0$, that is $x^2 \neq 4$. Both 2 and -2 square to 4, so $x \neq \pm 2$, and

$$D = (-\infty, -2) \cup (-2, 2) \cup (2, \infty),$$

the whole number line with two points removed.

3. Two conditions. The root needs $x \geq 0$, which is $[0, \infty)$. The denominator needs $x - 1 \neq 0$, so $x \neq 1$. Both must hold, so intersect: start from $[0, \infty)$ and remove the single point 1,

$$D = [0, 1) \cup (1, \infty).$$

#2 is worth a second look: the equation $x^2 = 4$ has *two* solutions, not one, so two points are excluded. Forgetting the negative solution $x = -2$ is the usual slip.

4.8 Exponential Functions

Definition: Exponential function

$$f(x) = a b^x, \quad b > 0, b \neq 1$$

a is the starting value at $x = 0$, and b is the **growth factor** per unit step.

Here the variable sits in the *exponent*, which is what separates this family from the polynomials, where x is the base. Since $b^0 = 1$, we get $f(0) = a$, which is why a is the starting value. The base must be positive and different from 1: with $b = 1$ every power is 1 and the function is constant, and a negative base breaks down at fractional exponents, since $(-4)^{1/2}$ would be $\sqrt{-4}$.

- If $b > 1$ the function shows **growth**: each unit step **multiplies** the output by b . If $0 < b < 1$, each step multiplies by a number smaller than 1, and the function shows **decay**.
- $D = \mathbb{R}$, and $R = (0, \infty)$ when $a > 0$: an exponential never reaches 0, however far it decays.
- The natural base is $e \approx 2.718$, and e^x is the default choice throughout mathematics and machine learning.

Linear vs. exponential, in one sentence

A linear function **adds** the same amount each step; an exponential **multiplies** by the same factor each step.

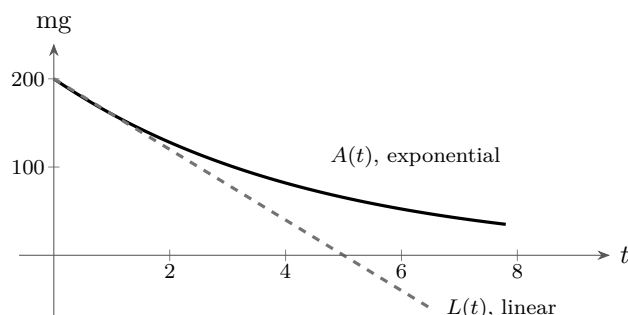
Drug clearance, bacterial growth, and compound interest are all exponential.

Example 4.16: Clearance: a fixed fraction per hour. A patient receives 200 mg of a drug, and the body clears 20% of whatever is present each hour. After each hour, 80% remains, so the amount is **multiplied** by 0.8 every hour:

$$A(t) = 200 \cdot (0.8)^t \text{ mg.}$$

Compare it with a drug that loses a fixed 40 mg every hour, $L(t) = 200 - 40t$, which also starts at 200 and also loses 40 mg in the first hour:

hour t	0	1	2	3	4	5
exponential $A(t) = 200(0.8)^t$	200	160	128	102.4	81.92	65.536
linear $L(t) = 200 - 40t$	200	160	120	80	40	0



After the first hour the two part ways. The linear rule subtracts the same 40 mg forever, reaches 0 at $t = 5$, and would go negative after that, which is impossible. The exponential rule subtracts 20% of a shrinking amount, so it loses less and less each hour and never reaches 0: that is the range $(0, \infty)$ in clinical form. Real first-order clearance behaves like the exponential.

Watch out: a 20% loss is a factor of 0.8, not 0.2

“Clears 20% per hour” means 80% **remains**, so the growth factor is $b = 1 - 0.20 = 0.8$. Writing $200 \cdot (0.2)^t$ instead would claim that only 20% remains each hour: 40 mg after one hour instead of 160. Likewise “grows 5% per year” is a factor of 1.05, not 0.05.

Example 4.17: Half-life. Many drugs are described by their **half-life**, the time for the amount to fall by half. If the half-life is 4 hours and the dose is 200 mg, the amount is multiplied by $\frac{1}{2}$ once every 4 hours, that is $t/4$ times in t hours:

$$A(t) = 200 \cdot \left(\frac{1}{2}\right)^{t/4}.$$

Then $A(4) = 100$, $A(8) = 50$, and $A(12) = 25$ mg: three half-lives, three halvings. In between, $A(2) = 200 \cdot (\frac{1}{2})^{1/2} \approx 141.4$ mg, which is **not** 150. Halfway through the first half-life, about 58.6 of the 100 mg that will be lost in it are already gone: the loss is fastest at the start, when there is the most drug to lose. Dividing t by 4 inside the exponent is the edit $f(bx)$ with $b = \frac{1}{4}$, a horizontal stretch that turns “halve every hour” into “halve every 4 hours.”

Example 4.18: Growth: doubling every three days. Early in an outbreak, a clinic sees 40 cases, and cases double every 3 days. After d days there have been $d/3$ doublings:

$$N(d) = 40 \cdot 2^{d/3}.$$

So $N(3) = 80$, $N(9) = 40 \cdot 2^3 = 320$, and $N(30) = 40 \cdot 2^{10} = 40,960$. Ten doublings multiply by 1024, which is why exponential growth looks harmless for a week and overwhelming a month later.

Example 4.19: Using the natural base. A drug’s concentration is modeled as $C(t) = 80e^{-0.2t}$ mg/L, with t in hours. At $t = 0$, $C(0) = 80e^0 = 80$. At $t = 5$, the exponent is -1 , so $C(5) = 80e^{-1} \approx 80 \times 0.368 \approx 29.4$ mg/L. The negative sign in the exponent makes this decay: $e^{-0.2t} = (e^{-0.2})^t$, and $e^{-0.2} \approx 0.82$ is a factor below 1.

Watch out: exponent rules people misapply

- $2 \cdot 3^x$ is **not** 6^x : the power applies only to the 3. At $x = 2$, $2 \cdot 3^2 = 18$ while $6^2 = 36$.
- A negative exponent does **not** make the output negative: $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$, a small positive number.

Exercise 4.5

1. Give D and R for $f(x) = 3^x$.
2. A colony starts at 200 cells and doubles every hour. Write $N(t)$, the count after t hours.
3. Is $c(x) = 3^{-x}$ growth or decay? Give its range.

For #3, ask yourself whether the output can ever be zero or negative.

Solution to Exercise 4.5

1. Any real exponent is allowed, so $D = \mathbb{R}$. A positive base raised to any power stays positive, and every positive value is achieved somewhere, so $R = (0, \infty)$: the curve approaches the x -axis but never touches it.
2. Doubling each hour means **multiplying** by 2 each step, so the model is exponential, with starting value $a = 200$ and growth factor $b = 2$:

$$N(t) = 200 \cdot 2^t.$$

Check it: $N(0) = 200 \cdot 1 = 200$ and $N(1) = 200 \cdot 2 = 400$, as required.

3. Rewrite it: $3^{-x} = \frac{1}{3^x} = \left(\frac{1}{3}\right)^x$, an exponential with growth factor $\frac{1}{3}$, which is below 1. So it is **decay**. The range is unchanged, $R = (0, \infty)$: flipping the sign of the exponent is the edit $f(-x)$, which reflects the curve left to right, and a horizontal edit cannot let the output reach 0.

4.9 Logarithmic Functions

Definition: Logarithm

$$\log_b(y) = x \quad \text{means exactly} \quad b^x = y.$$

The logarithm answers the question “what exponent?”

So $\log_2(8) = 3$, because $2^3 = 8$: the logarithm base 2 of 8 asks “2 to what power gives 8?” The logarithm **undoes** the exponential, and it is our first real example of an inverse (Section 7).

- $D = (0, \infty)$ and $R = \mathbb{R}$: exactly the exponential’s domain and range, swapped. The exponential only ever produces positive numbers, so the log can only be asked about positive numbers.
- Common bases: $b = e$ (written $\ln$, the **natural log**), $b = 10$, and $b = 2$, the base of the bits

of computing (the 1s and 0s that Unit 1-1 used as truth values).

- The useful rules:

$$\log(uv) = \log u + \log v, \quad \log(u^p) = p \log u.$$

In words, the log turns multiplication into addition, and powers into multiplication.

The first rule is why we **log-transform** skewed data: effects that multiply become effects that add.

Example 4.20: Domain and range of $h(x) = \ln(x + 1)$. The argument must be positive: $x + 1 > 0$, so $D = (-1, \infty)$. The log still climbs and falls without bound, so $R = \mathbb{R}$. Shifting the input moves D , not R , just as Exercise 4.1 found for horizontal edits in general.

Example 4.21: Evaluating logarithms by asking “what exponent?” Each of these is answered by turning it into a question about powers.

Log	Question	Answer
$\log_{10}(1000)$	$10^? = 1000$	3
$\log_3(81)$	$3^? = 81$	4
$\log_5(5)$	$5^? = 5$	1
$\ln(1)$	$e^? = 1$	0
$\log_{10}(0.01)$	$10^? = \frac{1}{100}$	-2
$\log_2(\frac{1}{8})$	$2^? = \frac{1}{8}$	-3

Two patterns are worth keeping: $\log_b(1) = 0$ for every base, since $b^0 = 1$; and $\log_b(b) = 1$. A log is negative exactly when its argument is between 0 and 1, because producing a number smaller than 1 from a base bigger than 1 takes a negative exponent.

Example 4.22: Using the rules. The product rule lets one known log produce others. Since $\log_2(32) = 5$ and $\log_2(4) = 2$,

$$\log_2(128) = \log_2(32 \cdot 4) = \log_2(32) + \log_2(4) = 5 + 2 = 7,$$

and indeed $2^7 = 128$. The power rule gives $\log_{10}(10^6) = 6 \log_{10}(10) = 6$, and $\log_b(\frac{1}{u}) = \log_b(u^{-1}) = -\log_b(u)$: taking the reciprocal flips the sign of the log.

Example 4.23: pH is a logarithm. Acidity is reported as $\text{pH} = -\log_{10}[\text{H}^+]$, where $[\text{H}^+]$ is the hydrogen-ion concentration in mol/L. Normal arterial blood has $[\text{H}^+] \approx 4.0 \times 10^{-8}$ mol/L, so

$$\text{pH} = -\log_{10}(4.0 \times 10^{-8}) = -(\log_{10} 4.0 + \log_{10} 10^{-8}) \approx -(0.602 - 8) \approx 7.40.$$

The product rule did the work: it split the scientific notation into a small log and a whole number. Because the scale is logarithmic, a step of 1 in pH is a tenfold change in $[\text{H}^+]$, and a drop from 7.4 to 7.1, which looks small, means $[\text{H}^+]$ has been multiplied by $10^{0.3} \approx 2$: the blood has become about twice as acidic.

Example 4.24: Log scale for viral load. Viral loads span many powers of ten, so clinicians report them as $\log_{10}$ copies/mL. A patient whose load falls from 100,000 to 1,000 copies/mL has

gone from $\log_{10} = 5$ to $\log_{10} = 3$: a “2-log drop,” which means a $10^2 = 100$ -fold decrease. On the log scale the change is a simple subtraction, $5 - 3 = 2$, even though the raw numbers differ by 99,000. This is the log-transform at work: multiplicative changes become additive ones.

Example 4.25: Solving for an unknown exponent. The drug of Example 4.19 has $C(t) = 80e^{-0.2t}$ mg/L. When does it fall to 20 mg/L?

$$\begin{aligned} 80e^{-0.2t} &= 20 \\ e^{-0.2t} &= 0.25 && \text{divide by 80} \\ -0.2t &= \ln(0.25) && \text{take } \ln \text{ of both sides: } \ln \text{ undoes } e \\ t &= \frac{\ln(0.25)}{-0.2} = \frac{\ln 4}{0.2} \approx \frac{1.386}{0.2} \approx 6.93 \text{ hours.} \end{aligned}$$

(The step $\ln(0.25) = -\ln 4$ is the reciprocal rule from the previous example.) Check: $80e^{-0.2 \times 6.93} \approx 80 \times 0.25 = 20$. ✓ Whenever the unknown sits in an exponent, a logarithm is the tool that brings it down to where ordinary algebra can reach it.

Example 4.26: Domains with a logarithm. The recipe applies to the whole argument of the log.

- $\log(5 - 2x)$ needs $5 - 2x > 0$, so $x < 2.5$: $D = (-\infty, 2.5)$.
- $\ln(x^2)$ needs $x^2 > 0$, which fails only at $x = 0$: $D = \mathbb{R} \setminus \{0\}$.
- $\ln(x^2 - 1)$ needs $x^2 > 1$. That holds for $x > 1$ **or** $x < -1$ (for example $x = -2$ gives $4 - 1 = 3 > 0$), so $D = (-\infty, -1) \cup (1, \infty)$.

The last one is worth a second look: a single condition, $x^2 > 1$, produced two separate pieces joined by OR. The squaring is what creates the second piece, exactly as $x^2 = 4$ had two solutions in Exercise 4.4.

Watch out: logs of sums, and which log your software means

- $\log(u+v) \neq \log u + \log v$. The product rule is about $\log(uv)$. For example $\log_{10}(10+10) = \log_{10}(20) \approx 1.301$, not $1 + 1 = 2$.
- $\log(0)$ and the log of a negative number are undefined, so a column containing zeros cannot be log-transformed as it stands.
- In Python, `np.log` is the **natural** log $\ln$; base 10 is `np.log10` and base 2 is `np.log2`. Many calculators use “log” for base 10. Always check which one a formula means.

Exercise 4.6

1. Give D and R for $h(x) = \log_{10}(x)$.
2. Give D and R for $\ell(x) = \ln(3 - x)$.
3. Solve $\log_2(x) = 5$, and say which function you used to undo the log.

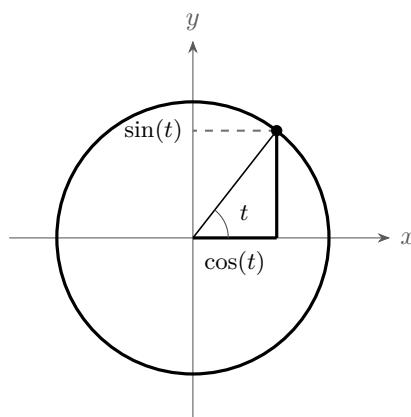
For #2, the recipe applies to whatever sits *inside* the logarithm, not to x itself.

Solution to Exercise 4.6

1. The argument must be strictly positive, so $D = (0, \infty)$. The log climbs and falls without bound, so $R = \mathbb{R}$: exactly the exponential's domain and range, swapped.
2. Now the *argument* $3 - x$ must be positive: $3 - x > 0$, so $x < 3$ and $D = (-\infty, 3)$. As x climbs toward 3 the argument shrinks toward 0 and the log dives toward $-\infty$; as x runs off to the left the argument grows without bound, and so does the log. So $R = \mathbb{R}$ still: shifting and flipping the input moves D , not R .
3. The exponential undoes the log. By the definition, $\log_2(x) = 5$ means exactly $2^5 = x$, so $x = 32$. (Equivalently, raise 2 to the power of each side: $2^{\log_2(x)} = 2^5$.) Check: $\log_2(32) = 5$ because $2^5 = 32$.

4.10 Trigonometric Functions: the Unit Circle

Sine and cosine come from one picture. Walk a point counter-clockwise around the **unit circle**, the circle of radius 1 centered at the origin, and let t be the angle turned, measured from the positive x -axis. Angles here are measured in **radians**, in which one full turn is 2π (about 6.28) rather than 360° ; so a half turn is π and a quarter turn is $\pi/2$.



Definition: Sine and cosine

For the point at angle t on the unit circle,

- $\cos(t)$ is its **x -coordinate**,
- $\sin(t)$ is its **height**, the y -coordinate.

In the figure, $\cos(t)$ is marked along the x -axis, and the height $\sin(t)$ is marked where the dashed line from the point meets the y -axis. The point, the origin, and the foot of the point on the x -axis form a right triangle: its thick horizontal leg has length $\cos(t)$, its thick vertical leg has length $\sin(t)$, and its long side is the radius, 1. Pythagoras on those two legs gives, for free,

$$\cos^2(t) + \sin^2(t) = 1,$$

where $\cos^2(t)$ is shorthand for $(\cos(t))^2$. Two more facts fall straight out of the picture:

- Going once around is 2π and lands you where you started, so both functions are **periodic**: $\sin(t + 2\pi) = \sin(t)$, and likewise for cosine.
- The point never leaves the circle, so neither coordinate can be larger than 1 or smaller than -1 : both functions are **bounded**, with $R = [-1, 1]$.

Example 4.27: Converting between degrees and radians. A full turn is 360° and also 2π radians, so one degree is $\frac{2\pi}{360} = \frac{\pi}{180}$ radians. Converting is therefore the linear function $r(d) = \frac{\pi}{180} d$:

$$r(90) = \frac{\pi}{2}, \quad r(180) = \pi, \quad r(45) = \frac{\pi}{4}, \quad r(270) = \frac{3\pi}{2}.$$

Going back, one radian is $\frac{180}{\pi} \approx 57.3^\circ$: the angle whose arc along the unit circle is exactly one radius long.

Example 4.28: Reading sine and cosine off the circle. At each quarter turn the point sits on an axis, and its coordinates can be read off without a calculator:

angle t	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
where the point is	(1, 0)	(0, 1)	(-1, 0)	(0, -1)	(1, 0)
$\cos(t)$, the x -coordinate	1	0	-1	0	1
$\sin(t)$, the height	0	1	0	-1	0

The last column repeats the first: that is periodicity. Each row also satisfies $\cos^2 + \sin^2 = 1$, since one coordinate is ± 1 and the other is 0.

Example 4.29: Using $\cos^2 + \sin^2 = 1$. The point at some angle t is at height $\sin(t) = 0.6$. What is $\cos(t)$?

From the identity, $\cos^2(t) = 1 - 0.6^2 = 1 - 0.36 = 0.64$, so $\cos(t) = 0.8$ or $\cos(t) = -0.8$. Both answers are real: a horizontal line at height 0.6 crosses the unit circle twice, once on the right, at $(0.8, 0.6)$, and once on the left, at $(-0.8, 0.6)$. Knowing the height alone does not tell you which side of the circle you are on; you also need to know which quarter of the turn t is in.

Watch out: software works in radians

In Python, `np.sin(90)` is **not** 1: it treats 90 as 90 radians, about fourteen full turns, and returns roughly 0.894. The sine of a right angle is `np.sin(np.pi/2)`, which is 1. Convert degrees with $\frac{\pi}{180}$ (or `np.radians`) before calling any trigonometric function.

4.11 The General Wave

Plotting the height $\sin(t)$ against the angle t unrolls the circle into a wave. Three constants bend that wave into any shape you need.

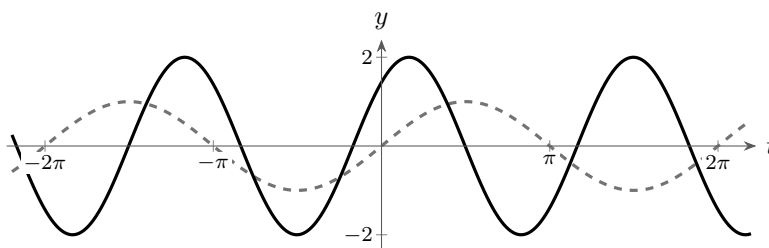
The general wave: three knobs

$$f(t) = A \sin(\omega t + \phi)$$

- A is the **amplitude**: how tall the wave is.

- ω (“omega”) is the **angular frequency**: how fast it cycles. The **period**, the length of one full cycle, is $2\pi/\omega$.
- ϕ (“phi”) is the **phase**: where in the cycle it starts. It slides the wave left by ϕ/ω .

Each knob is one of the edits from earlier in this section. The amplitude is a vertical stretch. The frequency is a horizontal squeeze: ωt runs through one full turn, 2π , while t runs through only $2\pi/\omega$. The phase is a horizontal shift, which you can see by writing $\omega t + \phi = \omega(t + \phi/\omega)$.



The dashed curve is plain $\sin(t)$; the solid one has all three knobs turned, $2\sin(1.5t + 0.8)$: twice as tall, cycling 1.5 times as fast, and slid a little to the left. For any wave of this form $D = \mathbb{R}$ and $R = [-A, A]$. Anything cyclic is modeled this way, and **fitting** a wave to data means choosing A , ω , and ϕ . The `U1-3_Functions-4_FormulaExplorer` notebook uses exactly this formula as its example, with a slider for each knob.

Example 4.30: From a rate to the frequency knob. A resting adult breathes about 15 times a minute. One breath is one cycle, so the period is $60/15 = 4$ seconds. The period is $2\pi/\omega$, so

$$\frac{2\pi}{\omega} = 4 \implies \omega = \frac{2\pi}{4} = \frac{\pi}{2} \text{ radians per second.}$$

A breathing signal is therefore modeled with $\sin(\frac{\pi}{2}t)$, with t in seconds. Faster breathing means a shorter period and a larger ω : the frequency knob and the period move in opposite directions.

Example 4.31: A daily rhythm in body temperature. Core body temperature rises and falls on a roughly 24-hour cycle. A simple model, with t in hours after midnight, is

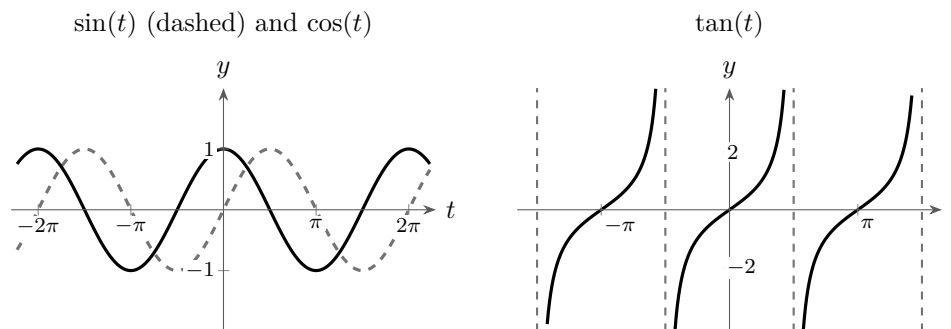
$$T(t) = 37 + 0.5 \sin\left(\frac{\pi}{12}(t - 10)\right) \text{ } ^\circ\text{C.}$$

Read off each knob:

- **Amplitude** $A = 0.5$: the temperature swings 0.5°C above and below its center line.
- The $+37$ outside is a vertical **shift**, one of the five edits: the wave oscillates around 37°C instead of around 0. So $R = [37 - 0.5, 37 + 0.5] = [36.5, 37.5]$.
- **Frequency** $\omega = \frac{\pi}{12}$, so the period is $2\pi/\frac{\pi}{12} = 24$ hours, as intended.
- The $(t - 10)$ inside is a horizontal shift **right** 10 hours. (Multiplied out, the phase is $\phi = -\frac{10\pi}{12}$, and sliding “left by $\phi/\omega = -10$ ” is the same as right by 10.)

When is the temperature highest? The sine is largest, 1, when its angle is $\frac{\pi}{2}$: $\frac{\pi}{12}(t - 10) = \frac{\pi}{2}$ gives $t - 10 = 6$, so $t = 16$, that is 4 pm, with $T(16) = 37.5^\circ\text{C}$. The lowest point comes half a period earlier, at $t = 4$, 4 am, with $T(4) = 36.5^\circ\text{C}$. Fitting such a model to a patient’s temperature log means choosing these four numbers.

4.12 Cosine and Tangent



- **Cosine** is sine with a phase of $\pi/2$:

$$\cos(t) = \sin\left(t + \frac{\pi}{2}\right).$$

Same shape, same $D = \mathbb{R}$ and $R = [-1, 1]$; it simply starts at its peak ($\cos(0) = 1$) instead of at zero.

- **Tangent** is their ratio,

$$\tan(t) = \frac{\sin(t)}{\cos(t)},$$

a **division**, so the domain rules of Section 3 apply: it is undefined wherever $\cos(t) = 0$. On the unit circle that happens at the very top and bottom of the circle, where the x -coordinate is zero, that is at $t = \frac{\pi}{2} + k\pi$ for every integer k (the dashed lines).

- So $D_{\tan} = \{t \in \mathbb{R} : t \neq \frac{\pi}{2} + k\pi \text{ for every integer } k\}$ and $R_{\tan} = \mathbb{R}$. Unlike sine and cosine, tangent is **not bounded**, and its period is π , not 2π .

Example 4.32: A few values of tangent. Tangent is “height divided by x -coordinate” for the point on the circle. At $t = 0$ the point is $(1, 0)$, so $\tan(0) = 0/1 = 0$. At $t = \pi$ it is $(-1, 0)$, so $\tan(\pi) = 0/(-1) = 0$ as well. At $t = \frac{\pi}{4}$, halfway to the top, the point sits on the diagonal line $y = x$, where its two coordinates are equal, so $\tan(\frac{\pi}{4}) = 1$. And at $t = \frac{\pi}{2}$ the x -coordinate is 0, so the division fails: that input is not in the domain.

Watch out: a number inside is not a number outside

$\sin(2t)$ is **not** $2\sin(t)$. The first speeds the wave up (a horizontal squeeze, period π) and stays between -1 and 1 ; the second makes it taller (a vertical stretch) and reaches 2 . At $t = \frac{\pi}{2}$: $\sin(\pi) = 0$, while $2\sin(\frac{\pi}{2}) = 2$. This is the “ $f(2x)$ is not $2f(x)$ ” warning of Section 2 again.

Exercise 4.7

Work with $k(t) = 4 \sin(2t + \pi)$.

1. Name the amplitude, the angular frequency, and the phase, then give the period.
2. Give D and R for k .
3. Why is $\tan(t)$ undefined at $t = \pi/2$, and what does its graph do there?
4. On the unit circle, what are $\cos(0)$ and $\sin(0)$? Read them off the picture rather than a calculator.

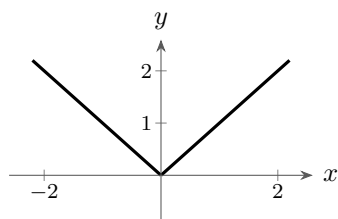
Solution to Exercise 4.7

1. Match $A \sin(\omega t + \phi)$ term by term: the amplitude is $A = 4$, the angular frequency is $\omega = 2$, and the phase is $\phi = \pi$. The period is $2\pi/\omega = 2\pi/2 = \pi$, so the wave completes a full cycle twice as fast as $\sin(t)$. (The phase slides it left by $\phi/\omega = \pi/2$.)
2. Every real t is a legal angle, so $D = \mathbb{R}$. The sine is trapped between -1 and 1 before the amplitude scales it, so k is trapped between -4 and 4 and reaches both: $R = [-4, 4]$.
3. $\tan(t) = \sin(t)/\cos(t)$ is a division. A quarter turn puts the point at the top of the circle, $(0, 1)$, so $\cos(\pi/2) = 0$ and the rule asks for a division by zero: $t = \pi/2$ is excluded from D . The graph tears there. Just to the left of $\pi/2$, $\sin(t)$ is close to 1 and $\cos(t)$ is small and positive, so $\tan(t)$ runs off toward $+\infty$; just to the right, $\cos(t)$ is small and negative, so $\tan(t)$ comes up from $-\infty$. The dashed vertical line it approaches is an **asymptote**.
4. At angle 0 the point has not moved: it sits at the far right of the circle, at coordinates $(1, 0)$. Its x -coordinate gives $\cos(0) = 1$, and its height gives $\sin(0) = 0$.

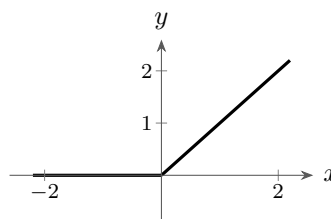
4.13 Piecewise Functions

A function may be defined by **different rules on different parts of its domain**, as long as each input still gets exactly one output. Such a function is called **piecewise**. Read the conditions on the right first: they say *which* rule applies to a given input.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$



$$\text{ReLU}(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$



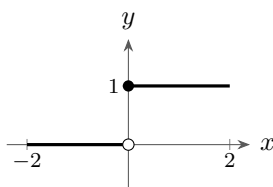
The **absolute value** $|x|$ keeps a non-negative input as it is and flips the sign of a negative one, so its output is always the distance from 0 . In both functions the two pieces meet at the origin, so these functions are **continuous** (informally, the graph can be drawn without lifting the pen),

but each has a **corner** at the origin. ReLU, “keep it if positive, otherwise zero,” is the workhorse activation function inside neural networks.

4.14 More Piecewise Examples

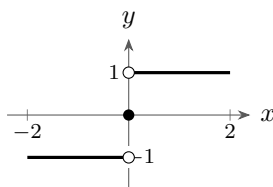
unit step (Heaviside)

$$H(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



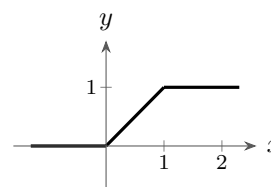
sign

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$



clipping (clamp)

$$c(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$



- The first two **jump** at $x = 0$: the pieces do not meet. The **filled** dot is the value the function takes there; a **hollow** dot is a value it does not take. So $H(0) = 1$ and $\text{sgn}(0) = 0$.
- Clipping doesn't jump, but it flattens. It is how you cap a dose, a pixel value, or a probability at its legal limits.

A threshold rule, “treat if the reading is above the cutoff,” is a step function: its output jumps from “don't treat” to “treat” as the reading crosses the cutoff.

Example 4.33: Evaluating a piecewise rule. Let

$$f(x) = \begin{cases} x^2, & x < 1 \\ 3 - x, & x \geq 1 \end{cases}$$

To evaluate, first decide which condition the input satisfies, then use *that* piece only.

- $f(-2)$: is $-2 < 1$? Yes, so use x^2 : $f(-2) = 4$.
- $f(1)$: is $1 < 1$? No. Is $1 \geq 1$? Yes, so use $3 - x$: $f(1) = 2$. The boundary point belongs to whichever piece has the “or equal to.”
- $f(4)$: $4 \geq 1$, so $f(4) = 3 - 4 = -1$.

Just below the boundary, $f(0.99) = 0.9801$, close to 1; at the boundary, $f(1) = 2$. So this graph **jumps** at $x = 1$, from heights near 1 up to 2, like the step functions above.

Example 4.34: A capped dose. A drug is dosed at 20 mg/kg, but never more than 1000 mg. As a piecewise rule on weights $w > 0$,

$$D(w) = \begin{cases} 20w, & w \leq 50 \\ 1000, & w > 50 \end{cases}$$

The switch happens at 50 kg because that is where $20w$ reaches the cap: $20 \times 50 = 1000$. So $D(30) = 600$ mg, $D(50) = 1000$ mg, and $D(80) = 1000$ mg (not 1600). The two pieces meet at

$w = 50$, so there is no jump, only a corner, exactly like the clipping function c . The same rule is often written in one line as $D(w) = \min(20w, 1000)$, “the smaller of the two.” Its range is $(0, 1000]$: the cap *chooses* the top of the range.

Example 4.35: A category as a function. A simplified, systolic-only version of the usual adult blood pressure categories assigns each reading s (in mmHg) one label:

$$\text{cat}(s) = \begin{cases} \text{normal}, & s < 120 \\ \text{elevated}, & 120 \leq s < 130 \\ \text{stage 1}, & 130 \leq s < 140 \\ \text{stage 2}, & s \geq 140 \end{cases}$$

Then $\text{cat}(118)$ is normal, $\text{cat}(129)$ is elevated, $\text{cat}(130)$ is stage 1, and $\text{cat}(145)$ is stage 2. Every reading falls under exactly one condition, so this is a function; its range is the finite set of four labels. It is also very much a many-to-one function: 131 and 138 both become “stage 1,” and once they have, nothing can tell them apart again. Section 7 explains why that makes it impossible to run backward.

Example 4.36: Absolute value as a distance. Since $|x|$ is the distance of x from 0, $|x - a|$ is the distance of x from a . For example $|2 - 5| = |-3| = 3$: the numbers 2 and 5 are 3 apart, whichever order you subtract in. This is why absolute values appear in error measures: $|y - \hat{y}|$ is how far a prediction $\hat{y}$ landed from the truth y , ignoring which side it missed on. Section 9 averages exactly these distances.

Watch out: the conditions must cover every input exactly once

The conditions of a piecewise rule have to fit together with no overlap and no gap.

- **Overlap:** $\begin{cases} x, & x \leq 1 \\ 2, & x \geq 1 \end{cases}$ gives the input 1 two outputs, 1 and 2. Not a function: “at most one” fails.
- **Gap:** $\begin{cases} x, & x < 0 \\ 1, & x > 0 \end{cases}$ gives the input 0 no output at all. It is still a function, but 0 is not in its domain.

An overlap is harmless only when the pieces agree at the shared point. Writing ReLU as “ x for $x \geq 0$, and 0 for $x \leq 0$ ” overlaps at 0, but both pieces give 0 there, so the input 0 still has one output. When writing a rule, check each boundary value: which piece claims it, and do all claimants agree?

Exercise 4.8

1. Give D and R for ReLU.
2. Give D and R for the clipping rule c , which returns 0 below 0, x on $[0, 1]$, and 1 above 1.
3. Give D and R for the unit step H . What is unusual about this range?
4. Using the five edits, describe the graph of $\text{ReLU}(x - 2) + 1$ and give its D and R .

A piecewise rule is still one function, so it still has exactly one D and one R : collect the pieces, then take stock of everything they produce together.

Solution to Exercise 4.8

1. Every real number falls under one of the two conditions, so every input gets an answer: $D = \mathbb{R}$. Inputs up to 0 return 0, and positive inputs return themselves, which covers every positive number. Together the outputs are 0 and all positive numbers: $R = [0, \infty)$.
2. Again every real number falls under one of the three conditions, so $D = \mathbb{R}$. The outer pieces produce only 0 and 1, and the middle piece produces every value between them, so the output is held between the two caps: $R = [0, 1]$. This is the point of clipping: it *chooses* the range.
3. $D = \mathbb{R}$, and the only values ever produced are 0 and 1, so $R = \{0, 1\}$: two isolated points. That is what is unusual. A range is a **set** of outputs, and nothing requires it to be an interval.
4. $\text{ReLU}(x - 2) + 1$ is the same bent line shifted **right** 2 (the -2 is inside, so it runs opposite to its sign) and **up** 1, so its corner moves from the origin to $(2, 1)$. Written out, it equals 1 for $x \leq 2$ and $x - 1$ for $x > 2$. The domain is untouched, $D = \mathbb{R}$, while the vertical shift carries the range with it: $R = [1, \infty)$.

4.15 More Practice: the Families

Exercise 4.9

Give D and R for each function. For each one, name the family and the edits applied to its basic member.

1. $f(x) = 2^x + 3$
2. $g(x) = \ln(x + 4) - 1$
3. $h(x) = 3 - \sqrt{x - 1}$
4. $k(x) = 5 \cos(x) + 2$
5. $m(x) = \frac{1}{x - 1} - 2$

Solution to Exercise 4.9

1. Exponential 2^x , shifted **up** 3. Any exponent is allowed, so $D = \mathbb{R}$. The outputs of 2^x fill $(0, \infty)$, and adding 3 slides them to $R = (3, \infty)$: the curve approaches height 3 far to the left but never reaches it.
2. Logarithm $\ln x$, shifted **left** 4 and **down** 1. The argument must be positive: $x + 4 > 0$, so $D = (-4, \infty)$. The log already takes every real value, and sliding every real number down by 1 still gives every real number: $R = \mathbb{R}$.
3. Square root $\sqrt{x}$, shifted **right** 1, flipped upside down by the minus sign in front, and shifted **up** 3. The radicand needs $x - 1 \geq 0$, so $D = [1, \infty)$. The root produces $[0, \infty)$; subtracting it from 3 produces everything at most 3, with $h(1) = 3$ the largest: $R = (-\infty, 3]$.
4. Cosine, stretched vertically by 5 and shifted **up** 2. $D = \mathbb{R}$. The cosine fills $[-1, 1]$; times

5 gives $[-5, 5]$; plus 2 gives $R = [-3, 7]$. Check the extremes: at $x = 0$, $k = 5 + 2 = 7$; at $x = \pi$, $k = -5 + 2 = -3$.

5. Reciprocal $\frac{1}{x}$, shifted **right** 1 and **down** 2. The denominator vanishes at $x = 1$, so $D = \mathbb{R} \setminus \{1\}$. The fraction takes every value except 0, so m takes every value except $0 - 2 = -2$: $R = \mathbb{R} \setminus \{-2\}$.

In every case the same pattern held: horizontal edits (inside) moved D , and vertical edits (outside) moved R .

Exercise 4.10

A 400 mg dose of a drug has a half-life of 6 hours.

1. Write the amount remaining, $A(t)$, after t hours.
2. How much remains after 18 hours?
3. Roughly how long until less than 10 mg remains? Use a logarithm.

Solution to Exercise 4.10

1. The amount halves once every 6 hours, so $t/6$ times in t hours:

$$A(t) = 400 \cdot \left(\frac{1}{2}\right)^{t/6}.$$

2. 18 hours is 3 half-lives: $A(18) = 400 \cdot (\frac{1}{2})^3 = 400/8 = 50$ mg.
3. Solve $400 \cdot (\frac{1}{2})^{t/6} = 10$. Divide by 400: $(\frac{1}{2})^{t/6} = \frac{1}{40}$, which says $2^{t/6} = 40$. Take $\log_2$ of both sides:

$$\frac{t}{6} = \log_2(40), \quad t = 6 \log_2(40) \approx 6 \times 5.32 \approx 31.9 \text{ hours.}$$

Sanity check: 5 half-lives (30 hours) leave $400/32 = 12.5$ mg, and 6 half-lives (36 hours) leave 6.25 mg, so the answer must lie between 30 and 36 hours, and 31.9 does. A common rule of thumb says a drug is essentially gone after about five half-lives; here that leaves about 3% of the dose.

5 Combining Functions

5.1 The Easy Combinations

Given two functions f and g , we can build new ones point by point, by doing arithmetic on their outputs:

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x), \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

To evaluate $f + g$ at some x , evaluate f there, evaluate g there, and add the two numbers.

Watch the domain

The new function is only defined where **both** pieces are, so

$$D_{f+g} = D_f \cap D_g,$$

another appearance of the intersection from Unit 1-2. The same holds for fg . Division adds one more condition, $g(x) \neq 0$, so $D_{f/g}$ is $D_f \cap D_g$ with the zeros of g removed.

Example 5.1: Sum, product, and quotient, with their domains. Let $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{x-3}$, so $D_f = [0, \infty)$ and $D_g = \mathbb{R} \setminus \{3\}$. Then

$$(f+g)(x) = \sqrt{x} + \frac{1}{x-3}, \quad D_{f+g} = D_f \cap D_g = [0, 3) \cup (3, \infty).$$

At $x = 4$: $(f+g)(4) = 2 + 1 = 3$, $(fg)(4) = 2 \times 1 = 2$, and $(\frac{f}{g})(4) = \frac{2}{1} = 2$.

The quotient deserves care. Dividing by $\frac{1}{x-3}$ is multiplying by $x-3$, so $\frac{f}{g}$ simplifies to $(x-3)\sqrt{x}$, a formula that happily accepts $x = 3$. But g itself is undefined at 3, so $\frac{f}{g}$ is too: $D_{f/g} = [0, 3) \cup (3, \infty)$. (The extra rule “remove the zeros of g ” removes nothing here, since $\frac{1}{x-3}$ is never 0.) This is the “find the domain before simplifying” trap of Section 3 in a new setting.

Example 5.2: Fluid balance is a difference of functions. A patient receives IV fluid at 125 mL per hour and produces urine at about 80 mL per hour. Over t hours the intake is $I(t) = 125t$ and the output is $O(t) = 80t$, and the net fluid balance is the difference, point by point:

$$N(t) = (I - O)(t) = I(t) - O(t) = 45t \text{ mL}.$$

After an 8-hour shift, $N(8) = 1000 - 640 = 360$ mL: the patient is 360 mL ahead. To evaluate a combination you never need a new idea: evaluate each piece, then do the arithmetic.

These combinations are useful but not deep. The combination that really matters is the next one. The companion notebook for the rest of this section is U1-3_Functions-2_Composition.

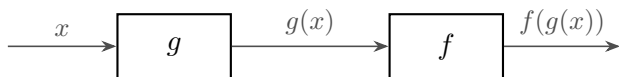
5.2 Composition: Feeding One Function into Another

Definition: Composition

The **composition** of f with g is

$$(f \circ g)(x) = f(g(x)),$$

read “ f of g of x ” or “ f after g .”



Work from the inside out. The output of g becomes the input of f , so g runs first, even though f is written first. The parentheses tell you the order, not the reading direction; that is why “ f after g ” is the better way to read the symbol $\circ$.

5.3 A Worked Composition

Let $f(x) = x^2$ and $g(x) = x + 3$. Evaluate $(f \circ g)(2)$ step by step:

$$\begin{aligned}
 (f \circ g)(2) &= f(g(2)) && \text{write it out} \\
 &= f(2 + 3) && \text{inner function first} \\
 &= f(5) \\
 &= 5^2 = 25 && \text{outer function second}
 \end{aligned}$$

Now do the same for every x at once, to get a formula:

$$(f \circ g)(x) = f(x + 3) = (x + 3)^2.$$

Substitute the *whole* inner expression wherever the outer function had its input, parentheses included. The rule for f is “square the input,” and the input is now $x + 3$, all of it. Leaving the parentheses off would give $x + 3^2 = x + 9$, a different function. As a check, the formula at $x = 2$ gives $(2 + 3)^2 = 25$, matching the step-by-step answer.

5.4 Order Matters

With the same $f(x) = x^2$ and $g(x) = x + 3$, swap the order:

	Formula	At $x = 2$
$(f \circ g)(x) = f(g(x))$	$(x + 3)^2$	25
$(g \circ f)(x) = g(f(x))$	$x^2 + 3$	7

For the second row, f runs first: $(g \circ f)(2) = g(f(2)) = g(4) = 4 + 3 = 7$.

Composition is not commutative

In general, $f \circ g \neq g \circ f$.

(Recall from Unit 1-1 that an operation is **commutative** when the order of its two inputs does not matter, as with $\wedge$ and $\vee$.) The plain-language version: “add 3, then square” is not the same instruction as “square, then add 3.” Real pipelines have the same hazard: removing outliers and then rescaling gives a different result from rescaling and then removing outliers.

5.5 Composition, Worked

Example 5.3: Both orders, at a number and as formulas. Let $f(x) = 3x - 2$ and $g(x) = x^2 + 1$.

At $x = 2$. For $f \circ g$, g runs first: $g(2) = 5$, then $f(5) = 13$. For $g \circ f$, f runs first: $f(2) = 4$, then $g(4) = 17$.

As formulas. Substitute the whole inner expression, in parentheses, for the outer function's input:

$$\begin{aligned}(f \circ g)(x) &= f(x^2 + 1) = 3(x^2 + 1) - 2 = 3x^2 + 1, \\(g \circ f)(x) &= g(3x - 2) = (3x - 2)^2 + 1 = 9x^2 - 12x + 5.\end{aligned}$$

Check both formulas at $x = 2$: $3(4) + 1 = 13$ and $36 - 24 + 5 = 17$, matching the step-by-step values. ✓ Two different functions, from the same two ingredients.

Example 5.4: Composing functions given as tables. When functions are only known as tables, composition is a double lookup.

x	1	2	3	4
$f(x)$	3	1	4	2
$g(x)$	2	3	4	1

- $f(g(1))$: look up $g(1) = 2$ in the g row, then look up $f(2) = 1$ in the f row. So $f(g(1)) = 1$.
- $g(f(1))$: look up $f(1) = 3$, then $g(3) = 4$. So $g(f(1)) = 4$, a different answer: order matters here too.
- $f(g(3))$: $g(3) = 4$, then $f(4) = 2$. And $g(f(3))$: $f(3) = 4$, then $g(4) = 1$.
- $f(f(2))$: $f(2) = 1$, then $f(1) = 3$. A function can be composed with itself.

In every lookup the inner function is read first, and its output becomes the column you look up next. This double lookup is exactly what a database **join** does when one table's output column is another table's key.

Example 5.5: A unit-conversion pipeline. A weight-based dose, $D(w) = 15w$ mg with w in kg, must be computed from a chart that records weight in pounds. The conversion is $k(p) = p/2.2046$ kg. The dose for a weight p in pounds is the composition

$$(D \circ k)(p) = D(k(p)) = 15 \cdot \frac{p}{2.2046} \approx 6.804p.$$

For a 154 lb patient: $k(154) \approx 69.85$ kg, then $D(69.85) \approx 1047.8$ mg. The order is forced by the units: D expects kilograms, so the conversion must run first. The reverse, $k(D(154))$, would treat 154 as kilograms and then “convert” a dose in mg as if it were pounds: arithmetic that runs without complaint and means nothing, the units trap of Section 3.

Example 5.6: Composing a function with itself. The body clears 20% of a drug each hour, so one hour sends an amount A to $f(A) = 0.8A$. Two hours is one hour applied twice:

$$(f \circ f)(A) = f(0.8A) = 0.8 \times 0.8A = 0.64A,$$

and three hours is $0.8^3A = 0.512A$. Composing “multiply by 0.8” with itself t times gives 0.8^tA : the exponential of Section 4 is repeated composition of one simple step.

Example 5.7: The domain of a composition. Let $f(x) = \sqrt{x}$ and $g(x) = 4 - x^2$.

- $(f \circ g)(x) = \sqrt{4 - x^2}$. The inner function accepts everything, but its output must land in the root's domain: $4 - x^2 \geq 0$, so $x^2 \leq 4$ and $-2 \leq x \leq 2$. So $D_{f \circ g} = [-2, 2]$.

- $(g \circ f)(x) = 4 - (\sqrt{x})^2 = 4 - x$. The simplified formula $4 - x$ accepts every real number, but the first stage, $\sqrt{x}$, only accepts $x \geq 0$. So $D_{g \circ f} = [0, \infty)$.

An input belongs to the domain of a composition when it passes **both** checkpoints: it must be a legal input for the inner function, and the inner function's output must be a legal input for the outer one.

Watch out: composing is not multiplying

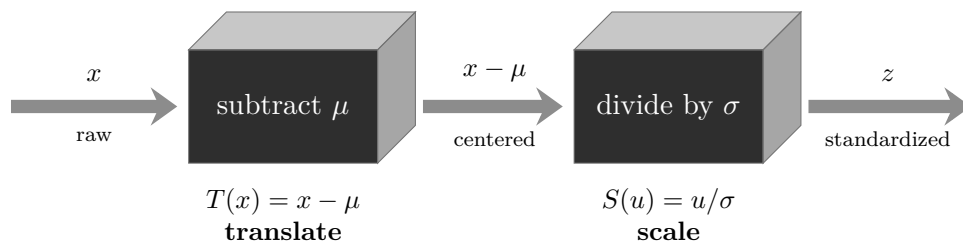
$f \circ g$ and fg look alike and are completely different. With $f(x) = x^2$ and $g(x) = x + 3$ at $x = 2$:

$$(f \circ g)(2) = f(5) = 25, \quad (fg)(2) = f(2)g(2) = 4 \times 5 = 20.$$

The small circle means “feed one into the other”; writing two functions side by side means “multiply their outputs.”

5.6 Chains of Steps: the z -Score

Composition is what a **pipeline** is. Standardizing a measurement, rewriting it as “how many standard deviations from the mean,” is two machines in a row. Here μ (“mu”) is the mean of the measurements and σ (“sigma”) is their standard deviation, a measure of their typical spread; both are developed in Unit 4, and here they are just two fixed numbers computed from the data.



$$z = (S \circ T)(x) = S(T(x)) = \frac{x - \mu}{\sigma}$$

T for **translate** and S for **scale** are the usual names for these two moves. Translate first, then scale. After T , the average input sits at 0; after S , one unit of output means one standard deviation. To standardize **new** data, re-run both boxes in the same order with the *same* μ and σ , not with a mean and standard deviation recomputed from the new data.

Example 5.8: Standardizing glucose readings. In a clinic's records, fasting glucose has mean $\mu = 100$ mg/dL and standard deviation $\sigma = 12$ mg/dL. Push two readings through the pipeline:

raw x	after T : $x - 100$	after S : $\div 12$	meaning
124	24	2.0	two standard deviations above average
94	−6	−0.5	half a standard deviation below average

The z -score has no units: mg/dL divided by mg/dL. That is the point of standardizing. A glucose z of 2.0 and a blood-pressure z of 2.0 can be compared directly (both are “unusually high by the same amount”), even though the raw numbers live on different scales.

Now run the boxes in the wrong order, scale first and then translate: $124/12 - 100 \approx 10.33 - 100 = -89.67$. That number means nothing at all. The $S \circ T$ order is part of the definition, not a matter of taste.

5.7 Reading a Composition Backward

Just as useful as building compositions is **spotting** them: given a complicated function, name the inside and the outside. Write the outer rule in its own variable y , which stands for the value the inner function handed it.

Function	Inner $y = g(x)$	Outer $f(y)$
$\sqrt{x^2 + 1}$	$x^2 + 1$	$\sqrt{y}$
$\ln(3x)$	$3x$	$\ln y$
$\sin^2(x)$	$\sin(x)$	y^2
e^{-x^2}	$-x^2$	e^y

To check a row, substitute the inner expression into the outer rule: in the first row, putting $y = x^2 + 1$ into $\sqrt{y}$ gives back $\sqrt{x^2 + 1}$.

The domain of a composition needs both stages to be valid. For $\ln(3x)$, the inner value $3x$ must land in the log's domain, so $3x > 0$, which forces $x > 0$.

Example 5.9: Spotting the inside and the outside. A reliable way to find the inner function: ask what you would compute *first* if you evaluated the expression at a number by hand. That first computation is the inner function; what you do to its result is the outer one.

Function	Inner $y = g(x)$	Outer $f(y)$	Domain
$(2x + 1)^5$	$2x + 1$	y^5	$\mathbb{R}$
e^{3x}	$3x$	e^y	$\mathbb{R}$
$\ln(x^2 + 1)$	$x^2 + 1$	$\ln y$	$\mathbb{R}$
$\sqrt{\ln x}$	$\ln x$	$\sqrt{y}$	$[1, \infty)$

The domains come from the two checkpoints. In $\ln(x^2 + 1)$ the inner value is at least 1, always positive, so the log never objects and $D = \mathbb{R}$. In $\sqrt{\ln x}$ there are two conditions: the inner $\ln x$ needs $x > 0$, and the outer root needs $\ln x \geq 0$, which holds when $x \geq 1$ (since $\ln 1 = 0$ and $\ln$ rises). Intersecting, $D = [1, \infty)$.

Example 5.10: The logistic function, a three-stage chain. The function

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

is the **logistic** (or sigmoid) function, the heart of logistic regression, which turns any real number into a probability. (This σ is a function; it has nothing to do with the standard deviation σ above. Greek letters get reused.) It is a chain of three machines:

$$x \xrightarrow{\text{negate}} u = -x \xrightarrow{\text{exponentiate}} v = e^u \xrightarrow{1/(1+v)} \sigma(x).$$

At $x = 0$: $u = 0$, $v = e^0 = 1$, and $\sigma(0) = \frac{1}{2}$. Because $v = e^u$ is always positive, the last stage divides 1 by something bigger than 1, so every output lies strictly between 0 and 1: $R = (0, 1)$, just what a probability needs.

Hold onto this skill. In Unit 3 the **chain rule** differentiates a composition by handling the inner and outer pieces separately, and it starts by asking exactly this question.

Exercise 5.1

Use the functions

$$f(x) = 2x - 1, \quad g(x) = x^2, \quad h(x) = \sqrt{x}.$$

1. Compute $(f \circ g)(3)$ and $(g \circ f)(3)$.
2. Write formulas for $(f \circ g)(x)$ and $(g \circ f)(x)$.
3. Give a formula for $(h \circ g \circ f)(x)$, and state its domain D .
4. Identify an inner $y = g(x)$ and an outer $f(y)$ for $1/\sqrt{x-4}$.

Work #1 and #2 **inside out**: evaluate the inner function first, then feed its output to the outer one.

Solution to Exercise 5.1

1. For $f \circ g$, g runs first: $g(3) = 9$, then $f(9) = 2(9) - 1 = 17$. For $g \circ f$, f runs first: $f(3) = 5$, then $g(5) = 25$. So

$$(f \circ g)(3) = 17, \quad (g \circ f)(3) = 25.$$

Different, as expected: composition is not commutative.

2. Substitute the whole inner expression into the outer rule:

$$(f \circ g)(x) = f(x^2) = 2x^2 - 1, \quad (g \circ f)(x) = g(2x - 1) = (2x - 1)^2.$$

At $x = 3$ these give 17 and 25, matching #1.

3. Inside out: f first, then g , then h :

$$(h \circ g \circ f)(x) = h(g(f(x))) = h((2x - 1)^2) = \sqrt{(2x - 1)^2} = |2x - 1|.$$

Domain: $(2x - 1)^2$ is a square, so it is never negative, and the root is always valid. So $D = \mathbb{R}$. The absolute value appears because a square root returns the *non-negative* root: when $2x - 1$ is negative, squaring it and taking the root gives back its size without its sign.

4. Take the inner function $y = x - 4$ and the outer function $f(y) = \frac{1}{\sqrt{y}}$. (You can also split the outer part further into a chain of three stages: $y = x - 4$, then $\sqrt{y}$, then $1/y$.) For the domain, the root needs $x - 4 \geq 0$, and because $\sqrt{x - 4}$ also sits in a denominator it must not be 0, so $x - 4 > 0$, strictly. So $D = (4, \infty)$.

5.8 More Practice: Combining Functions

Exercise 5.2

Let $f(x) = x + 5$ and $g(x) = \frac{1}{x}$.

1. Find formulas for $(f \circ g)(x)$ and $(g \circ f)(x)$, and the domain of each.
2. Evaluate both at $x = 1$.

Solution to Exercise 5.2

1. $(f \circ g)(x) = f(\frac{1}{x}) = \frac{1}{x} + 5$. The inner stage $\frac{1}{x}$ needs $x \neq 0$, and the outer stage accepts anything, so $D_{f \circ g} = \mathbb{R} \setminus \{0\}$.
 $(g \circ f)(x) = g(x + 5) = \frac{1}{x + 5}$. The inner stage accepts anything, but its output must be a legal input for g , that is nonzero: $x + 5 \neq 0$, so $D_{g \circ f} = \mathbb{R} \setminus \{-5\}$.
2. $(f \circ g)(1) = 1 + 5 = 6$, while $(g \circ f)(1) = \frac{1}{6}$. Different formulas, different domains, different values: order matters.

Exercise 5.3

Resting heart rates in a study have mean 70 bpm and standard deviation 8 bpm.

1. Find the z -scores of readings of 86 and 62 bpm, and say what each means.
2. Write the standardizing function as a composition $S \circ T$, naming T and S .

Solution to Exercise 5.3

1. Translate, then scale: $86 \mapsto 86 - 70 = 16 \mapsto 16/8 = 2$, and $62 \mapsto 62 - 70 = -8 \mapsto -8/8 = -1$. So 86 bpm is two standard deviations above the study average, and 62 bpm is one standard deviation below it.
2. $T(x) = x - 70$ centers the reading, and $S(u) = u/8$ rescales it, so $z = (S \circ T)(x) = \frac{x - 70}{8}$.
 T is written second in $S \circ T$ but runs first.

Exercise 5.4

Write $h(x) = \sqrt{3x - 6}$ as a composition $f(g(x))$, naming f and g , and use the two checkpoints to find its domain.

Solution to Exercise 5.4

By hand you would compute $3x - 6$ first, so that is the inner function: $g(x) = 3x - 6$, and the outer function is $f(y) = \sqrt{y}$. Checkpoint one: g accepts every real x . Checkpoint two: $g(x)$ must be a legal input for the root, so $3x - 6 \geq 0$, that is $x \geq 2$. So $D = [2, \infty)$. At the edge, $h(2) = \sqrt{0} = 0$.

6 A Tool from Logic: Quantifiers

6.1 Statements That Depend on a Variable

Unit 1-1 dealt with **propositions**: sentences that are definitely true or definitely false. But “ $x > 3$ ” is neither, until you say something about x .

Definition: Predicate

A **predicate** $p(x)$ is a statement whose truth depends on the value of x . It becomes a proposition once x is pinned down.

There are two ways to pin it down. You can **substitute** a value: with $p(x) = “x > 3”$, $p(5)$ is true and $p(1)$ is false. Or you can say **how many** values make it true. That second move is called a **quantifier**, and there are only two of them:

Symbol	Name	Read as
$\forall$	universal	“for all”, “for every”
$\exists$	existential	“there exists”, “for some”

The definition of a function in Section 2, “*every* input gets *exactly one* output,” is two quantifiers in one breath. Inverses, in Section 7, are where writing that down carefully pays off.

Example 6.1: Predicates, and pinning them down. Let $p(x)$ be “ $x^2 = 4$.” Substituting: $p(2)$ is true, $p(-2)$ is true, $p(3)$ is false. On its own, $p(x)$ is neither true nor false; each substitution produces a proposition.

Predicates are everywhere in clinical data. Let $d(\text{patient})$ be “the patient’s HbA1c is at least 6.5%,” a common diagnostic threshold for diabetes. For a patient with HbA1c 7.1%, d is true; for one with 5.8%, it is false. A database filter such as `WHERE hba1c >= 6.5` is exactly this: a predicate, evaluated row by row, keeping the rows where it comes out true.

The two quantifiers then ask questions about the *whole* set of rows at once: “is d true for **every** patient in the cohort?” and “is d true for **at least one**?”

6.2 The Universal Quantifier: $\forall$

Notation: $\forall$

$\forall x \in S, p(x)$ means “ $p(x)$ is true for **every** x in S .”

Three examples, with their truth values:

- $\forall x \geq 0, \sqrt{x}$ is defined **true**
- $\forall x \in \mathbb{R}, x^2 \geq 0$ **true**
- $\forall x \in \mathbb{R}, \sqrt{x}$ is defined **false**

(In the first, “ $\forall x \geq 0$ ” is shorthand for “for every real x with $x \geq 0$.”)

Naming the set is not optional. The first and third statements differ only in *where x is allowed to live*, and that difference flips the truth value. The single input $x = -4$ is the whole objection to the third: $\sqrt{-4}$ is not a real number.

One counterexample destroys a $\forall$. No matter how many cases work, a single failure makes the universal claim false. That is why “it worked on my test set” is not a proof of anything.

Example 6.2: Deciding a universal claim. For each claim, either give a reason that covers *every* case, or find *one* counterexample.

- $\forall x \in \mathbb{R}, x + 1 > x$. **True:** adding 1 to any number makes it larger, and this reason does not depend on which x you picked.
- $\forall x \in \mathbb{R}, 2x > x$. **False:** at $x = 0$ both sides are 0, and $0 > 0$ fails. (Every $x \leq 0$ fails; one is enough.) Restricted to $x > 0$ the claim becomes true.
- $\forall x \in \mathbb{R}, x^2 \geq x$. **False:** at $x = \frac{1}{2}$, $x^2 = \frac{1}{4}$, which is less than $\frac{1}{2}$. Squaring a number between 0 and 1 makes it smaller.
- $\forall n \in \mathbb{Z}, n^2 \geq n$. **True.** For $n \leq 0$, $n^2 \geq 0 \geq n$. For $n \geq 1$, $n^2 = n \cdot n \geq n \cdot 1 = n$. The counterexample $\frac{1}{2}$ from the previous claim is not an integer, so it is not allowed here.

The last two claims have the same predicate and different sets, and different truth values. The counterexample hunt in the third one succeeded by looking at an unusual kind of number, a fraction between 0 and 1: that is where claims about squares most often break.

6.3 The Existential Quantifier: $\exists$

Notation: $\exists$

$\exists x \in S$ such that $p(x)$ means “**at least one** x in S makes $p(x)$ true.”

Two examples:

- $\exists x \in \mathbb{R}$ such that $x^2 = 2$ **true:** take $x = \sqrt{2}$.
- $\nexists x \in \mathbb{R}$ such that $x^2 = -1$ **true:** no real number squares to a negative.

The struck-through symbol $\nexists$ says “there does not exist.” Notice the **asymmetry of effort**: exhibiting one example settles an $\exists$, while denying one means ruling out every candidate. The first example above needed one number; the second needed a reason covering all real numbers at once.

Example 6.3: Deciding an existential claim.

- $\exists x \in \mathbb{R}$ such that $x^2 = 5x$. **True:** $x = 5$ works, since $25 = 25$. (So does $x = 0$, but one witness is all an $\exists$ needs.)
- $\exists n \in \mathbb{Z}$ such that $2n = 7$. **False:** the only number with $2n = 7$ is 3.5, which is not an integer.
- $\exists x \in \mathbb{R}$ such that $2x = 7$. **True:** $x = 3.5$. Same predicate as the previous claim, bigger set, opposite answer.

Example 6.4: Quantifiers over a data table. Over a finite set, such as the rows of a table, a

quantifier is just a scan. The six BMIs in the table of Section 9 are 27.4, 22.8, 31.5, 24.1, 33.2, and 28.6.

- “ $\exists$ a patient with BMI > 30 ”: scan until one is found. Patient C has 31.5. **True**.
- “ $\forall$ patients, BMI > 20 ”: check every row; the smallest is 22.8. **True**.
- “ $\exists$ a patient with BMI > 40 ”: every row fails. **False**, and it took all six checks to say so.

Python has these two quantifiers built in: `any(b > 30 for b in bmi)` is the $\exists$, and `all(b > 20 for b in bmi)` is the $\forall$. Notice the asymmetry of effort again: `any` can stop at the first success, `all` at the first failure, and each must read every row to reach its other answer.

Exactly one: $\exists!$

$\exists! x \in S$ such that $p(x)$ means there is **one and only one** such x : one exists, and no second one does. This is the quantifier the definition of a function is built on.

6.4 Order Matters When Quantifiers Stack

Two quantifiers in a row are read **left to right**, and swapping them changes the meaning entirely. Here are the same symbols making two different claims.

- $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y > x$ **true**
 “Every real number has some real number above it.” For each x , pick $y = x + 1$. The y is allowed to depend on the x .
- $\exists y \in \mathbb{R}$ such that $\forall x \in \mathbb{R}, y > x$ **false**
 “Some one real number sits above every real number.” This demands one fixed y that beats *every* real number at once. There is none: whatever y you propose, $x = y + 1$ beats it.

The plain-English version: “everyone has a mother” versus “someone is everyone’s mother.” Both use “every” and “some” about the same relationship; only the order differs, and only one is true.

The rule for stacked quantifiers

A variable introduced *later* may depend on the ones before it, never the other way round.

Example 6.5: Swapping the order, three times.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x + y = 0$. **True**: given x , choose $y = -x$. The y depends on the x , which the order allows.
 Swapped: $\exists y \in \mathbb{R}$ such that $\forall x \in \mathbb{R}, x + y = 0$. **False**: one fixed y cannot cancel every x . If y cancels $x = 1$, then $y = -1$, and it fails to cancel $x = 2$.
2. $\exists y \in \mathbb{R}$ such that $\forall x \in \mathbb{R}, x \cdot y = 0$. **True**: take $y = 0$, since $x \cdot 0 = 0$ for every x . So an $\exists\forall$ statement is not automatically false. It is a *stronger* claim than its $\forall\exists$ partner, demanding one answer that works for everyone, and occasionally such an answer exists.
3. In a hospital: “every patient has a nurse assigned to them” is $\forall$ patient $\exists$ nurse, and it is what a staffing plan should guarantee. “Some nurse is assigned to every patient” is $\exists$ nurse

$\forall$ patient: a single nurse covering the whole hospital. Same words, same relationship, very different staffing.

6.5 Negating a Quantifier

Denying a quantified statement **flips the quantifier** and negates what is inside.

Negating a quantifier

$$\begin{aligned}\neg(\forall x, p(x)) &\equiv \exists x \text{ such that } \neg p(x) \\ \neg(\exists x \text{ such that } p(x)) &\equiv \forall x, \neg p(x)\end{aligned}$$

The symbol $\equiv$ means “is logically equivalent to,” as in Unit 1-1. The first line says: if it is false that *every* x works, then at least one x fails. The second says: if no x works, then every x fails.

Read them in words.

- “Not every patient responded” says exactly “**some** patient did not respond.” It does not say “no patient responded,” which is a far stronger claim and a common misreading.
- “There is no cure” says “**every** candidate fails to cure.”

This is the same shape as De Morgan’s laws from Unit 1-1: push the negation inward, and every connective turns into its partner. There AND and OR were the partners; here $\forall$ and $\exists$ are. It is also how you **disprove** a universal claim in practice: negating it turns the task into finding a single counterexample.

Example 6.6: Negating, step by step. The recipe: flip each quantifier, keep the set, and negate the predicate at the end.

1. $\neg(\forall x \in \mathbb{R}, x^2 \geq 0) \equiv \exists x \in \mathbb{R} \text{ such that } x^2 < 0$. The negation is false (no real square is negative), so the original is true.
2. “There is a patient with glucose above 300” negates to “**every** patient has glucose **at most** 300.” The negation of $>$ is $\leq$, not $<$: a glucose of exactly 300 is not above 300.
3. “All lab results in the panel are normal” negates to “**some** lab result in the panel is abnormal,” which is the condition that triggers a review.
4. A stacked statement flips every quantifier in turn:

$$\neg(\forall x, \exists y \text{ such that } y > x) \equiv \exists x \text{ such that } \forall y, y \leq x.$$

In words: “not every number has something above it” means “some number has nothing above it,” a largest number. Since the original statement is true, this negation is false: the real numbers have no largest element.

Watch out: the negation of “all” is not “none”

The negation of “all patients responded” is “at least one patient did not respond,” **not** “no patient responded” and **not** “all patients failed to respond.” Negating a $\forall$ gives an $\exists$: it only

takes one exception. Likewise the negation of $x > 3$ is $x \leq 3$, not $x < 3$: the boundary value switches sides too.

Exercise 6.1

1. Write in symbols: “every real number has a cube root” and “some integer is even.”
2. True or false: $\forall x \in \mathbb{R}, x^2 > 0$. Justify your answer.
3. Negate: “ $\forall$ patients p , the drug lowered p ’s blood pressure.” What would you have to find to prove the original false?
4. One of these is true and one is false. Which is which?

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ s.t. } y = x^2 \qquad \exists y \in \mathbb{R} \text{ s.t. } \forall x \in \mathbb{R}, y = x^2$$

(“s.t.” abbreviates “such that.”)

Solution to Exercise 6.1

1. A cube root of x is a number y with $y^3 = x$, and “every real number has one” puts the $\forall$ first:

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } y^3 = x.$$

“Some integer is even” is a single $\exists$ over the integers $\mathbb{Z}$ of Unit 1-2:

$$\exists n \in \mathbb{Z} \text{ such that } n \text{ is even.}$$

2. **False**, by the single counterexample $x = 0$: there $x^2 = 0$, which is not > 0 . One failure is enough to sink a $\forall$. The claim becomes true if you either weaken the predicate to $x^2 \geq 0$, or shrink the set to the real numbers with $x \neq 0$. **The set and the predicate are both part of the statement.**
3. Flip the $\forall$ to an $\exists$ and negate the inside: “ $\exists$ a patient p such that the drug did **not** lower p ’s blood pressure.” So to disprove the original you need to produce **one** such patient, not a whole failing trial.
4. The first is **true**: every real number has a square, and y is allowed to depend on x (take $y = x^2$). The second is **false**: it asks for one fixed y equal to x^2 for *every* x at once. That fails already at $x = 1$ and $x = 2$, which would need $y = 1$ and $y = 4$ at the same time.

6.6 Why We Needed This

Every definition in this lecture so far has been a quantified sentence in disguise. Here are three of them written properly; the last two are the ones the next section runs on.

Three definitions, in symbols

- **Function** $f : A \rightarrow B$: every input, exactly one output.

$$\forall a \in A, \exists! b \in B \text{ such that } f(a) = b$$

- **One-to-one (injective)**: no output is used twice.

$$\forall x_1, x_2 \in A, f(x_1) = f(x_2) \implies x_1 = x_2$$

- **Onto (surjective)**: nothing in B is left unhit.

$$\forall b \in B, \exists a \in A \text{ such that } f(a) = b$$

Read them in words. The one-to-one condition says: if two inputs give the same output, they were really the same input. The onto condition says: every element of B is the output of at least one input, so the range is all of B . The arrow $\implies$ is the IF-THEN of Unit 1-1.

A function that is both one-to-one and onto is called a **bijection**, and bijections are exactly the functions that can be **inverted**, which is the next section.

Example 6.7: Checking the definitions on small functions. Take $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, and three functions from A to B :

Function	One-to-one?	Onto B ?	Bijection?
$f: 1 \mapsto a, 2 \mapsto c, 3 \mapsto b$	yes	yes	yes
$g: 1 \mapsto a, 2 \mapsto a, 3 \mapsto b$	no: $g(1) = g(2)$	no: c unhit	no
$h: 1 \mapsto b, 2 \mapsto b, 3 \mapsto b$	no	no: a, c unhit	no

For g , the one-to-one definition fails because $g(1) = g(2)$ while $1 \neq 2$; the onto definition fails because no $a \in A$ has $g(a) = c$. And h , a constant function, is as far from both as a function can get: every input shares one output, and two of the three targets are never hit.

Example 6.8: Proving one-to-one and onto with algebra. Is $f(x) = 2x + 3$, from $\mathbb{R}$ to $\mathbb{R}$, a bijection?

- **One-to-one.** Follow the definition: suppose $f(x_1) = f(x_2)$. Then $2x_1 + 3 = 2x_2 + 3$; subtract 3, divide by 2, and $x_1 = x_2$. Same output forces same input. ✓
- **Onto.** Given any target $b \in \mathbb{R}$, we need some a with $2a + 3 = b$. Solving gives $a = \frac{b-3}{2}$, a real number for every b . So every b is hit. ✓

So f is a bijection. Notice that the onto argument just *solved for the input*, the same algebra that will produce the inverse in the next section.

Compare $g(x) = x^2$ from $\mathbb{R}$ to $\mathbb{R}$. It is not one-to-one: $g(2) = g(-2) = 4$ with $2 \neq -2$. It is not onto: $b = -1$ is never hit, since no real square is negative. But shrink both sets to $[0, \infty)$ and it becomes both: on non-negative numbers, equal squares force equal numbers, and every $b \geq 0$ is hit by $a = \sqrt{b}$. The formula did not change; the sets did, and the sets are part of the function.

Example 6.9: A bijection in the medical record. Among the patients of a hospital, “patient $\mapsto$ MRN” is one-to-one (no two patients share a number) and onto the set of MRNs in use (every number in use belongs to someone). It is a bijection, which is exactly why an MRN can be looked up *backward* to find the patient. “Patient $\mapsto$ birthday” is not one-to-one, which is exactly why a birthday cannot.

6.7 More Practice: Quantifiers

Exercise 6.2

Consider the claim “every patient in the ICU has a recorded weight.”

1. Write it with a quantifier, using ICU for the set of ICU patients.
2. Write its negation, in symbols and in words.
3. A data-quality script is meant to test the original claim. What should it search for, and when can it stop?

Solution to Exercise 6.2

1. $\forall p \in \text{ICU}, p$ has a recorded weight.
2. $\exists p \in \text{ICU}$ such that p does **not** have a recorded weight. In words: at least one ICU patient is missing a weight.
3. It should search for a **counterexample**: a single ICU patient with no weight. It can stop, and report the claim false, as soon as it finds one. To report the claim true it must check every patient. That is the asymmetry of effort: in code, `all(...)` over the ICU rows.

Exercise 6.3

True or false? Justify each answer with a reason or a counterexample.

1. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y^2 = x$.
2. $\forall x \in [0, \infty), \exists y \in \mathbb{R}$ such that $y^2 = x$.
3. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, x \leq y^2$.

Solution to Exercise 6.3

1. **False.** Counterexample: $x = -1$. No real y has $y^2 = -1$. One failing x sinks the $\forall$.
2. **True.** For each $x \geq 0$, choose $y = \sqrt{x}$; then $y^2 = x$. Shrinking the set removed every counterexample.
3. **True.** Take $x = 0$: then $0 \leq y^2$ for every real y , since squares are never negative. (Any negative x works too.) This $\exists\forall$ claim asks for one number below every square, and 0 is one.

Claims 1 and 2 differ only in the set x lives in, and that alone flips the answer, just as in the $\sqrt{x}$ examples of this section.

7 Inverse Functions

7.1 Undoing a Function

Temperature conversion runs in both directions:

$$F(c) = \frac{9}{5}c + 32 \quad \text{and back} \quad C(f) = \frac{5}{9}(f - 32).$$

The first turns a Celsius temperature c into Fahrenheit; the second turns a Fahrenheit temperature f (here just a number, not a function) back into Celsius. Run one and then the other and you are back where you started:

$$C(F(c)) = \frac{5}{9}\left(\frac{9}{5}c + 32\right) - 32 = \frac{5}{9} \cdot \frac{9}{5}c = c \quad \text{for every } c.$$

Definition: Inverse function

If $f : A \rightarrow B$, its **inverse** $f^{-1} : B \rightarrow A$ is the function that undoes it:

$$\begin{aligned} f^{-1}(f(x)) &= x && \text{for every } x \text{ in the domain,} \\ f(f^{-1}(y)) &= y && \text{for every } y \text{ in the range.} \end{aligned}$$

An inverse reverses every arrow in the mapping diagram. The whole question is whether the result is still a function; Exercise 2.1 showed that it need not be (#4 and #5 there).

Example 7.1: Round trips. Run each function and then its inverse, and you land back where you began.

- **Temperature.** $F(37) = \frac{9}{5}(37) + 32 = 98.6$, and back, $C(98.6) = \frac{5}{9}(98.6 - 32) = \frac{5}{9}(66.6) = 37$. A fever of 104°F is $C(104) = \frac{5}{9}(72) = 40^\circ\text{C}$.
- **The IV bag.** $m(c) = 500c$ has inverse $m^{-1}(m) = m/500$: divide instead of multiply. A 1500 mg vial gives $m^{-1}(1500) = 3$ mg/mL, the answer to the pharmacy's backward question in Section 1; and $m(3) = 1500$ closes the loop.
- **Weight-based dosing.** $D(w) = 15w$ has inverse $D^{-1}(d) = d/15$: a 900 mg dose corresponds to a 60 kg patient, and $D(60) = 900$.

In each case the inverse answers the question “what input produced this output?” That is also what a clinician does when reading a lab result backward to a cause, or what an analyst does when turning a model's prediction back into original units.

The companion notebook for this section is `U1-3_Functions-3_Inverse`.

7.2 A Warning About the Notation

f^{-1} does not mean $1/f$

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

The superscript -1 here means “the undoing function,” not “raised to the power -1 .”

For $f(x) = 3x$:

- the **inverse** is $f^{-1}(x) = x/3$: divide by three, undoing the multiplication, and indeed $f^{-1}(f(x)) = 3x/3 = x$;
- the **reciprocal** is $1/f(x) = 1/(3x)$, a completely different function.

Unfortunately the notation is not consistent. $\sin^2(x)$ *does* mean $(\sin(x))^2$, an ordinary power, while $\sin^{-1}(x)$ means the inverse of the sine function. The exponent -1 on a function name is the exception, and you simply have to remember it.

Example 7.2: Three different things at one input. Let $f(x) = x + 4$. At $x = 6$, compare:

$f(6) = 10$	the function itself,
$f^{-1}(6) = 6 - 4 = 2$	the inverse: subtract 4, undoing the $+4$,
$\frac{1}{f(6)} = \frac{1}{10}$	the reciprocal.

Check the inverse: $f(2) = 6$, so f^{-1} really did recover the input that produces 6. The reciprocal $\frac{1}{10}$ has nothing to do with running f backward.

7.3 When Does an Inverse Exist?

Reversing the arrows can break the “exactly one output” rule, so not every function has an inverse. Go through the two demands of Section 2 for the reversed arrows, starting with “at most one.”

Definition: One-to-one (injective)

f is **one-to-one** if different inputs always give different outputs:

$$\forall x_1, x_2 \in A, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2).$$

If two inputs shared an output, reversing the arrows would send that output to **two** places: one-to-many, not a function. This is the same condition as the one-to-one line of the previous section, written the other way round: “different inputs give different outputs” is the **contrapositive** of “the same output means the same input,” and by Unit 1-1 a statement and its contrapositive are equivalent.

The other demand, “at least one,” takes care of itself. f^{-1} only ever has to answer for values that f **actually produces**, that is, for its **range**, and every one of those came from some input.

When an inverse exists

One-to-one is exactly the condition. A one-to-one f has an inverse, and that inverse carries the range of f back to its domain.

A note on the target set, since Section 6 said an invertible function is a bijection (one-to-one *and* onto). The definition above wrote $f^{-1} : B \rightarrow A$. If f is not onto B , some elements of B are never produced, and f^{-1} would have nothing to send them to. The fix is simply to take the range R_f as the target set. Every function is onto its own range, so once that choice is made, one-to-one is the only condition left, and the two statements agree.

Example 7.3: Which of these can be run backward? Check one-to-one, either by algebra or by exhibiting two inputs that share an output.

- $f(x) = 5 - 2x$. If $5 - 2x_1 = 5 - 2x_2$, then $-2x_1 = -2x_2$ and $x_1 = x_2$. **One-to-one**, so it has an inverse.
- $f(x) = \frac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. If $\frac{1}{x_1} = \frac{1}{x_2}$, take reciprocals: $x_1 = x_2$. **One-to-one**.
- $f(x) = x^2 + 1$. $f(1) = f(-1) = 2$. **Not** one-to-one, so no inverse on all of $\mathbb{R}$.
- $f(x) = |x|$. $f(3) = f(-3) = 3$. **Not** one-to-one.
- $\text{cat}(s)$, the blood pressure category of Section 4. $\text{cat}(131) = \text{cat}(138) = \text{"stage 1."}$ **Not** one-to-one.

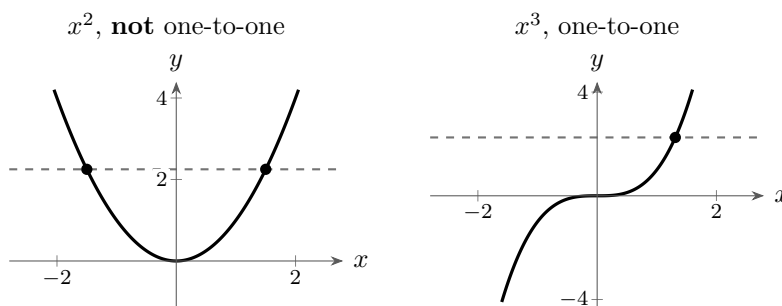
The last one carries a data-science lesson. Grouping a measurement into categories (“binning”) is a many-to-one function, so it has no inverse: once a reading is stored only as “stage 1,” the original number cannot be recovered. Binning throws information away, which is why raw values are kept whenever possible.

7.4 The Horizontal Line Test

One-to-one has a picture, just as “is a function” did. Every point on a horizontal line has the same height, that is, the same output; so a horizontal line crossing the graph twice would mean two inputs sharing an output.

The horizontal line test

A function is one-to-one exactly when **no horizontal line crosses its graph twice**.



Squaring loses the sign. The dashed line at height 2.25 meets x^2 at both $x = -1.5$ and $x = 1.5$, since both square to 2.25, and the same happens at every positive height. So two inputs share each positive output, and the square has no inverse on all of $\mathbb{R}$. The cube passes: each horizontal line meets it exactly once.

Restrict the square's domain to $x \geq 0$ and it becomes one-to-one, since only the right half of the bowl is left. That is why $\sqrt{\cdot}$ is defined as the **non-negative** root: it is the inverse of the square on $x \geq 0$.

Example 7.4: The horizontal line test on the families. Every graph in Section 4 can be put through the test by eye.

Function	A horizontal line that hits twice?	One-to-one?
$mx + b, m \neq 0$	none: a slanted line meets each height once	yes
$e^x, 2^x$	none: always rising	yes
$\ln x$	none: always rising	yes
$\frac{1}{x - 3}$	none: each height other than 0 met once	yes
$\sin(t)$	height 0, at $t = 0, \pi, 2\pi, \dots$	no
$\text{ReLU}(x)$	height 0, at every $x \leq 0$	no
$\text{clip } c(x)$	heights 0 and 1, along the flat pieces	no

A pattern: a function that is always rising, or always falling, passes automatically, because it can never come back to a height it has already left. Anything that turns around (a parabola, a wave) or goes flat (ReLU, clipping, a step) fails.

Watch out: flat pieces fail the test

It is easy to check a graph for “U-turns” and miss flat stretches. A function that is constant on an interval, like ReLU for $x \leq 0$ or a capped dose above 50 kg, sends infinitely many inputs to one output, so a horizontal line along the flat piece meets the graph infinitely often. A capped dose cannot be run backward: a 1000 mg dose could belong to any patient over 50 kg.

7.5 Finding an Inverse by Algebra

Procedure: finding an inverse

1. Write $y = f(x)$.
2. **Solve for x** in terms of y .
3. Swap the names: the expression in y is $f^{-1}(y)$; rename y to x if you prefer.
4. Check: does $f^{-1}(f(x))$ simplify to x ?

Example 7.5: Inverting $f(x) = 3x + 5$. Write $y = 3x + 5$ and peel away what was done to x :

$$y = 3x + 5$$

$$y - 5 = 3x \quad \text{undo the } + 5$$

$$\frac{1}{3}(y - 5) = x \quad \text{undo the } \times 3$$

So $f^{-1}(x) = \frac{x - 5}{3}$. Check:

$$f^{-1}(3x + 5) = \frac{(3x + 5) - 5}{3} = \frac{3x}{3} = x. \checkmark$$

7.6 Inverses, Worked

The procedure is always the same four steps. What changes from one function to the next is the algebra in step 2, so here it is on a range of functions, from easy to harder. In each, the check at the end is not optional: it is the only way to be sure the algebra was right.

Example 7.6: A linear function, divided. Find the inverse of $f(x) = \frac{x - 2}{5}$.

$$y = \frac{x - 2}{5}$$

$$5y = x - 2 \quad \text{undo the } \div 5$$

$$5y + 2 = x \quad \text{undo the } - 2$$

So $f^{-1}(x) = 5x + 2$. Check: $f^{-1}(f(x)) = 5 \cdot \frac{x - 2}{5} + 2 = (x - 2) + 2 = x. \checkmark$

Example 7.7: The reciprocal from Section 4. Find the inverse of $f(x) = \frac{1}{x - 3}$, whose domain is $\mathbb{R} \setminus \{3\}$ and range $\mathbb{R} \setminus \{0\}$.

$$y = \frac{1}{x - 3}$$

$$y(x - 3) = 1 \quad \text{multiply both sides by } x - 3$$

$$x - 3 = \frac{1}{y} \quad \text{divide by } y \text{ (allowed: } y \neq 0 \text{ in the range)}$$

$$x = 3 + \frac{1}{y}.$$

So $f^{-1}(x) = 3 + \frac{1}{x}$, with domain $\mathbb{R} \setminus \{0\}$, which is exactly the range of f . Check at a number: $f(5) = \frac{1}{2}$, and $f^{-1}(\frac{1}{2}) = 3 + 2 = 5. \checkmark$

Example 7.8: An exponential, undone by a logarithm. Find the inverse of $f(x) = 2e^x + 1$.

The range of f comes first: e^x fills $(0, \infty)$, doubling keeps it $(0, \infty)$, and adding 1 gives $R_f = (1, \infty)$. Now solve:

$$y = 2e^x + 1$$

$$\frac{y - 1}{2} = e^x \quad \text{undo the } + 1, \text{ then the } \times 2$$

$$\ln\left(\frac{y - 1}{2}\right) = x \quad \text{undo the exponential with } \ln$$

So $f^{-1}(x) = \ln(\frac{x-1}{2})$. Its domain, from the recipe, needs $\frac{x-1}{2} > 0$, that is $x > 1$: $(1, \infty)$, which matches R_f , as it must.

Example 7.9: A fraction with x on top and bottom. Find the inverse of $f(x) = \frac{2x+1}{x-1}$.

Here x appears twice, so “peel off the operations” does not work directly. Instead, clear the fraction and collect every x on one side:

$$\begin{array}{ll} y = \frac{2x+1}{x-1} & \\ y(x-1) = 2x+1 & \text{multiply by } x-1 \\ yx - y = 2x+1 & \text{expand} \\ yx - 2x = y+1 & \text{gather the } x \text{ terms on the left} \\ x(y-2) = y+1 & \text{factor out } x \\ x = \frac{y+1}{y-2} & \text{divide by } y-2 \end{array}$$

So $f^{-1}(x) = \frac{x+1}{x-2}$. Check at a number: $f(3) = \frac{7}{2} = 3.5$, and $f^{-1}(3.5) = \frac{4.5}{1.5} = 3$. ✓ The last step divided by $y-2$, which fails when $y = 2$. That is not a flaw in the method: 2 is the one value f never produces, so $D_{f^{-1}} = \mathbb{R} \setminus \{2\} = R_f$.

Example 7.10: How long until the drug is down to 50 mg? The clearance model $A(t) = 200 \cdot (0.8)^t$ of Section 4 is always falling, so it is one-to-one and can be run backward, from an amount to a time:

$$\begin{array}{ll} A = 200 \cdot (0.8)^t & \\ \frac{A}{200} = (0.8)^t & \\ \ln\left(\frac{A}{200}\right) = t \ln(0.8) & \text{take } \ln; \text{ power rule} \\ t = \frac{\ln(A/200)}{\ln(0.8)}. & \end{array}$$

This is the inverse function: amount in, time out. For $A = 50$ mg, $t = \frac{\ln(0.25)}{\ln(0.8)} \approx \frac{-1.386}{-0.223} \approx 6.21$ hours. The two logarithms are both negative, so their ratio is positive, as a time should be. Check: $200 \cdot 0.8^{6.21} \approx 50$. ✓

Watch out: an inverse of a root needs its domain stated

Take $f(x) = \sqrt{x-2}$ on $[2, \infty)$, with range $[0, \infty)$. Solving $y = \sqrt{x-2}$ gives $y^2 = x-2$, so the algebra suggests $f^{-1}(x) = x^2 + 2$. But that formula, on all of $\mathbb{R}$, is a whole parabola and is not one-to-one. The inverse is $x^2 + 2$ **only on** $[0, \infty)$, the range of f . Forget the restriction and you get nonsense: the formula gives $(-1)^2 + 2 = 3$, but $f(3) = 1$, not -1 . Whenever squaring appears in step 2, write the inverse’s domain explicitly.

7.7 Undo in the Opposite Order

Notice what the algebra actually did. To build $f(x) = 3x + 5$ you **multiply by 3, then add 5**. To undo it you **subtract 5, then divide by 3**: the opposite operations, in the opposite order.

$$x \xrightarrow{\times 3} 3x \xrightarrow{+5} 3x + 5$$

$$\frac{y-5}{3} \xleftarrow{\div 3} y-5 \xleftarrow{-5} y$$

It is the same idea as taking off a coat and then a sweater, after putting on the sweater and then the coat. Reversing a composition reverses the order of its steps:

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

Since $f \circ g$ runs g first and f second, undoing it must undo f first and g second, and $g^{-1} \circ f^{-1}$ (read inside out) does exactly that.

This is also why a preprocessing pipeline has to be undone stage by stage, backward, to turn a prediction back into the original units. The z -score of Section 5 translated and then scaled, so getting back to raw units means un-scaling and then un-translating: $x = \sigma z + \mu$.

Example 7.11: Undoing a four-step recipe. Find the inverse of $g(x) = 2(x+3)^3 - 1$ by listing its steps.

To build $g(x)$ from x : **add 3**, then **cube**, then **multiply by 2**, then **subtract 1**. To undo it, reverse the list and undo each step: **add 1**, then **divide by 2**, then **take the cube root**, then **subtract 3**:

$$\begin{aligned} \text{build: } x &\xrightarrow{+3} x+3 \xrightarrow{\text{cube}} (x+3)^3 \xrightarrow{\times 2} 2(x+3)^3 \xrightarrow{-1} 2(x+3)^3 - 1 \\ \text{undo: } y &\xrightarrow{+1} y+1 \xrightarrow{\div 2} \frac{y+1}{2} \xrightarrow{\text{cube root}} \sqrt[3]{\frac{y+1}{2}} \xrightarrow{-3} \sqrt[3]{\frac{y+1}{2}} - 3 \end{aligned}$$

So $g^{-1}(x) = \sqrt[3]{\frac{x+1}{2}} - 3$. Check at a number: $g(-1) = 2(2)^3 - 1 = 15$, and $g^{-1}(15) = \sqrt[3]{8} - 3 = 2 - 3 = -1$. ✓ No restriction was needed, because every step (including cubing) is one-to-one.

Example 7.12: The coat-and-sweater rule, checked. Let $f(x) = 2x$ and $g(x) = x + 3$, so $(f \circ g)(x) = 2(x+3) = 2x + 6$. Solving $y = 2x + 6$ directly gives $(f \circ g)^{-1}(x) = \frac{x-6}{2}$. The rule says this should equal $g^{-1} \circ f^{-1}$: halve first (undoing f), then subtract 3 (undoing g), which gives $\frac{x}{2} - 3$. And indeed $\frac{x-6}{2} = \frac{x}{2} - 3$. At $x = 10$ both give 2. ✓

Example 7.13: Turning a z -score back into mg/dL. With the glucose pipeline of Section 5 ($\mu = 100$, $\sigma = 12$), a model outputs a standardized prediction of $z = 1.5$. Undo the scale, then the translation:

$$x = \sigma z + \mu = 12 \times 1.5 + 100 = 118 \text{ mg/dL}.$$

Undoing in the wrong order, $(1.5 + 100) \times 12 = 1218$, is absurd, and absurd in a way that is easy to miss if nobody looks at the numbers.

7.8 Domain and Range Trade Places

Reversing every arrow swaps the roles of the two sets:

$$D_{f^{-1}} = R_f, \quad R_{f^{-1}} = D_f.$$

Function	Domain D	Range R
e^x	$\mathbb{R}$	$(0, \infty)$
$\ln x$	$(0, \infty)$	$\mathbb{R}$

This is the cleanest way to see why $\ln$ refuses non-positive inputs: e^x never *produces* a non-positive value, so $\ln$, its inverse, is never *asked* about one.

The same story holds for $\sqrt{\cdot}$ as the inverse of x^2 restricted to $x \geq 0$: the restricted square has $R = [0, \infty)$, and that is exactly the square root's D .

Example 7.14: Reading the inverse's domain and range for free. Let $f(x) = \sqrt{x-2} + 1$. Its domain is $[2, \infty)$ (the radicand must be non-negative) and its range is $[1, \infty)$ (a root is at least 0, shifted up 1). Without doing any algebra, we already know the inverse has

$$D_{f^{-1}} = [1, \infty), \quad R_{f^{-1}} = [2, \infty).$$

The algebra confirms it: $y = \sqrt{x-2} + 1$ gives $y - 1 = \sqrt{x-2}$, then $(y-1)^2 = x-2$, so $f^{-1}(x) = (x-1)^2 + 2$, valid on $[1, \infty)$ only (the square in step 2 again). Check: $f(6) = \sqrt{4} + 1 = 3$ and $f^{-1}(3) = 2^2 + 2 = 6$. ✓

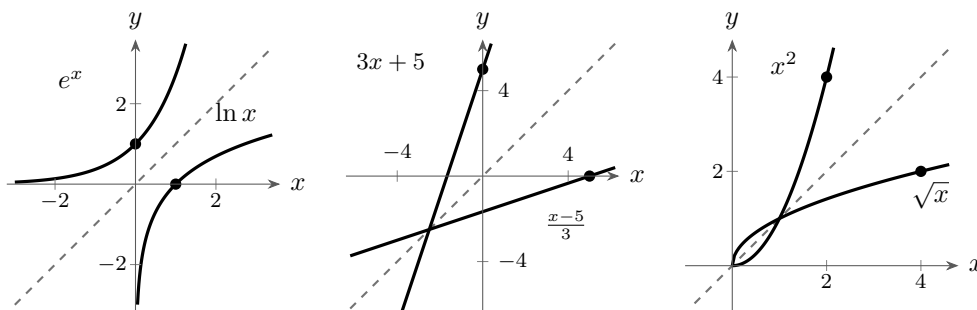
Example 7.15: Two ways to restrict a parabola. The function $h(x) = x^2 - 4$ is not one-to-one on $\mathbb{R}$. Restricting it to $x \geq 0$ gives the inverse $\sqrt{x+4}$, as Exercise 7.1 at the end of this section works out. But the **left** half, $x \leq 0$, is an equally good choice. On it, solving $y = x^2 - 4$ gives $x^2 = y + 4$, and now x must be the *non-positive* root:

$$h^{-1}(x) = -\sqrt{x+4}, \quad D = [-4, \infty), \quad R = (-\infty, 0].$$

Check: $h(-3) = 5$, and $-\sqrt{5+4} = -3$. ✓ The two choices give two different inverse functions, and the domain restriction is what decides which one you mean. A formula alone does not determine an inverse; the formula plus its domain does.

7.9 An Inverse Is a Reflection Across $y = x$

Swapping input and output swaps the coordinates of every point: (a, b) on the graph of f becomes (b, a) on the graph of f^{-1} . The line $y = x$ (dashed below) is where the two coordinates are equal, and a point and its swap sit at the same distance on opposite sides of it. So swapping is a **mirror across** $y = x$.



Each pair of dots is a point and its mirror image: $(0, 1)$ on e^x and $(1, 0)$ on $\ln x$; $(0, 5)$ on $3x + 5$ and $(5, 0)$ on its inverse; $(2, 4)$ on x^2 and $(4, 2)$ on $\sqrt{x}$. The last panel draws x^2 only for $x \geq 0$: reflecting the **restricted** half is what produces $\sqrt{x}$.

Example 7.16: Building a graph of $\log_2$ from a graph of 2^x . List a few points of 2^x , swap each pair, and you have points of its inverse $\log_2 x$:

on 2^x	$(-1, \frac{1}{2})$	$(0, 1)$	$(1, 2)$	$(2, 4)$	$(3, 8)$
on $\log_2 x$	$(\frac{1}{2}, -1)$	$(1, 0)$	$(2, 1)$	$(4, 2)$	$(8, 3)$

Each swapped pair is a true statement about logarithms: $(8, 3)$ on the graph says $\log_2 8 = 3$, which is the same fact as $2^3 = 8$. So a table of powers, read backward, is a table of logs, and plotting the second row and joining the dots gives the log curve, the mirror image of the first row's curve across $y = x$.

Exercise 7.1

Use the functions

$$f(x) = 4x - 7, \quad g(x) = x^3 + 1, \quad h(x) = x^2 - 4.$$

1. Find f^{-1} , and check it.
2. Find g^{-1} .
3. Does h have an inverse on all of $\mathbb{R}$? If not, restrict its domain so that it does, and give the inverse there.
4. Solve $2^x = 32$ for x . (Which function undoes an exponential?)

Solution to Exercise 7.1

1. Write $y = 4x - 7$. Undo the -7 , then the $\times 4$: $y + 7 = 4x$, so $x = \frac{y + 7}{4}$, and

$$f^{-1}(x) = \frac{x + 7}{4}.$$

$$\text{Check: } f^{-1}(4x - 7) = \frac{(4x - 7) + 7}{4} = \frac{4x}{4} = x. \quad \checkmark$$

2. Write $y = x^3 + 1$. Undo the $+1$, then the cube: $y - 1 = x^3$, so $x = \sqrt[3]{y - 1}$, and

$$g^{-1}(x) = \sqrt[3]{x - 1}.$$

No restriction is needed: cubing *is* one-to-one (it passed the horizontal line test above), and a cube root is defined for every real number.

3. No: $h(-3) = 9 - 4 = 5$ and $h(3) = 9 - 4 = 5$, two inputs sharing an output, so h fails the horizontal line test. Restrict to $x \geq 0$, where h is one-to-one. Then $y = x^2 - 4$ gives $x^2 = y + 4$, and since $x \geq 0$ we take the non-negative root, $x = \sqrt{y + 4}$. So

$$h^{-1}(x) = \sqrt{x + 4}, \quad \text{with domain } [-4, \infty).$$

That domain is precisely the range of the restricted h : on $x \geq 0$ the smallest value of $x^2 - 4$ is -4 , at $x = 0$, and it rises without limit from there.

4. The unknown sits in an exponent, and the logarithm undoes the exponential. Apply the logarithm base 2 to both sides: $x = \log_2(32) = 5$, since $2^5 = 32$. Whenever the unknown sits in an **exponent**, the log is the tool that brings it down.

7.10 More Practice: Inverses

Exercise 7.2

Find each inverse, give its domain, and check it at one number.

1. $f(x) = 7 - 3x$
2. $g(x) = 10^x$
3. $k(x) = \frac{3}{x+1}$

Solution to Exercise 7.2

1. $y = 7 - 3x$ gives $3x = 7 - y$, so $f^{-1}(x) = \frac{7-x}{3}$, with domain $\mathbb{R}$ (the range of a non-constant line). Check: $f(2) = 1$ and $f^{-1}(1) = \frac{6}{3} = 2$. ✓
2. $y = 10^x$ means exactly $x = \log_{10} y$, by the definition of the logarithm. So $g^{-1}(x) = \log_{10} x$, with domain $(0, \infty)$, the range of 10^x . Check: $g(3) = 1000$ and $\log_{10}(1000) = 3$. ✓
3. $y = \frac{3}{x+1}$ gives $y(x+1) = 3$, so $x+1 = \frac{3}{y}$ and $x = \frac{3}{y} - 1$. So $k^{-1}(x) = \frac{3}{x} - 1$, with domain $\mathbb{R} \setminus \{0\}$: the fraction $\frac{3}{x+1}$ is never 0, so 0 is not in the range of k . Check: $k(2) = 1$ and $k^{-1}(1) = 3 - 1 = 2$. ✓

Exercise 7.3

1. Many labs report serum creatinine in $\mu\text{mol/L}$, found by multiplying the value in mg/dL by 88.4. Write the conversion $u(c)$ and its inverse, and convert 110 $\mu\text{mol/L}$ back to mg/dL .
2. For a patient of height 1.75 m, BMI is a function of weight alone: $B(w) = w/1.75^2$. Find B^{-1} , and use it to find the weight at which this patient's BMI would be exactly 25.

Solution to Exercise 7.3

1. $u(c) = 88.4c$, so the inverse divides: $u^{-1}(v) = v/88.4$. Then $u^{-1}(110) = 110/88.4 \approx 1.24$ mg/dL . Check: $88.4 \times 1.244 \approx 110$. ✓
2. $1.75^2 = 3.0625$, so $B(w) = w/3.0625$ and $B^{-1}(b) = 3.0625b$. Then $B^{-1}(25) = 3.0625 \times 25 \approx 76.6$ kg. In words: at 1.75 m tall, a weight of about 76.6 kg is the boundary between the “normal” and “overweight” BMI categories. A clinician uses exactly this inverse when advising a patient on a target weight.

8 Functions of Several Variables

8.1 One Input Is Rarely Enough

Almost nothing worth predicting depends on a single number. A risk score depends on age *and* blood pressure *and* weight; a dose depends on weight *and* kidney function.

Definition: Function of p variables

$$f(x_1, x_2, \dots, x_p) = \text{one output}$$

The rule takes an **ordered list** of p numbers and returns a single value.

Everything in these notes so far has been the case $p = 1$. The ordered list is a p -tuple, from Unit 1-2.

Example 8.1: Body mass index.

$$\text{BMI}(w, h) = \frac{w}{h^2}$$

Two inputs (weight w in kg, height h in m), one output. The input is a **pair**, and the order matters: $\text{BMI}(70, 1.8) = 70/1.8^2 \approx 21.6$, a typical adult value, while $\text{BMI}(1.8, 70) = 1.8/70^2$ is a meaningless tiny number.

“Exactly one output per input” is unchanged. What changed is that one input is now a whole list.

Example 8.2: Evaluating BMI. Evaluating works exactly as before, one slot at a time: the first number goes wherever w appears, the second wherever h appears. For a patient weighing 95 kg at 1.75 m,

$$\text{BMI}(95, 1.75) = \frac{95}{1.75^2} = \frac{95}{3.0625} \approx 31.0,$$

above the usual obesity threshold of 30. Swap the slots, $\text{BMI}(1.75, 95) = 1.75/95^2 \approx 0.0002$, and the arithmetic still runs, which is exactly why the order of arguments has to be fixed by the definition and respected every time.

Example 8.3: Kidney function: three inputs. Drug doses are often adjusted for kidney function, estimated by the Cockcroft-Gault formula for creatinine clearance (in mL/min):

$$\text{CrCl}(a, w, s) = \frac{(140 - a) \cdot w}{72 \cdot s},$$

with age a in years, weight w in kg, and serum creatinine s in mg/dL (the result is multiplied by 0.85 for women). For a 60-year-old man of 80 kg with $s = 1.2$:

$$\text{CrCl}(60, 80, 1.2) = \frac{(140 - 60) \cdot 80}{72 \cdot 1.2} = \frac{6400}{86.4} \approx 74.1 \text{ mL/min.}$$

For a woman with the same three values, $0.85 \times 74.1 \approx 63.0$ mL/min. For a 30-year-old man of 70 kg with $s = 0.9$: $\frac{110 \cdot 70}{64.8} \approx 118.8$ mL/min. So $p = 3$ (four, if sex is counted as an input), and each tuple of inputs gives one clearance.

Example 8.4: Body surface area, and a dose built on it. Chemotherapy doses are often given per square meter of body surface area (BSA). One common estimate, the Mosteller formula, is

$$\text{BSA}(h, w) = \sqrt{\frac{h \cdot w}{3600}} \text{ m}^2,$$

with height h in cm and weight w in kg. For $h = 170$ and $w = 70$: $\text{BSA} = \sqrt{11900/3600} = \sqrt{3.306} \approx 1.82 \text{ m}^2$. A drug ordered at 75 mg/m^2 then gets $75 \times 1.82 \approx 136 \text{ mg}$. The dose is the **composition** of a one-variable function ($b \mapsto 75b$) with a two-variable one (BSA): Section 5's machines-in-a-row, with a two-slot machine at the front.

Notebook U1-3_Functions-5_SeveralVariables, §1: evaluating a two-input rule

8.2 The Letters Are Ours to Choose

These three lines all say the same thing, **two inputs and one output**:

$$y = f(x_1, x_2), \quad z = f(x, y), \quad \text{BMI} = f(w, h).$$

Two conventions are both standard.

- **Subscripts**, $f(x_1, x_2, \dots, x_p)$, scale to any number of inputs, so this is what data science uses once p is large.
- **Separate letters**, $f(x, y)$ or $f(w, h)$, are handy when p is 2 or 3, because each input gets its own name and its own axis in a picture.

Watch out: a letter does not carry a job. In $y = f(x_1, x_2)$ the letter y is the **output**, plotted upward. In $z = f(x, y)$ that same letter y is an **input**, plotted along a horizontal axis, and z is the output.

Position tells you the role, not the letter

Whatever sits inside the parentheses is an input, and whatever f equals is the output.

Example 8.5: Naming the roles. For each statement, name the inputs, the output, and p .

- $V = f(r, h) = \pi r^2 h$, the volume of a cylindrical vial: inputs r and h , output V , $p = 2$.
- $x = g(t, y)$: inputs t and y , output x , $p = 2$. The letters x and y have swapped their usual jobs, and that is allowed.
- $\hat{y} = h(x_1, x_2, \dots, x_{10})$: ten inputs, one output $\hat{y}$, $p = 10$. This is the shape of a model with ten features, in Section 9.

Notebook U1-3_Functions-5_SeveralVariables, §2: the same function, three ways

8.3 Where the Inputs Live: the Cartesian Product

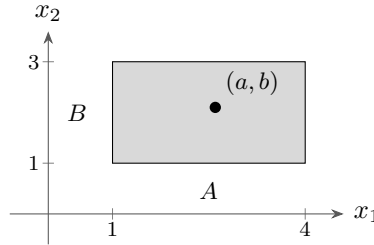
Unit 1-2 introduced ordered pairs and tuples. Collecting *every* ordered pair that can be drawn from two sets gives a new set, and it is exactly where a two-variable function's inputs live.

Definition: Cartesian product

$$A \times B = \{ (a, b) : a \in A, b \in B \}$$

and, for p sets, $A_1 \times A_2 \times \cdots \times A_p$ is the set of ordered p -tuples whose i -th entry comes from A_i .

$A \times B$ is read “ A cross B ” (the colon in the set-builder notation means “such that,” like the vertical bar). For example, take $A = [1, 4]$ and $B = [1, 3]$, two intervals:



Here $A \times B$ is the shaded **rectangle**: every allowed x_1 paired with every allowed x_2 . Taking $A = B = \mathbb{R}$ gives the whole plane, written $\mathbb{R}^2$, and p copies of $\mathbb{R}$ give $\mathbb{R}^p$.

Example 8.6: Blood types are a Cartesian product. The eight common blood types come from pairing an ABO group with an Rh factor:

$$\{A, B, AB, O\} \times \{+, -\} = \{ (A, +), (A, -), (B, +), (B, -), \\ (AB, +), (AB, -), (O, +), (O, -) \}.$$

That is $4 \times 2 = 8$ elements: each of the 4 groups paired with each of the 2 factors. The label “O–” is just a compact way of writing the ordered pair $(O, -)$.

Example 8.7: The arms of a factorial trial. A trial tests a drug against placebo, each given either in the morning or in the evening. The set of study arms is

$$\{\text{drug, placebo}\} \times \{\text{morning, evening}\},$$

four arms in all, and every participant is assigned one element of this product. A design that crosses every level of one factor with every level of another is called a **factorial** design, and its list of arms is literally a Cartesian product.

Watch out: $A \times B$ is not $B \times A$

The pairs are *ordered*, so $(O, -)$ and $(-, O)$ are different pairs, and $\{1, 2\} \times \{x, y\}$ contains $(1, x)$ while $\{x, y\} \times \{1, 2\}$ contains $(x, 1)$ instead. The two products have the same *number* of elements, $|A| \cdot |B|$, but not the same elements. For a function of several variables this is the “order of the arguments matters” point once more.

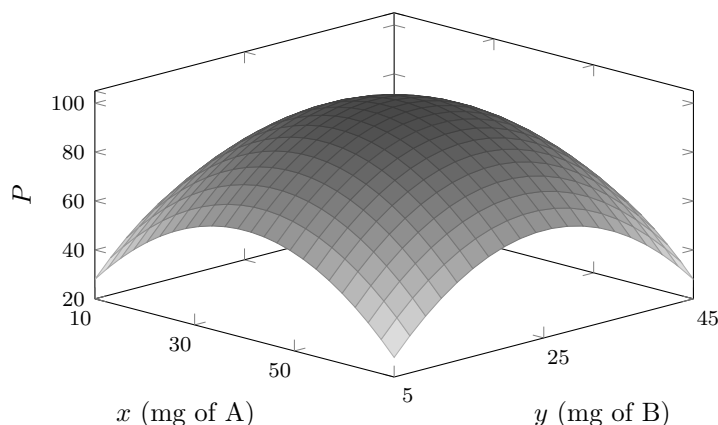
Notebook U1-3_Functions-5_SeveralVariables, §3: building a product set

8.4 The Graph of $f(x, y)$ Is a Surface

Two drugs are given together. Let x be the dose of drug A in mg, y the dose of drug B in mg, and $P(x, y)$ the percent of symptoms relieved. Too little of either does nothing, and too much brings side effects that undo the benefit, so a simple model of the response is

$$P(x, y) = 100 - 0.04(x - 40)^2 - 0.09(y - 25)^2.$$

A one-variable function graphs as a curve over a line. A two-variable function graphs as a **surface** over a plane:



A dose pair (x, y) is a **point on the floor**, the flat plane of all dose combinations, and the **height of the sheet** above that point is the response to that pair. Both squared terms are subtracted from 100, so the sheet is a **dome**: it rises toward a peak and falls away in every direction.

Example 8.8: Reading heights off the dome. Each dose pair is a point on the floor; evaluating P there gives the height of the sheet above it.

$$P(30, 20) = 100 - 0.04(-10)^2 - 0.09(-5)^2 = 100 - 4 - 2.25 = 93.75,$$

$$P(10, 5) = 100 - 0.04(-30)^2 - 0.09(-20)^2 = 100 - 36 - 36 = 28.$$

The first pair is a little below both ideal doses and still relieves nearly 94% of symptoms. The second is far below both and relieves only 28%, a point near the lowest corner of the plotted sheet.

Watch out: a model has a practical domain too

Evaluate at no drug at all: $P(0, 0) = 100 - 0.04(1600) - 0.09(625) = 100 - 64 - 56.25 = -20.25$. A *negative* percent of symptoms relieved is meaningless. The formula is only a sensible model over the dose window it was built for (the plot shows 10 to 70 mg of A and 5 to 45 mg of B). Outside that window the algebra still runs, and the answer is nonsense: the practical-domain lesson of Section 3, now in two dimensions.

Notebook U1-3_Functions-5_SeveralVariables, §4: **drag to rotate the dome**

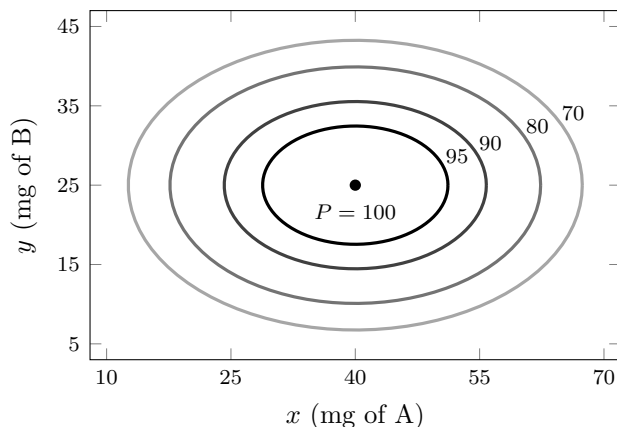
8.5 Level Curves: the Surface Seen from Above

A surface is hard to draw on flat paper. A contour map solves the same problem for mountains, and it works for any function of two variables.

Definition: Level curve

A **level curve** of P collects every input pair with the *same* output c :

$$\{(x, y) : P(x, y) = c\}.$$



Here the level curves for $P = 95, 90, 80$, and 70 are drawn on the floor of dose pairs. Each one is an **ellipse**, an oval, and they shrink toward a single point as c climbs. That point is the **maximum**: the best dose combination, $(40, 25)$, with $P = 100$.

This is how a contour map shows a mountain: rings closing in on the summit. Crowded rings mean a steep climb; rings far apart mean a flat plateau, where a dosing error costs little.

Reading that peak off the picture is what **optimization** does with algebra, in Unit 3.

Example 8.9: Finding points on a level curve. Which dose pairs give exactly 96% relief? The level curve is $P(x, y) = 96$, that is

$$0.04(x - 40)^2 + 0.09(y - 25)^2 = 4.$$

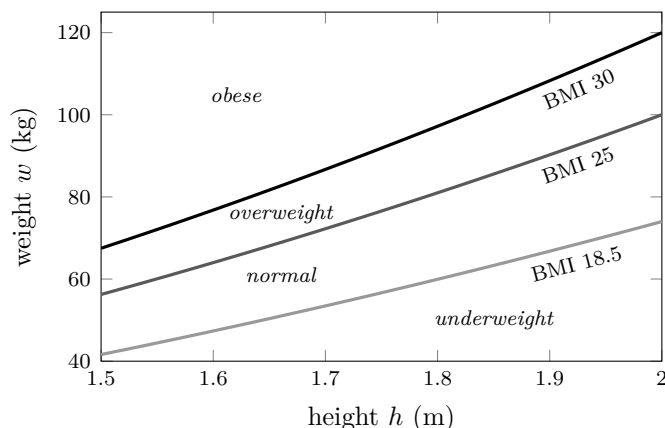
Find where it crosses the two lines through the peak.

- Hold $y = 25$: then $0.04(x - 40)^2 = 4$, so $(x - 40)^2 = 100$ and $x - 40 = \pm 10$. The curve passes through $(30, 25)$ and $(50, 25)$.
- Hold $x = 40$: then $0.09(y - 25)^2 = 4$, so $(y - 25)^2 = 44.4 \dots$ and $y - 25 \approx \pm 6.67$. The curve passes through about $(40, 18.3)$ and $(40, 31.7)$.

So the 96 ellipse reaches 10 mg to either side of the peak in drug A but only about 6.7 mg in drug B: the dose of B must be held more tightly for the same result, the same conclusion Exercise 8.1 reaches another way.

Example 8.10: Level curves of BMI: the category boundaries. BMI has level curves too, and clinicians use them constantly without calling them that. The level curve $\text{BMI}(w, h) = 25$ is

every (weight, height) pair with BMI exactly 25. Solving $w/h^2 = 25$ for w gives the curve $w = 25h^2$: at $h = 1.6$ m it passes through $w = 64$ kg, at 1.8 m through 81 kg. The curves for 18.5, 25, and 30 are the boundaries between the underweight, normal, overweight, and obese categories:

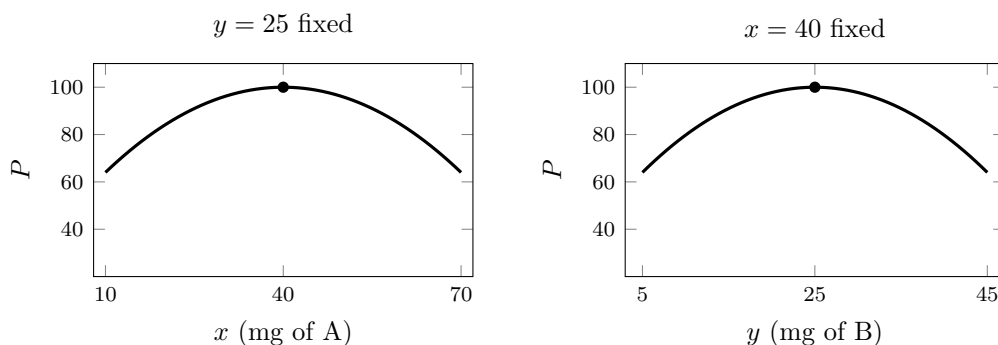


A patient is a point in this plane, and the category is read off by seeing which band the point falls in. The boundaries curve upward because weight has to grow with the *square* of height to keep BMI fixed: a 2.0 m patient reaches BMI 25 only at 100 kg.

Notebook U1-3_Functions-5_SeveralVariables, §5: contour maps, and finding the peak

8.6 Slices: Hold One Input Fixed

Fix one variable at a number and a two-variable function collapses to an ordinary one-variable function, one of the families from Section 4.



Both slices are parabolas, and both peak at 100. Holding drug B at $y = 25$ makes the last term zero and leaves

$$P = 100 - 0.04(x - 40)^2,$$

a downward parabola with its vertex (its highest point) at $x = 40$. Holding drug A at $x = 40$ leaves

$$P = 100 - 0.09(y - 25)^2,$$

with its vertex at $y = 25$. A slice is a vertical cut through the dome, and clinically it asks: **“I have fixed one drug; how much of the other?”**

Example 8.11: Two slices of BMI, from two different families. Slicing BMI shows that the slices of one function need not look alike.

- **Fix the height** at $h = 1.75$ m. Then $\text{BMI} = w/3.0625 \approx 0.327 w$: a **linear** function of weight through the origin, with slope about 0.33 BMI units per kg. At $w = 70$, $\text{BMI} \approx 22.9$. For a given adult, every extra kilogram adds the same amount to BMI.
- **Fix the weight** at $w = 70$ kg. Then $\text{BMI} = 70/h^2$: a **reciprocal** of a square, falling as height grows. At $h = 1.6$, $\text{BMI} \approx 27.3$; at $h = 1.8$, ≈ 21.6 ; at $h = 2.0$, 17.5. The same 70 kg is overweight on a short frame and underweight on a tall one.

One two-variable function, two slices, two different families from Section 4. That is typical: which family a slice belongs to depends on which variable was fixed.

Notebook U1-3_Functions-5_SeveralVariables, §6: slide the slice yourself

Exercise 8.1

Use the two-drug model $P(x, y) = 100 - 0.04(x - 40)^2 - 0.09(y - 25)^2$ for #1 and #2.

1. Why is $P(40, 25) = 100$ the largest value P can take? Argue from the formula, not the picture.
2. Compute $P(40, 35)$ and $P(50, 25)$. Which 10 mg overshoot costs the patient more, and why?
3. For $f(x, y) = x + y$, describe the surface and its level curves. Does it have a highest point?
4. For $g(x, y) = x^2 + y^2$, what shape are the level curves, and is the point at the center a maximum or a minimum?

Solution to Exercise 8.1

1. A square is never negative, so $0.04(x - 40)^2 \geq 0$ and $0.09(y - 25)^2 \geq 0$, and both are *subtracted* from 100. Any dose pair other than $(40, 25)$ makes at least one of those terms positive and so subtracts something. So $P \leq 100$ always, with equality only when both squares are zero, at $x = 40$ and $y = 25$.
2. Substitute:

$$P(40, 35) = 100 - 0.04(0)^2 - 0.09(10)^2 = 100 - 0 - 9 = 91,$$

$$P(50, 25) = 100 - 0.04(10)^2 - 0.09(0)^2 = 100 - 4 - 0 = 96.$$

The same 10 mg overshoot costs more on **drug B** (9 points of relief lost instead of 4), because its coefficient 0.09 is larger than 0.04. The response falls off faster in the y direction, which is exactly why the level-curve ellipses are shorter in that direction.

3. $f(x, y) = x + y$ is a flat, tilted **plane**. Its level curves $x + y = c$ are the lines $y = -x + c$: **parallel lines** of slope -1 , evenly spaced because the tilt never changes. It has **no highest point**: keep increasing x and it keeps rising.
4. The level curve $x^2 + y^2 = c$ (for $c > 0$) is a **circle** of radius $\sqrt{c}$ centered at the origin, so the level curves are concentric rings and the surface is a **bowl**. Here the squares are *added*, so $g \geq 0$ everywhere with $g(0, 0) = 0$: the center is a **minimum**, the dome turned upside down.

8.7 Domain and Range, with Several Inputs

Nothing about D and R changes, except that D is now a set of **tuples**, carved out of a Cartesian product:

$$f : D \rightarrow R, \quad D \subseteq A_1 \times A_2 \times \cdots \times A_p.$$

(The symbol $\subseteq$ is “is a subset of,” from Unit 1-2.) Two examples:

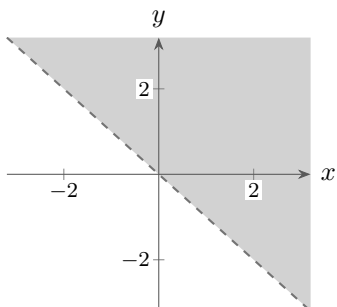
$\text{BMI}(w, h) = w/h^2$. Weight and height are both positive, and h sits in a denominator, so $D = (0, \infty) \times (0, \infty)$: every pair of positive numbers, the open first quadrant of the plane. Every such pair gives a positive value, so $R = (0, \infty)$.

$g(x, y) = \sqrt{9 - x^2 - y^2}$. The root forces $9 - x^2 - y^2 \geq 0$, that is $x^2 + y^2 \leq 9$: D is the **disk** of radius 3 centered at the origin, the circle together with everything inside it, and not a rectangle. The outputs run from $\sqrt{0} = 0$ on the rim up to $\sqrt{9} = 3$ at the center, so $R = [0, 3]$.

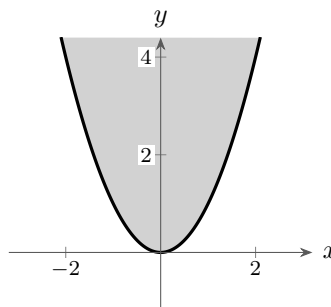
A domain in $\mathbb{R}^2$ is a **region**, not an interval, but it is still “the inputs the formula accepts.”

Example 8.12: Shading a domain in the plane. The recipe of Section 3 works unchanged; only the answer is now a region to shade.

- $f(x, y) = \ln(x + y)$. The log needs $x + y > 0$, that is $y > -x$: every point strictly **above** the line $y = -x$. The line itself is excluded (there $x + y = 0$), so it is drawn dashed.
- $g(x, y) = \sqrt{y - x^2}$. The root needs $y - x^2 \geq 0$, that is $y \geq x^2$: every point **on or above** the parabola $y = x^2$. The boundary is included, so it is drawn solid.



D for $\ln(x + y)$: above $y = -x$



D for $\sqrt{y - x^2}$: on or above $y = x^2$

A third example mixes the two kinds of condition: $k(x, y) = \frac{\sqrt{x}}{y - 1}$ needs $x \geq 0$ (the root) and $y \neq 1$ (the denominator). The domain is the right half-plane, boundary included, with the horizontal line $y = 1$ removed from it: $D = [0, \infty) \times (\mathbb{R} \setminus \{1\})$. When each condition involves only one coordinate, as here, the domain is a Cartesian product; when a condition ties the coordinates together, as $x + y > 0$ does, it is not.

Notebook U1-3_Functions-5_SeveralVariables, §7: shading the allowed region

Exercise 8.2

1. $A = \{1, 2\}$ and $B = \{x, y, z\}$. List $A \times B$, and say how many elements it has.
2. Give D for $f(x, y) = \frac{1}{x - y}$, described in words.

3. Give D and R for $g(x, y) = \sqrt{x} + \ln y$.
4. A room's temperature depends on where you stand and when you measure. What are the inputs, and what is p ?

Solution to Exercise 8.2

1. Pair each element of A with each element of B :

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\},$$

which has $2 \times 3 = 6$ elements. In general $|A \times B| = |A| \cdot |B|$: each of the $|A|$ first entries can be paired with each of the $|B|$ second entries.

2. The only restriction is the division: $x - y \neq 0$, that is $x \neq y$. So D is **the whole plane except the line** $y = x$:

$$D = \{(x, y) \in \mathbb{R}^2 : x \neq y\}.$$

3. The root needs $x \geq 0$, and the log needs $y > 0$. These conditions act on separate coordinates, so the domain is a product:

$$D = [0, \infty) \times (0, \infty).$$

For the range: $\sqrt{x}$ covers $[0, \infty)$ and $\ln y$ covers all of $\mathbb{R}$. Holding $x = 0$ already lets $\ln y$ alone reach every real value, so the sum reaches every real value: $R = \mathbb{R}$.

4. Three space coordinates and one time coordinate, $T(x, y, z, t)$, so $p = 4$, with $D \subseteq \mathbb{R}^4$ and one temperature out.

8.8 More Practice: Several Variables

Exercise 8.3

Use the Cockcroft-Gault formula $\text{CrCl}(a, w, s) = \frac{(140 - a)w}{72s}$ mL/min.

1. Compute CrCl for an 80-year-old man weighing 55 kg with serum creatinine 1.5 mg/dL.
2. Holding weight and creatinine fixed, what kind of function of age is CrCl? What does its slope mean?
3. What practical domain would you impose on s , and why does the formula itself demand part of it?

Solution to Exercise 8.3

1. $\text{CrCl}(80, 55, 1.5) = \frac{(140 - 80) \cdot 55}{72 \cdot 1.5} = \frac{3300}{108} \approx 30.6$ mL/min. Many drug labels adjust the dose when clearance falls below thresholds such as 50 or 30 mL/min, so this patient sits right where the formula starts to matter.
2. With w and s fixed, $\text{CrCl} = \frac{w}{72s}(140 - a)$: a constant times $(140 - a)$, which is a **linear**

function of age with a negative slope, $-\frac{w}{72s}$. For this patient the slope is $-\frac{55}{108} \approx -0.51$: each additional year of age lowers the estimated clearance by about half a mL/min. This is a slice of a three-variable function.

3. The formula divides by s , so $s = 0$ is excluded mathematically, and a negative creatinine would give a negative clearance, which is meaningless. Practically, s must be positive and within a plausible laboratory range; a value like 0.01 mg/dL is almost certainly a data-entry error, and would produce an absurdly large clearance.

Exercise 8.4

1. List $\{\text{low, high}\} \times \{\text{drug, placebo}\} \times \{\text{am, pm}\}$ and count its elements.
2. For the dome $P(x, y) = 100 - 0.04(x - 40)^2 - 0.09(y - 25)^2$, find the slice at $y = 35$ and the dose of A that maximizes it. What is the best possible relief when B is fixed at 35 mg?

Solution to Exercise 8.4

1. A product of three sets is a set of ordered triples, one entry from each set:

$\{ (\text{low, drug, am}), (\text{low, drug, pm}), (\text{low, placebo, am}), (\text{low, placebo, pm}),$
 $(\text{high, drug, am}), (\text{high, drug, pm}), (\text{high, placebo, am}), (\text{high, placebo, pm}) \}.$

That is $2 \times 2 \times 2 = 8$ elements: the eight arms of a three-factor trial.

2. Put $y = 35$: the last term is $0.09(10)^2 = 9$, so the slice is

$$P = 100 - 9 - 0.04(x - 40)^2 = 91 - 0.04(x - 40)^2,$$

a downward parabola with its vertex at $x = 40$. The best relief with B stuck at 35 mg is 91%, reached at 40 mg of A. The overshoot in B costs 9 points that no choice of A can win back, because the two squared terms are separate: adjusting x only changes the first one.

9 Functions in Machine Learning

9.1 What Machine Learning Is

In ordinary programming, **you** write the rule, the way the IV bag rule $m(c) = 500c$ was written from what we know about concentrations. But nobody can write the rule that turns a patient's chart into next year's glucose.

Definition: Machine learning

Machine learning is choosing a function from data. You supply a *family* of candidate rules and a pile of solved examples; the computer picks the member of the family that best reproduces them, and you use it on unsolved cases.

The rule being chosen is a **function**, mapping **features** to a **label**:

$$\underbrace{y}_{\text{label}} = \underbrace{f}_{\text{model}} \left(\underbrace{x_1, x_2, \dots, x_p}_{\text{features}} \right)$$

- A **feature** is a measured quantity used as an **input**: age, BMI, a lab value, a pixel.
- The **label** is the one quantity you want **out**. Learning from examples that already carry their label is called **supervised** learning.
- Which column plays which role is a **decision**, and a feature must exist **when you predict**. A feature measured afterward is a **leak**: it would not be available when the model is actually used, so a model that relies on it looks excellent in testing and fails in practice.
- “Learning” is not understanding; it is **minimization**. The computer picks the member of the family that makes an error as small as possible, as Section 9.6 shows.

Example 9.1: Features, label, and a leak. A hospital wants to predict, on the day of discharge, whether a patient will be readmitted within 30 days.

- **Label:** readmitted within 30 days, yes or no.
- **Reasonable features:** age, number of admissions in the past year, length of the current stay, discharge diagnosis, number of medications at discharge. All are known on the day of discharge.
- **A leak:** “number of clinic visits in the 30 days after discharge.” It may predict readmission very well in historical data, since sicker patients visit more, but on the day of discharge it has not happened yet. A model that uses it cannot be used when it is actually needed.

The test for every candidate feature is the same question: *would I have this number at the moment I make the prediction?*

Example 9.2: Regression or classification? The kind of output, that is, the range of the model, decides which it is.

Predict	Range of the model	Task
length of hospital stay, in days	a set of numbers	regression
next month’s HbA1c	a set of numbers	regression
readmitted within 30 days	{yes, no}	classification
blood type from a lab image	the 8 blood types	classification

Some tasks can be posed either way: “HbA1c at least 6.5%, yes or no” turns the second regression into a classification, by composing the numeric prediction with a threshold, a step function from Section 4.

9.2 Start with a Table

Six patients, three features each, and one label: their fasting glucose six months from now.

	features, the inputs x			label, the output y
patient	age	BMI	systolic BP	glucose in 6 mo.
A	54	27.4	138	110
B	61	22.8	122	101
C	47	31.5	145	117
D	58	24.1	118	102
E	43	33.2	150	116
F	66	28.6	131	116

- There are three features here, so $p = 3$, and six patients, so $n = 6$.
- Each **row** is one patient: three numbers in, one number out.
- The double line is the whole problem. Everything left of it we have on the day we predict; the column right of it we will not see for six months.

A model is a rule that turns the left of that table into the right, for patients who are not in the table.

Notebook U1-3_Functions-5_SeveralVariables, §8: the table as a set of tuples

9.3 That Table Describes a Function

Read one row of the table as a single input and a single output. For patient A,

$$f(\underbrace{54}_{\text{age}}, \underbrace{27.4}_{\text{BMI}}, \underbrace{138}_{\text{BP}}) \approx \underbrace{110}_{\text{glucose}}.$$

The sign is $\approx$ (“approximately equal”), not $=$: we hope for a rule that comes close, not one that is exact.

- Three inputs and one output, so f is a function of $p = 3$ variables, the subject of Section 8.
- Its **domain** is the set of measurement triples that make sense, the ages, BMIs, and pressures a real patient could have: a region inside $\mathbb{R}^3$.
- Its **range** says what kind of answer comes out. Here it is a number, which makes this **regression**; a label from a finite set such as {diabetic, not diabetic} would make it **classification**.

The goal: find one rule f that comes close on *all six* rows at once and, far more importantly, on patients we have never measured.

Notebook U1-3_Functions-5_SeveralVariables, §9: features in, label out

9.4 Naming the Rows and Columns

Six rows is small enough to point at. A real table has thousands, so we need names, and each name carries **two** indices: which row, and which column.

	feature 1	feature 2	feature 3	label
patient 1	x_{11}	x_{12}	x_{13}	y_1
patient 2	x_{21}	x_{22}	x_{23}	y_2
$\vdots$				$\vdots$
patient n	x_{n1}	x_{n2}	x_{n3}	y_n

- x_{ij} is one **cell**: row i , column j . In the table of Section 9.2, $x_{12} = 27.4$, patient A’s BMI (row 1, feature 2).
- A whole **row** is the tuple $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})$: one input to f .
- y_i is that row’s label, so the n rows give the pairs $(x_1, y_1), \dots, (x_n, y_n)$, called the **training data**.

Read (x_i, y_i) out loud as “patient i ’s measurements, and patient i ’s answer.” The goal is $f(x_i) \approx y_i$ for every i .

Example 9.3: Reading the indices. In the six-patient table, with patients A to F as rows 1 to 6 and the features in the order age, BMI, systolic BP:

- x_{32} is row 3, column 2: patient C’s BMI, 31.5.
- x_{53} is row 5, column 3: patient E’s systolic pressure, 150.
- y_4 is patient D’s label, a glucose of 102.
- x_2 , with a single index, is patient B’s whole row, the triple (61, 22.8, 122).

The first index always picks the row (the patient) and the second the column (the feature). Getting them backward is the table version of swapping the arguments of BMI: x_{23} is patient B’s blood pressure, 122, while x_{32} is patient C’s BMI.

9.5 Inputs vs. Parameters

Look again at the linear function, now with its two roles labelled:

$$f(x) = \underbrace{m}_{\text{parameter}} \underbrace{x}_{\text{input}} + \underbrace{b}_{\text{parameter}}$$

- The **input** x changes from patient to patient.
- The **parameters** m and b are fixed once the model is trained: they identify *which* line, out of the whole family of lines, we are using.

Two decisions, not one

Choosing a family (linear? exponential? something with a bend?) is **modeling**. **Choosing the parameters within that family** is **fitting**.

The families of Section 4 are the menu you are choosing from.

Example 9.4: Spotting the parameters. Which symbols are inputs and which are parameters?

- The IV bag, written generally as $m(c) = Vc$: the concentration c is the input; the bag volume V is a parameter, fixed at 500 for one bag and 250 for another.
- The clearance model $A(t) = a \cdot b^t$: the time t is the input; the starting amount a and the factor b are parameters. Fitting it to a patient's drug levels means choosing a and b .
- The wave $A \sin(\omega t + \phi)$: t is the input; A , ω , and ϕ are parameters. The formula-explorer notebook's sliders are exactly the parameters.

A useful rule: the input is what changes from one case (patient, moment, row) to the next; a parameter is what stays fixed once the model has been chosen.

9.6 Error Is a Function Too

To keep the formulas short, take a single feature, so each row's input x_i is one number, and use the linear model. Write the model's answer for row i as $\hat{y}_i$, read “ y hat,” to keep it apart from the truth y_i . The error is then a function of the *parameters*, not of x :

$$\begin{aligned}\hat{y}_i &= mx_i + b && \text{the prediction for row } i \\ E(m, b) &= \sum_{i=1}^n \left(\underbrace{y_i}_{\text{truth}} - \underbrace{\hat{y}_i}_{\text{prediction}} \right)^2 && \text{total squared miss} \\ &= \sum_{i=1}^n (y_i - (mx_i + b))^2 && \text{the same thing, in full}\end{aligned}$$

- In the last line every x_i and y_i is a fixed *number* from the table; the only symbols still free are m and b . So E is a function of two variables, m and b .
- Squaring makes every miss count as positive, so errors above and below the truth cannot cancel out.
- **Training** is finding the m and b that make E smallest, and “find the input that minimizes a function” is **optimization**, the subject of Unit 3.
- Notice the reversal: the symbols that were *unknowns to estimate* in the previous subsection are the *inputs* here. That is the central move in all of model fitting.

Example 9.5: The error of three different lines. Take a tiny data set of three points, $(x_i, y_i) = (1, 2), (2, 3), (3, 5)$, and compute $E(m, b)$ for three candidate lines. For each line, predict every row, subtract from the truth, square, and add.

Line	Predictions at $x = 1, 2, 3$	Misses $y_i - \hat{y}_i$	$E(m, b)$
$m = 1, b = 1$	2, 3, 4	0, 0, 1	$0 + 0 + 1 = 1$
$m = 2, b = 0$	2, 4, 6	0, -1, -1	$0 + 1 + 1 = 2$
$m = 1.5, b = 0.3$	1.8, 3.3, 4.8	0.2, -0.3, 0.2	$0.04 + 0.09 + 0.04 = 0.17$

Three choices of the parameters, three values of E : that is what it means for E to be a function of (m, b) . The data never changed; only the line did. Of these three, the last fits best. The very best

line for these points, which Unit 3's methods find, has $m = 1.5$ and $b = \frac{1}{3}$, with $E \approx 0.167$, just below the 0.17 we stumbled on. Notice also the second line: it is exact on the first row but misses the other two, and squaring counts each of those misses as positive, so they cannot cancel.

Notebook U1-3_Functions-5_SeveralVariables, §10: the error bowl and its minimum

9.7 Many Features, Same Function

With p features, the linear model just grows one term per feature:

$$f(x_1, x_2, \dots, x_p) = w_1x_1 + w_2x_2 + \dots + w_px_p + b.$$

- The **inputs** are the x_j : one tuple from D , which is one row of the table.
- The **parameters** are now $w_1, \dots, w_p$ and b : $p+1$ numbers to fit instead of two, and the error $E(w_1, \dots, w_p, b)$ is a function of all of them.
- Nothing conceptual changed: one output per input, a domain, a range, composition, and sometimes an inverse.

Example 9.6: Counting what has to be fitted. For our table, $p = 3$, so the model has w_1, w_2, w_3 and b : four parameters, fitted from six rows. A model built on 40 columns of an EHR extract has 41 parameters. With p features, the error E is a function of $p+1$ variables, and the “error surface” lives in a space with one more dimension than that. Nobody can draw it once p is more than 1, which is exactly why the rest of the course replaces pictures with algebra.

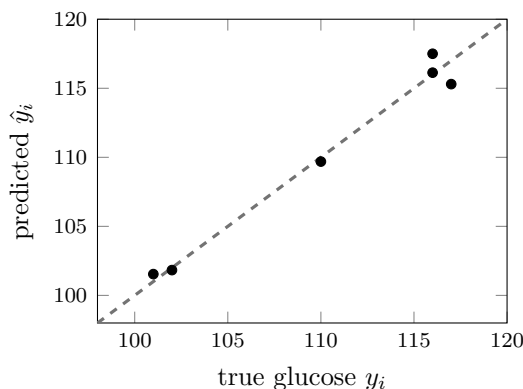
The bookkeeping is the only thing that grew. Keeping track of many numbers at once is exactly what Unit 2's **vectors** and **matrices** are for, and minimizing E over $p+1$ parameters is Unit 3's problem rather than this lecture's.

9.8 Solving It: One Rule for All Six Patients

Minimizing E over the six rows of our table returns one specific member of the linear family:

$$\hat{y} = 0.458x_1 + 1.841x_2 + 0.181x_3 + 9.57,$$

where x_1 is age, x_2 is BMI, and x_3 is systolic blood pressure. The weights say, for example, that each extra year of age adds about 0.46 to the predicted glucose when the other two features are held fixed.



Each dot is one patient, with the true label across and the prediction up. The dashed line is $\hat{y}_i = y_i$, **perfect prediction**, so a dot's vertical gap from that line is that patient's error.

How wrong is it? Three standard summaries of the misses:

$$\begin{aligned} \text{SSE} &= \sum_i (y_i - \hat{y}_i)^2 = 5.57 && \text{sum of squared errors, the } E \text{ we minimized} \\ \text{RMSE} &= \sqrt{\text{SSE}/n} = 0.96 \text{ mg/dL} && \text{root mean squared error} \\ \text{MAE} &= \frac{1}{n} \sum_i |y_i - \hat{y}_i| = 0.72 \text{ mg/dL} && \text{mean absolute error} \end{aligned}$$

RMSE is in the label's own units: a typical miss of about 1 mg/dL. That looks excellent, but be suspicious. Six rows against four parameters (w_1, w_2, w_3, b) is nearly an even match: with that much freedom, the rule can bend itself to fit these particular six patients almost exactly, so a small error here says little about how it will do on new patients. This fit flatters itself.

Example 9.7: Using the fitted rule on a new patient. A new patient is 50 years old with BMI 30 and systolic pressure 140. The fitted rule is an ordinary function of three variables, so evaluate it slot by slot:

$$\begin{aligned} \hat{y} &= 0.458(50) + 1.841(30) + 0.181(140) + 9.57 \\ &= 22.90 + 55.23 + 25.34 + 9.57 = 113.04 \text{ mg/dL}. \end{aligned}$$

So the model predicts a fasting glucose of about 113 mg/dL in six months. This is the whole point of fitting: the rule was chosen using patients A to F, and it is used on someone who was never in the table.

Example 9.8: What a weight means. The BMI weight is 1.841. Take two patients identical in age and blood pressure whose BMIs differ by 5. The model's predictions for them differ by $1.841 \times 5 \approx 9.2$ mg/dL, since every other term is the same for both and cancels. That is what “holding the other features fixed” means: a weight is the slope of one slice of the model.

A small detail: evaluating the printed rule at patient A's row gives 109.72, while the plotted dot sits at 109.69. The difference is rounding. The computer fitted the weights to many more digits than the three printed here, and the plot used the unrounded ones. A reported model is usually a rounded copy of the fitted one.

Example 9.9: Computing the three error summaries. For the tiny data set of Section 9.6 and the line $m = 1.5$, $b = 0.3$, the misses were 0.2, -0.3 , and 0.2, with $n = 3$:

$$\begin{aligned} \text{SSE} &= 0.2^2 + (-0.3)^2 + 0.2^2 = 0.04 + 0.09 + 0.04 = 0.17, \\ \text{RMSE} &= \sqrt{0.17/3} \approx \sqrt{0.0567} \approx 0.238, \\ \text{MAE} &= \frac{1}{3}(|0.2| + |-0.3| + |0.2|) = \frac{0.7}{3} \approx 0.233. \end{aligned}$$

The SSE grows with the number of rows, so it is hard to compare across data sets; RMSE and MAE are averages, in the label's own units, and are the numbers usually reported. MAE is the plain average distance of a miss, using the absolute value of Section 4. RMSE squares before averaging, so it counts one large miss more heavily than several small ones.

Watch out: the model inherits a practical domain from its data

The six patients were aged 43 to 66, with BMIs from 22.8 to 33.2. The fitted rule will happily return a number for a 20-year-old with BMI 45, but no patient remotely like that was used to choose it. Predictions far outside the ranges of the training data are extrapolation, the dosing-chart trap of Section 3 in a new setting.

Notebook U1-3_Functions-5_SeveralVariables, §11: fit it yourself, and plot these residuals

9.9 More Practice: Functions in Machine Learning

Exercise 9.1

A model predicts length of stay in days as $\hat{y} = 0.05a + 1.2c + 0.8$, where a is age in years and c is the number of chronic conditions.

1. Identify the inputs, the parameters, and p . Is this regression or classification?
2. Predict the stay for a 70-year-old with 3 chronic conditions.
3. Two patients have the same age; one has 2 more chronic conditions. How do their predictions differ?
4. For three patients, the true stays were 6, 4, and 9 days, and the model predicted 5.5, 4.5, and 7.5. Compute the MAE and the RMSE.

Solution to Exercise 9.1

1. Inputs (features): a and c , so $p = 2$. Parameters: the weights 0.05 and 1.2 and the intercept 0.8, three numbers fixed by training. The output is a number of days, so this is **regression**.
2. $\hat{y} = 0.05(70) + 1.2(3) + 0.8 = 3.5 + 3.6 + 0.8 = 7.9$ days.
3. The age terms are equal and cancel, so the predictions differ by $1.2 \times 2 = 2.4$ days, the patient with more conditions predicted to stay longer.
4. The misses $y_i - \hat{y}_i$ are 0.5, -0.5 , and 1.5. Then

$$\text{MAE} = \frac{1}{3}(0.5 + 0.5 + 1.5) = \frac{2.5}{3} \approx 0.83 \text{ days,}$$

$$\text{RMSE} = \sqrt{\frac{1}{3}(0.25 + 0.25 + 2.25)} = \sqrt{\frac{2.75}{3}} \approx \sqrt{0.917} \approx 0.96 \text{ days.}$$

RMSE is larger than MAE here because the single 1.5-day miss, once squared, dominates the average.

Exercise 9.2

Three rows of data are $(x_i, y_i) = (0, 1), (1, 3), (2, 4)$. Compute the total squared error $E(m, b)$ for the lines $m = 2, b = 1$ and $m = 1.5, b = 1$. Which fits better? Then compute the RMSE of the better one.

Solution to Exercise 9.2

- *Line* $m = 2$, $b = 1$. Predictions at $x = 0, 1, 2$ are 1, 3, 5; misses $y_i - \hat{y}_i$ are 0, 0, -1 ; so $E(2, 1) = 0 + 0 + 1 = 1$.
- *Line* $m = 1.5$, $b = 1$. Predictions are 1, 2.5, 4; misses are 0, 0.5, 0; so $E(1.5, 1) = 0 + 0.25 + 0 = 0.25$.

The second line has the smaller error, so it fits better. Its RMSE is $\sqrt{0.25/3} = \sqrt{0.0833\dots} \approx 0.29$, in the units of y . (Neither line is the best possible one: that turns out to be $m = 1.5$, $b \approx 1.17$, with $E \approx 0.167$. Finding it is Unit 3's job; comparing candidates, as here, is this unit's.)

10 Review

What we covered: one variable

1. A **relation** is any set of ordered pairs; a **function** pairs *each* input with *exactly one* output.
2. Many-to-one is allowed, one-to-many is not: the **vertical line test**.
3. **Domain** D is the set of valid inputs, and **range** R is the set of outputs that actually occur. Watch $1/x$, $\sqrt{x}$, and $\log x$.
4. Four **families** (polynomial, exponential, logarithmic, trigonometric), plus piecewise rules like ReLU, and the five **transformations**.
5. **Composition** $(f \circ g)(x) = f(g(x))$ runs inside out, and order matters; a pipeline is a composition.

What we covered: quantifiers, inverses, several variables

6. **Quantifiers**: $\forall$ (every), $\exists$ (some), $\exists!$ (exactly one); negating one flips it, and their order matters.
7. An **inverse** f^{-1} undoes f ; it exists exactly when f is **one-to-one**, and its graph is the reflection across $y = x$.
8. With **several variables** the input is a tuple and D lives in a **Cartesian product**; one row of a data table is one such tuple.
9. Two inputs graph as a **surface**; **level curves** show it from above, and a **slice** at a fixed input is a one-variable curve.
10. A **model** is a function from a table's features to its label; its **error** is a function of the parameters, and minimizing it is Unit 3.

Notebooks:

- U1-3_Functions-1_DomainRangeAndFamilies.ipynb
- U1-3_Functions-2_Composition.ipynb
- U1-3_Functions-3_Inverse.ipynb
- U1-3_Functions-4_FormulaExplorer.ipynb
- U1-3_Functions-5_SeveralVariables.ipynb

Previous: Unit 1-2, **Sets**.

Next: Unit 1-4, **Series**.